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Ion-Atom Charge Transfer Reactions at Ultra-Cold Temperatures

Abstract

We study radiative charge-transfer reactions and the radiative association of Yb with Ca^+ in the cold-to-ultracold temperature range. Our findings confirm the validity of Langevin behavior at temperatures in the single Kelvin range. However, we also show that this behavior breaks down at ultracold temperatures, resulting in a significant isotope effect.

We explore resonant charge transfer (RCT) reactions in two-dimensions for the $H^+ + H$ system. We introduce a toy model in which the collision partners are effective anyons and investigate how their introduction affects RCT rates.

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CHAPTER 1

INTRODUCTION

1.1 MOTIVATION

Charge transfer refers to the movement of electrons or ions from one species (atom, molecule, or ion) to another. This transfer can occur through various mechanisms, including chemical reactions, electrochemical processes, and interactions with electromagnetic fields. Charge transfer can occur through several mechanisms, including chemical reactions, Electrochemical processes, Photoinduced charge transfer, and ionization.

- Chemical reactions
  Charge transfer occurs when electrons are exchanged between the reactants.

- Electrochemical processes
  Electrochemical reactions involve the transfer of electrons between electrodes and electrolytes. This is the main process in batteries, where the

transfer of electrons creates an electric current.

- Photo-induced charge transfer

  In this scenario, the charge transfer is induced by light. It is the process in photosynthesis, photovoltaic cells, and optical devices.

- Ionization processes

  These processes often occur in plasmas or atmospheric chemistry. The charge transfer in this environment leads to the formation of ions. A large part of the visible universe is plasma, including stars, nebulae, interstellar space, and intergalactic space. Understanding charge transfer leads to a better understanding of the formation of these cosmic bodies.

One promising platform, topological quantum computing[1;2;3;4;5], relies on the unusual quantum properties of 2-dimensional (2D) matter. In the past few decades, there have been great strides in creating 2D materials that promise novel applications.[6] To facilitate this emergent technology, it is imperative that we understand, at a fundamental level, the processes that govern the behavior of matter at the atomic and molecular levels. In this thesis, we will explore ions' interactions with the neutral matter in a 2D environment. Specific topics include non-radiative and radiative charge transfer reactions. Such processes are important in 3D gases, as in astrophysical and laboratory plasma environments, and have attracted much attention[7]

In this thesis, we will calculate the rate of the radiative and charge transfer reactions between a positively charged ion and neutral 2D hydrogen at very low

collision energies. In our formalism, the particles are constrained to move in a 2D manifold, but the emitted radiation can propagate in 3D. This study will provide insight into how constrained 2D dynamics affect the rate coefficient for radiative charge transfer.

1.2 Some Applications of the 2D Quantum Mechanics

Quantum Mechanics was developed during the first quarter of the 20th century, and its methods have been applied to increasingly complex systems. QM has allowed the development of novel electronic components, lasers, high-speed telecommunications, new diagnostic technologies in medicine, and new materials.

In this introduction, we discuss two major applications. One is a creation of the novel, 2-dimensional materials. The second is the application of 2D materials to quantum computing.

1.3 New Materials

The application of QM to solid-state systems has led to a deeper understanding of the properties of materials. That understanding has led to the creation of new materials, including so-called two-dimensional (2D) materials.

2D materials can exhibit unique topological properties that do not exist in the 3D case. An exciting and important feature in this context is the emergence of anyons[8]. Anyons are neither Fermions nor Bosons. The exchange of two

anyons induces an arbitrary relative phase between the partners; an effect called fractional statistics. In 3D space, the relative phase can only take the integer values of 2π (Bosons) or π (Fermions).

1.4 On 2D Materials No-Go Theorems and Graphene

Landau and Pierls[9 10] initially argued that 2D materials could not exist. That argument posited that the displacement of atoms at finite temperatures induces thermal fluctuations that are an order of the atomic distance and lead to melting of the 2D material at temperatures higher than $0K$.

There is an even stronger statement against 2D materials, in the form of the Mermin-Wagner-Hohenberg (MWH) theorem[11 12]. In these papers, the authors show that 2D systems with short-range interactions are unstable and cannot exist. Using thermodynamics arguments, one can show that a 3D system is stable if its free energy is bounded from below and convex (upwards). In this way, the system can minimize the energy of the ground state. Because higher-order fluctuations are finite, the system is stable. The MWH argument posits that, in 2D systems, fluctuations around the ordered state (lattice in the 2D case) decorrelate over large distances, thus destroying the large-scale order. Therefore, it seemed that the possible existence and applications of the 2D materials were quite limited. The only 2D materials considered stable were thin molecular films, which could form on a solids' surface. Dash[13] predicted that 2D materials exist as a thin film either on the surface of the boundaries of 3D materials.

Despite those arguments, it has been speculated that stable 2D materials are possible[13]. The situation changed dramatically in 2004 when 2D carbon (graphene) was discovered and created in the laboratory[14;15] at the University of Manchester, Great Britain.

2D materials have qualitatively different topological properties[16]. Moreover, those properties do not depend on the type of microscopic interactions between particles. While this subject is beyond the scope of this thesis, it does show that the field of 2D materials is a fertile ground for both fundamental and applied research. Interestingly, it has been shown[17] that it takes ten layers of graphene for it to start behaving like a 'regular' 3D material. The existence of graphene does not contradict the above-mentioned no-go theorem[12], but it does circumvent it[18]. In graphene, thermal oscillations produce ripples or bending of the graphene sheet. That way, graphene behaves like an elastic membrane that allows phonons to propagate in 2 dimensions and couple in a plane stretching along the transverse fluctuations out of the plane. These phonons mediate a long-range interaction, thereby circumventing the Mermin-Wagner no-go theorem.

Additionally[14 19 20],2D materials exhibit several other interesting properties. Its quantum excitations behave in a manner described by relativistic dynamics. Usually, in condensed matter systems, the Schrodinger equation's solutions describe the material's electronic properties. In graphene, the electrons obey the non-relativistic Schrodinger equation, but interactions with the graphene lattice's periodic potential give rise to solutions referred to as quasiparticles

excitations, and they behave in the same manner as massless Dirac fermions. Even at low energies, these quasiparticles are described by a Dirac equation in 2+1 dimensions.

1.4.1 EXCITONS

These excitations represent a wide class of intrinsic electronic excitations (Excitons)[21;22] in crystals of semiconductors and dielectrics. In one model, Excitons represent a bound state of an electron and a hole, typically formed when an incident photon is absorbed, exciting an electron from the valence to the conduction band. The attractive Coulomb interaction between the excited electron and the newly created hole binds them together to form a bound neutral compound system of the two charge carriers similar to a hydrogen atom. The character of exciton motion depends on the strength of the exciton interaction with phonons. The electron-hole model is common in crystals of insulators and semiconductors. Ya. I. Frenkel (1931) introduced excitons to explain the light absorption in crystals, which does not lead to photoconductivity. In the Frenkel model, the exciton is considered as an electronic excitation of one crystal site with the energy close to, but a bit smaller than that necessary for the excitation of a free electron. Due to the crystal's translation symmetry, the exciton can move along the lattice sites transferring the energy to the electrically active or luminescence centers.

Similar excitations occur in 2D systems.[23;24]. 2D excitons promise to form a building block of the 2D electronics.[25]. Although exciton-based transistor

actions have been demonstrated successfully in bulk semiconductor-based coupled quantum wells, the low temperature required for their operation limits their practical application. The emergence of 2D semiconductors with large exciton binding energies may lead to excitonic devices and circuits that operate at room temperature.[25].

1.4.2 GRAPHENE

Graphene exhibits similar properties[14], such as its Quantum Electrodynamics (QED) like electronic spectrum, electron tunneling. Another interesting consequence of the QED like electron spectrum is the possibility of experimentally studying QED in curved space by controllable bending of a graphene sheet. Such study may offer a possibility to address a certain class of cosmological problems.

At present, since the discovery of graphene, several new 2D materials have been identified[26]. As the new 2D materials are getting discovered, people are investigating their possible applications. One of the very useful applications for any exotic materials is electronics. These new 2D materials promise that they can be used to build existing electronic components, but with potentially better performances[27 28].

Nanotubes are rolled-up graphene sheets, and depending on the application, graphene (as a sheet) is sometimes superior, sometimes inferior, and sometimes completely different. The other fascinating application is the appearance of high-temperature superconductivity in 2D materials[29].

2D Electronics

Driven by a demand for higher performance and lower consumption in electronic devices, 2D materials have found application in that arena[27],[28]. There is the hope that graphene can be used to build active electronic components, such as diodes, transistors (FET), and other electronic components. Alternatively, graphene may also be used as a conductor in electronic components and batteries due to its conductivity. graphene itself is a zero-gap semiconductor. There is another unique opportunity in 2D semiconductors. In a 3D semiconductor, electronic states are buried deep inside the material. Unlike the surface state, electron states inside the material are localized, i.e., the wave function has an exponentially decaying tail.

In 2D semiconductors, the electronic states exist on the surface. The surface states admit both localized and non-localized, Block bulk states[30]. Another interesting application of 2D electronics is the LED elements in 2D[31]. The electronics research is closely related to the research of new 2D materials. For example, graphene itself is not compatible with silicon. However, there is a new material, black phosphorus[32], which is both compatible with silicon and it holds promise for future electronic[33] devices. This fact relies on two key properties; the first is that black phosphorus has higher mobility, the speed at which it can carry an electrical charge, than silicon. The other is that it has a finite bandgap, whereas graphene does not. So, in essence, in the presence of an electric field, black phosphorus can act as a semiconductor[34].

QUANTUM COMPUTING

The current computer architecture is based on the so-called 'von Neumann' architecture. The algorithms based on that architecture have some inherent limits, and as it currently stands, the architecture cannot overcome those limits.

There are two main classes of computer algorithms, P and NP. There are subclasses within those classes, but those two classes capture the essential behavior of classical algorithms. The algorithms belonging to class P are solvable in polynomial time, with regard to the size of the input. Those algorithms belonging to class NP are those which require exponential time to complete[35].

Currently, the major unsolved problem in Computer Science is the P vs. NP problem[35]. While the hypothesis that $P \neq NP$ is generally accepted, no proof has been found. Applying quantum mechanical principles in engineering disciplines, such as computer science[36] promises to bring us the much qualitatively more powerful quantum computer.

Now all problems in the NP class (more precisely, NP-complete subclass) can be transformed into each other in polynomial time to each other[37], therefore finding a polynomial-time solution for one of them would amount to having a polynomial-time solution to all problems in NP class. So the goal is rather worthy.

Currently, it is known that the quantum computer will not be able to solve NP problems. They could solve the class of the bounded-error, quantum, polynomial time (BQP) problems[38]. Even if the NP problems remain intractable, just the possibility that according to Feynman[36], a quantum computer can be used to simulate the quantum processes, which will in itself be a step forward.

For example, Schor's algorithm[39] can provably solve integer factorization in a polynomial time, as opposed to the best classical algorithm (quadratic sieve), which works in sub-exponential time[40]. Unfortunately, the integer factorization problem, as well as other BQP problems, do not belong to the class of NP-complete problems.

Interestingly, according to Abrams and Lloyd[41],[42] if the small nonlinear term is added to Quantum Mechanics, quantum computers would be able to solve NP-complete problems in polynomial time.

Topological Quantum Computing

The concept of topological quantum computing[1] depends on the unique properties of 2D materials. In principle, there are no theoretical objections to building a quantum computer; there are serious technical issues, such as noise and scaling limitations, for current qubit technologies that need to be overcome[43]. The topological approach promises a physically realizable quantum computer[2;3].

ANYONS

Anyons are quasi-particles that can arise in 2D systems and whose quantum statistics in neither Fermionic nor Bosonic[44]. This fact has been proven[45], but arbitrary statistics are valid only in 2 dimensions[46]. After exchanging two identical particles, the wave function gains an arbitrary phase factor $\Psi = e^{i\theta}\Psi$, where $\theta = 2\pi\nu^s$. In contrast, for a 3D system, the exchange of two particles leads to either phase factor π or 0, for fermions and bosons, respectively.

In general, there are two kinds of Anyons, Abelian and non-Abelian. Abelian Anyons have been reliably detected in systems exhibiting the Fractional Quantum Hall Effect[47]. Whereas the computation with Abelian Anyons is theoretically possible[48], considerably more attention has been paid to non-Abelian Anyons. There are several possibilities for realizing non-abelian statistics experimentally:

- A two-dimensional electron gas in a large magnetic field (the fractional quantum Hall effect)
- Rapidly rotating Bose-Einstein condensates[49]
- Frustrated magnets[50].
- One of the anyons candidates the Majorana fermion[51].

Majorana fermion (particle) the particle that is its own antiparticle. A pair of localized Majorana states has been predicted to reside at the ends of a specially designed superconducting wire. Very recently (2020)[52] there is a pos-

sibility that Majorana particles has been observed for the first time. The other property of the Majorana fermion is that it is a non-Abelian anyon. This present the possibility for the realization of fault-tolerant topological quantum computing[53].

EXCITONS AND TRIONS

Other interesting quasiparticle phenomena are called excitons[23]. It is a bound state of a hole and an electron in a solid-state material. Excitons allow the transport of energy without the transport of an electric charge. There are two types of excitons:

1. Frenkel excitons are found in organic molecular crystals[54] and have binding energies on order of $1eV$ and radii 10Å.

2. Wannier excitons are found in semiconductors[22] and have binding energies on the order of $1meV$ with radii 100Å.

Excitons have been observed in 2D materials[25] as well as 2D excitons in 3D materials. The promise here is that these kinds of quasi-particles can help create a new kind of electronic components, typically opto-electronics.

Since the exciton system resembles either the hydrogen atom or the hydrogen molecule, and it seems conceivable that the dynamics of H_2^+ molecule is qualitatively similar and could be applied to the excitons systems as well.

The similar system is the trion, a bound state of three charged particles. It consists of either 2 electrons and a hole, or 2 holes and an electron, therefore

resembling the hydrogen molecular ion.

1.5 Ion - Atom Charge Transfer

Charge transfer refers to the movement of electrons or ions from one species (atom, molecule or ion) to another. This transfer can occur through various mechanisms, including chemical reactions, electrochemical processes and interactions with electromagnetic fields. The charge transfer can occur through several mechanisms, such as Chemical reactions, Electrochemical processes, Photo induced charge transfer, Ionization processes.

- Chemical reactions Charge transfer occurs when electrons are exchanged between the reactants.
- Electrochemical processes Electrochemical reactions involve the transfer of electrons between electrodes and electrolytes. This is the main process in batteries, where transfer of electrons create an electric current.
- Photo induced charge transfer In this scenario the charge transfer is induced by light. It the process in photosynthesis, photovoltaic cells and optical devices.
- Ionization processes These process often occur in plasmas or atmospheric chemistry. The charge transfer in these environment leads to the formations of ions. The large part of visible universe is plasma, including stars, nebulae interstellar space, and intergalactic space. Understanding charge transfer leads to better understanding of the formation of these

cosmic bodies.

The charge transfer is a major part of these processes: Chemical reactivity, Energy conversion and storage, Biological processes, Creating electronic devices.

In this thesis we shall consider a non-relativistic charge transfer that occurs at low energies in 2 dimensions (2D).

Given that the universe is 3 dimensional, obviously most of the processes and application listed above are not applicable in 2D. But we have already seen that there exist a 2D materials. Thus the 2D charge transfer calculations can be applied to those kind of materials. Those 2D materials also offer promises of a new electronic devices, whose behaviour is governed by the charge transfer.

We address the charge transfer processes in more details in chapter 2

Future Applications and Speculations

While the applications of the 2D materials above are certainly interesting, creating an exact analytical model of such material is extremely complex. Even with today's computing machinery, the many-body wave equation is intractable.

Equations for many bodies systems may formally be created by extending equations for a one or two bodies systems. But as soon as one tries that, one realizes that the complexity of such equations increases enormously, and than

assumptions made for a small systems are not valid any more. In this author's opinion, moving from a simple system to an order of magnitude more complex, requires a completely new mathematical and physical framework. Something analogous between the classical mechanics and thermodynamics.

OUTLINE

In this thesis, we will focus on investigating and solving the radiative charge transfer for the most straightforward molecular system, namely hydrogen molecular ion H_2^+ in 2 dimensions (2D in the further text), and solving such system in 2D exactly. Chapter 2 we will compare results applying our method to the already solved and known H_2^+ results in 3D. Then in chapter 3, we will introduce the procedure to analytically solve the H_2^+ in 2D. After providing an analytical solution, we will fall back to the numerical methods to calculate the energy levels and obtain the potential curves. In chapter 4, we use the results obtained to calculate the dipole moment and the cross section for the radiative charge transfer process.

In the following chapter, we re-calculate the electron energies for hydrogen molecular ion in 3D by using a different and arguably simpler approach.

We shall use analytic methods where possible and resort to the numerical approach when there is no hope of finding an analytic solution. Quantitative calculations have greatly enabled the advancement of computing technology, as it is now possible to solve partial differential equations relatively quickly us-

ing computers that are widely accessible. However, applying approximate methods and using only numerical methods allow limited insights. While the numerical methods provide results, they do not provide much insight into a quantum system's behavior. Furthermore, unless one is cautious, systemic numerical errors can accumulate and lead to neither insightful nor correct results. Therefore, analytic solutions are almost always desirable. However, for moderately complex systems, closed analytic solutions are usually impossible to obtain. Nevertheless, it is still desirable to find an exact analytic solution, if possible, and fall back to numerical methods only as a last step.

Finally, while naturally occurring 2D atoms and molecules do not exist, excitons are proven to exist, and they are analogous to the 2D atoms. So we explore the H_2 molecule so that we gain insights that can be translated into excitons.

CHAPTER 2

RADIATIVE CHARGE TRANSFER AND ASSOCIATION FOR THE $Yb^+ + Ca$ SYSTEM

2.1 DESCRIPTION

The analysis of the radiative and charge transfer processes is related to the analysis of the general scattering process. Typically, in scattering processes, we can distinguish between radiative and non-radiative processes. We can consider these charge transfers from an energetic perspective and observe that they are endothermic or exothermic.

In the endothermic case, the binding energy of the electron on the ion is less than that of the atom. In this case, to satisfy the energy conservation requirement, an incoming ion must supply the additional energy. However, if the ion approaches slowly, no additional energy is supplied. So when additional energy is not available, charge transfer can only occur if the final electron binding energy is greater than or equal to its binding energy on the atom.

If the energy is equal, the reaction is called resonant charge transfer.

If not, the reaction will be processed with the possibility of the emission of a photon. If radiation is emitted, the reaction is termed radiative charge transfer-/association; if not, it is termed direct charge transfer.

2.2 Radiative Charge Transfer and Association

Radiative charge transfer (abbreviated RCT) is an inelastic, irreversible quantum collision process in which an electron is transferred and a photon is emitted.

In general the process looks like: $A(n+1\,^1S) + B \longleftrightarrow A(n\,^1S) + B + \hbar\omega$
It is governed by the dipole matrix element $D(R) \;=\; \langle\psi_f|r|\psi_i\rangle$ where $|\psi_f\rangle$ and $|\psi_i\rangle$ are the electronic wavefunctions associated with the initial and final states, respectively.

In our case process is $Yb^+ + Ca \rightarrow Ca^+ + Yb + \gamma$, for radiative charge transfer.
In this reaction, the electron jumps from the upper adiabatic curve to the lower one by emitting a photon. The nuclei continue moving while the electron radiates and drops to the lower curve, after which the electron localizes on one proton. This is a true inelastic process, controlled by the dipole elements between initial and final states.

2.3 Radiative Association

Related radiative process is Radiative association:

$A(n+1\,^1S) + B \longleftrightarrow A(n\,^1S)B + \hbar\omega$.

There is also emission of a photon, but as a result, a molecule is formed. In this case, a photon removes excess electronic energy and the relative kinetic energy. So in this case, from a wavefunction point of view, there is a transition from a continuum state to a bound vibrational–rotational state.

This process is dominant at very low collision energies. This is a key mechanism for molecule formation in the interstellar medium.

There are several theoretical approaches and treatments of these classes of problems.[55],[56],[57].

Fairly recently, there have been advances in trapping, cooling, and manipulation of ultra-cold atoms[58]. The development of hybrid ion-atom traps has enabled researchers to explore competing pathways, including non-radiative and radiative charge transfer, as well as radiative association, in which ions and atoms recombine and rearrange at low temperatures.

Some possible applications include astrophysics[59], single-ion clocks, and, more recently, quantum computing. Ultracold atoms, cooled to nanokelvin temperatures near absolute zero $\approx \mu K$ using laser cooling and evaporative techniques, serve as a premier platform for quantum simulation and computing. By trapping atoms in optical lattices or tweezers, they act as stable qubits for quantum information processing and simulate complex, many-body systems.

There has been a recent[58] laboratory study that used a hybrid trap to investi-

gate the interactions among cold $^{1}74Yb^{+}$ ions with ^{40}Ca atoms. That study reported a large chemical reaction rate coefficient of $2 \times 10^{-10} cm^{3} s^{-1}$ and, based on their calculations, argued that radiative charge transfer is the dominant process contributing to this rate. This is several orders of magnitude larger than similar reactions for the lighter species. The paper by[60] repeated the calculation using the optical potential method[57] for the total of two reactions.

$$\begin{aligned} &Yb^{+} + Ca \rightarrow Yb + Ca^{+} + \hbar\omega \\ &Yb^{+} + Ca \rightarrow YbCa^{+} + \hbar\omega \end{aligned} \tag{2.1}$$

We will show that our results are several orders of magnitude smaller than those given in[58]. Based on these results, there are some doubts about the conclusion given in that paper concerning the role of radiative relaxation. Following the approach in the[61], we apply a semi-classical approach to the system of $YbCa$ molecules. This approach at ultra-low energies (temperatures) has not been tested before. In addition, we used molecular data obtained from the illustrations in[58]. At these energies, small details in the potential surfaces and dipole moments can be important. For these reasons, we have repeated the calculations using the original data provided in[58]. We compare the predictions of the local-optical potential with the results obtained using the Fermi-Golden-Rule prescription[57]. The result obtained by our calculation in the subsequent chapter is several orders of magnitude smaller than that given in[58].

The local optical potential method has its roots in the semi-classical theory

of radiative association first developed by Kramers and Ter-Haar[62]. In the subsequent discussion, based on that theory, we assume that a photon is emitted with energy equal to the energy difference between the two Born-Oppenheimer potential surfaces at the transition distance. The efficiency of a transition is determined by an Einstein-A coefficient at that internuclear distance. This leads to the Local Optical Potential Method, and following the semi-classical approach, the total rate is estimated as a classical integral over all localized transitions.

In our discussion below, we briefly summarize the various theoretical approaches and use them to calculate the rates for processes.

We provide a rigorous upper bound for the sum of rates given in (2.1).

2.3.1 THEORY

Here we follow the discussion in[61], and apply the theory to the $YbCa^+$ system. The cross section of the spontaneous radiative association is given by[59]. This can also be derived using Fermi's Golden Rule (FGR)(which was actually published by Dirac 20 years before Fermi)[63]: Considering the radiative association process:

$$Yb^+ + Ca \rightarrow YbCa^+ + \hbar\omega \tag{2.2}$$

where $\hbar\omega$ is the energy of the emitted photon. The cross-section for the radiative association process is given by:

$$\sigma_{RA} = \sum_J \sum_n \frac{8}{3} \frac{\pi^2 \omega_{nJ}^2}{c^3 k^2} \left[(J+1) M_{J+1,J}^2(k,k') + J\, M_{J-1,J}^2(k,k') \right]$$
$$M_{J,J'}(k,n) = \int_0^\infty dR\, f_J(kR) D(R) \phi_{J'n}(R) \tag{2.3}$$

where $D(R)$ is the transition dipole moment between $X^2\Sigma^+$ and $A^2\Sigma^+$ states of the $YbCa^+$ molecular ion. $\phi_{J'n}(R)$ is a rho-vibrational eigenstate of the $X^2\Sigma^+$ ground state, with energy eigenvalue ϵ_{nJ}. The J and n represent the angular and vibrational momentum quantum numbers, respectively. $f_J(kR)$ is s the wavefunction that satisfies the radial Schrodinger equation

$$f_J''(kR) - \frac{J(J+1)}{R^2} f_J(kR) + 2\mu V_A(R)) f_J(kR) + k^2 f_J(kR) = 0 \tag{2.4}$$

where $V_A(R)$ is the Born-Openheimer energy of the excited $A^2\Sigma^+$ state, μ is the reduced mass of the collision system, and k is the wavenumber of the incident collision partners in that channel. The asymptotic solution of the $f_J(kR)$ has the form:

$$f_J(kR) \longleftarrow \sqrt{\frac{2\mu}{\pi k}} \sin\left(kR - \frac{J\pi}{2} + \delta_J \right) \tag{2.5}$$

where δ_J is a phaseshift, as $R \rightarrow \infty$. The energy of the emitted photon is given by

$$\hbar\omega = \frac{\hbar k^2}{2\mu} + V_A(\infty) - \epsilon_{nJ} - V_X(\infty) \tag{2.6}$$

This process is governed by the selection rules,

- Rotational: $\Delta J = \pm 1$

- Electric Dipole: $\langle f|D|i\rangle$
- Electronic angular momentum: $\Delta\Lambda = 0, \pm 1$
  - where $\Lambda = 0, 1, 2$ corresponds to Σ, Π, Δ states

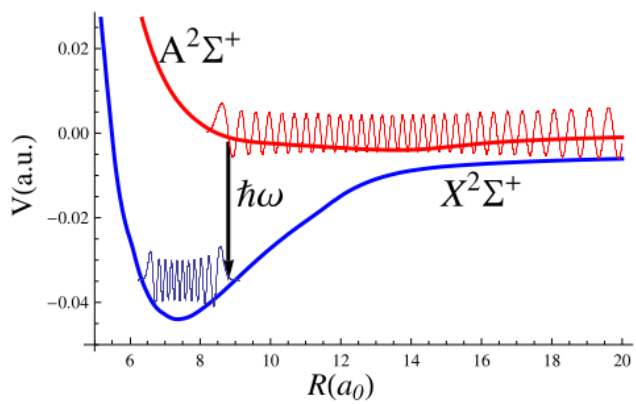


Figure 2.1: Illustration of the BO molecular potential curves (solid thick lines) participating in the radiative association process. In the $A^2\Sigma^+$ entrance channel, the wave function is shown by the light undulating line. The oscillations are due to the strong polarization force in the entrance channel, leading to a potential minimum at $R \approx 14a_0$. Association is precipitated by the emission of a photon of energy ω near the classical turning point. The final bound rho-vibrational state, in the $X^2\Sigma^+$ channel, is shown by the thin line.

2.4 Radiative Charge Transfer theory

Here, we consider the radiative charge transfer process

$$Yb^+ + Ca = Yb + Ca^+ + \hbar\omega \tag{2.7}$$

where $\hbar\omega$ is the energy of the emitted photon given by the expression (2.10).

The cross-section for the radiative charge transfer process is given by

$$\begin{aligned} \sigma_{CT} &= \int_0^{\omega_m ax} d\omega \, \frac{d\sigma}{d\omega}, \\ \frac{d\sigma}{d\omega} &= \sum_J \frac{8}{3} \frac{\pi^2 \omega_{nJ}^3}{c^3 k^2} \left[(J+1) M_{J+1,J}^2(k,k') + J \, M_{J-1,J}^2(k,k') \right] \end{aligned} \tag{2.8}$$

where

$$M_{J,J'}(k,n) = \int_0^\infty dR \, f_J(kR) D(R) f'_J(k'R \tag{2.9}$$

Here $f_J(kR)$ is a solution to (2.4) and $f'_J(k'R)$ obeys the corresponding equation for the $X^2\Sigma^+$ exit channel with wavenumber and partial wave k', J' respectively. The radial wavefunctions are normalized as in (2.5) and

$$\begin{aligned} \hbar\omega &= \frac{\hbar k^2}{2\mu} - \frac{\hbar k'^2}{2\mu} + \Delta E \\ \Delta E &= V_A(\infty) - V_X(\infty) \end{aligned} \tag{2.10}$$

From the (2.10) the maximum angular frequency $\omega_m ax$ is

$$\hbar\omega_{max} = \frac{\hbar k^2}{2\mu} + \Delta E \tag{2.11}$$

We avoid evaluating the sum (2.8).

Instead, we will compute an upper bound on the charge transfer cross section. From the equation above,e we get for the frequency of the emitted photon, during an RCT transition, to be

$$\hbar\omega = \frac{\hbar k^2}{2\mu} - \frac{\hbar k'^2}{2\mu} + V_A(\infty) - V_X(\infty) = \frac{\hbar k^2}{2\mu} - E' + V_A(\infty) - V_X(\infty) \tag{2.12}$$

Now, from the equation above, we can see that to achieve the maximum energy of the emitted photon, $\hbar\omega_{max}$, we need to have $E' = 0$. The maximum

value of E' is achieved when $\hbar\omega = 0$ therefore:

$$E_max^{'} = \frac{\hbar k^2}{2\mu} + V_A(\infty) - V_X(\infty) \tag{2.13}$$

Now the (2.8) can be written as:

$$\sigma = \frac{8}{3}\frac{\pi^2}{c^3k^2}\int_0^{E'_{max}} dE'\,\omega^3(E')\left[(J+1)M^2_{J,J+1}(k,E') + J\,M^2_{J,J-1}(k,E')\right] \tag{2.14}$$

where we have the inequality

$$\sigma_{CT} < \frac{8}{3}\frac{\pi^2\omega^3(E')_{max}}{c^3k^2}\int_0^{E'_{max}} dE'\,\left[(J+1)M^2_{J,J+1}(k,E') + J\,M^2_{J,J-1}(k,E')\right] \tag{2.15}$$

Now we can replace the expression for $M^2_{J+1,J}(k,E')$ in (2.15) with the expression in (2.9) and squaring it, we get the integral

$$\int_0^\infty dE'\,JM^2_{J,J-1}(k,E') = \int_0^\infty dE'\int_0^\infty dR\,f_J(kR)D(R)f_{J-1}(k'R\int_0^\infty dR'\,f_J(kR')D(R)f_{J-1}(k'R' \tag{2.16}$$

so the inequality becomes

$$\int_0^\infty dE'\,JM^2_{J,J-1}(k,E') < \sum_{E'} JM^2_{J,J-1}(k,E') = J\int_0^\infty dR\,f_J^2(kR)D^2(R) \tag{2.17}$$

and we get for the upper bound

$$\sigma_{CT} < \tilde{\sigma}_{CT}\frac{8}{3}\frac{\pi^2\omega^3(E')_{max}}{c^3k^2}\sum_J(2J+1)\int_0^\infty dR\,f_J^2(kR)D^2(R) \tag{2.18}$$

2.4.1 Optical Potential Approach

The alternative way to compute the cross section is to use the local optical potential method[57]. The loss of flux from the entrance channel equals the radiative probability, which is converted to a radiative cross section. In this method, the collision system in the incoming $A^2\Sigma^+$ state experiences an imaginary (ab-

sorptive) potential $iA(R)/2$ added to the entrance-channel BO potential. So we define a complex potential

$$\begin{aligned} V(R) &= V_{BO}(R) + V_{OPT}(R) = V_{BO}(R) + i\frac{A(R)}{2} \\ A(R) &= \frac{4}{3c^2}D(R)[V_A(R) - V_X(R)] \end{aligned} \tag{2.19}$$

$A(R)$ is the Einstein-A coefficient. We shall use a 2D version of the optical theorem to compute the radiative charge transfer in the 2D case in the next chapter. The next step is to solve the usual radial scattering equation, but with $V(R)$ complex. The resulting S matrix becomes non-unitary, $|S| < 1$, and the "missing probability" is the radiative loss.

The cross-section for radiative quenching is given by

$$\sigma = \frac{\pi}{k^2}\sum_J (2J+1)(1 - e^{-4\eta_j}) \tag{2.20}$$

where η_j is the imaginary part of the J'th partial wave phase shift δ_j for the radial wave $f_J(kR)$ that satisfies

$$f_J''(kR) - \frac{J(J+1)}{R^2}f_J(kR) + 2\mu[V_A(R) + V_{opt}(R)]f_J(kR) + k^2 f_J(kR) = 0 \tag{2.21}$$

The Local Optical Potential is generally applicable when the following conditions are satisfied, which aligns well with our case.

- Transition happens over a region small compared to relevant wavelengths
- Emission is relatively weak compared to the collision energies
- It is accurate enough for the computation of total probability

2.4.2 Ultra Cold Limit

In the limit of ultra-cold temperatures, only the s-wave state participates in the process, so the total cross section takes a simpler form.

$$\sigma = \sum_n \frac{16\mu\pi\omega_n^3}{3c^3k} \left| \int_0^{R_0} dR\, \phi(R) D(R) \phi_{J=1}^n(R) \right|^2 \tag{2.22}$$

where $\phi(R)$ is the s-wave solution to (2.4). The boundary condition for $\phi(R)$

$$\frac{d\phi(R)}{dR}|_{R_0} = 1 \tag{2.23}$$

is valid for a sufficiently large radius R and $\phi_{J=1}^n(R)$ are $J = 0$ rho-vibrational states of the $X^2\Sigma^+$ potential. Because there is only an s-wave, the integral (2.22) does not depend on the incoming wave number k. Therefore, the equation (2.22) predicts that the association cross section, in the ultra cold regime, scales as the inverse of the incoming velocity, i.e., the rate tends to a constant.

We shall show in the subsequent chapter the method used to compute the phase shift. The idea is to match the numerical solution for $f_J(kR)$ to its asymptotic form for large R.

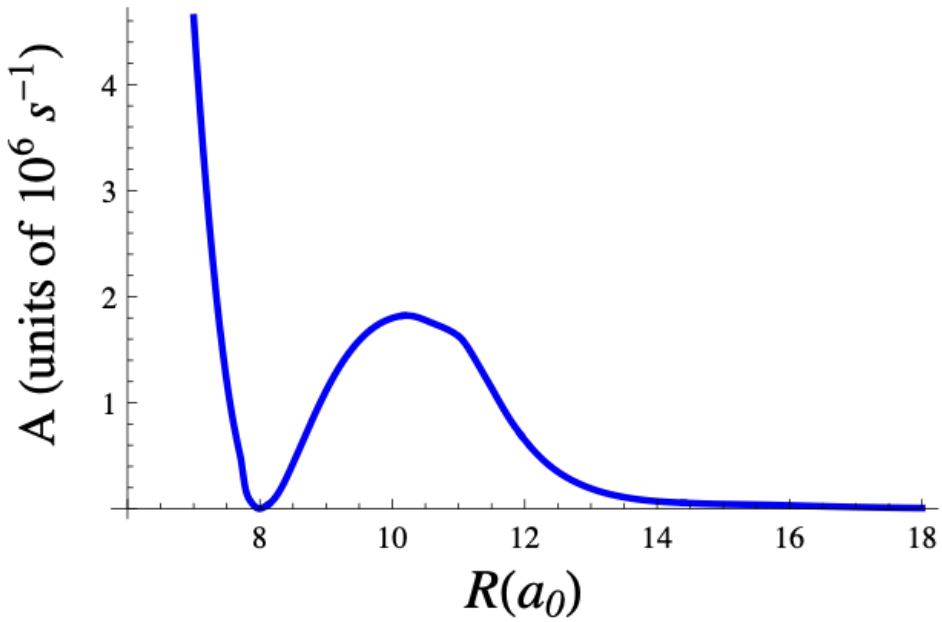


Figure 2.2: R-dependent Einstein A coefficient as a function of internuclear distance

2.5 Results

Figure 2.1 illustrates the mechanism for radiative association for the Yb^+ ion and Ca atom that approach in the $A^+\Sigma^2$ electronic BO state. The BO energies were taken from the data of the ab-initio calculations reported in[64]. At large internuclear distances, this potential has the form

$$V_A(R) \to \frac{C_4^A}{R^4} \quad C_4^A = 78.5 \tag{2.24}$$

In the case of Radiative association, the incident $A^2\Sigma^+$ channel can relax via photon emission. In that case, the final state is a bound rho-vibrational level of

the $X^2\Sigma^+$ channel. In radiative charge transfer, the collision partners can exit in that channel, as a re-arranged $Yb - Ca^+$ pair. In the exit channel

$$V_X(R) \to \frac{C_4^X}{R^4} + \Delta E \quad C_4^X = 71.5 \quad \Delta E = -0.0052 \tag{2.25}$$

as $R \to \infty$

In calculating the radiative association cross sections given by (2.3), we need to itemize all bound states supported by the $X^2\Sigma^+$ channel. The total number of bound states can be approximated using the JWKB expression

$$n = \text{Floor}\left[\int_{R_c}^{\infty} dR\,\sqrt{-2\mu(V_X(R) - \Delta E)} - 1/2\right] \approx 54,878 \tag{2.26}$$

where R_c is a classical turning point Floor[x] is the integer lower bound of x.

Here we have a large reduced mass μ, i.e., the participant's mass is large, which in turn produces a very large number of bound states. For lighter participants, the number of rho-vibrational states is typically in the order of a few hundred[59].

Here we have another factor to consider. In the ultra-cold temperatures, the participants have lower energies, so the centrifugal repulsion in the entrance channel limits the number of partial waves that participate. Since the selection rules $J \pm 1$ still apply, they constrain the possible rho-vibrational levels. For example, at a collision energy corresponding to a temperature of $1mK$, only levels with J up to the value ≈ 15 contribute to the association rate.

The following graph 2.3 shows the results of the calculation.

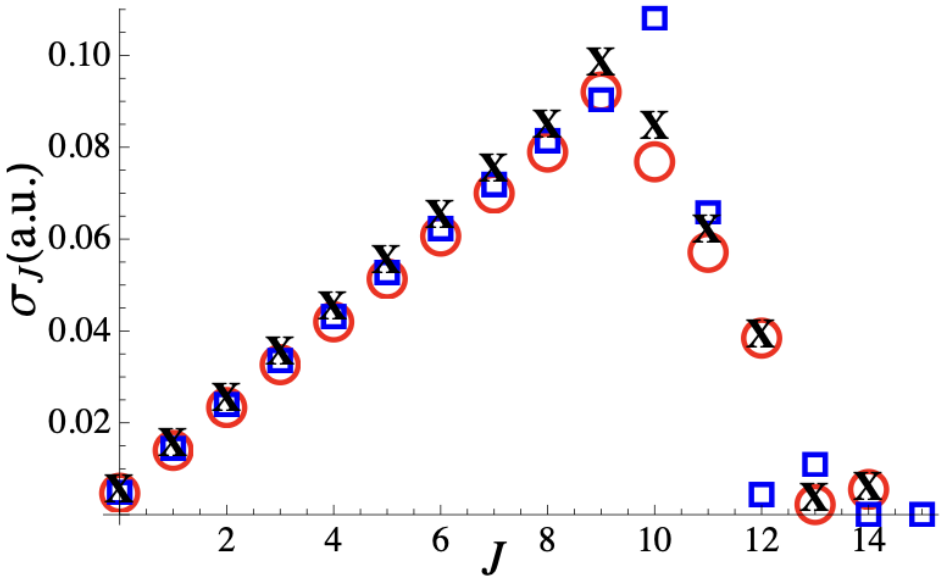


Figure 2.3: Plot of various cross sections as a function of incoming partial wave J. Circles represent data for the radiative association cross sections, X's represent an upper bound for the total (association + radiative charge transfer), and squares represent the data for the total radiative relaxation obtained using the optical potential method.

We have made the calculations two ways. First, starting from the equation (2.4), we computed the cross section using equation (2.8). In the figure 2.3 symbol X represents the upper limit for total radiative relaxation, which is obtained by adding the association cross sections (2.8) with those given by expression (2.18).

The square icons represent the cross sections predicted by expression (2.20),i.e., by the optical potential method.

So from figure 2.3 for values $J < 10$ we see that an optical potential method

provides a very good approximation to the total cross section.

However, this approximation is good only for values $J < J_{max}$[60].

$$J_{max} = \sqrt[4]{8\mu k^3 C_4^A} = \sqrt[4]{24\mu^2 k_B T C_4^A} \approx 12 \tag{2.27}$$

The collision energy of the collision system is given as $3/2k_B T$ where k_B is the Boltzman constant and T is the temperature in Kelvin

Value J_{max} is the critical angular momentum for which the collision system has sufficient collision energy to overcome the centrifugal potential barrier[60]. For larger J, tunneling resonances can access the inner region where the transition dipole moment is non-negligible and induce radiative transitions.

In Table I of[61], we present the various cross sections at several representative collision temperatures. The second column contains the association cross-section obtained using the FGR method described above. For the radiative charge-transfer cross sections, as listed in the third column, we used (2.18). Thus, the upper bound for the total radiative relaxation cross sections is given in column 4. The last column gives the results obtained using the local optical potential method. The table shows that, over the temperature range considered, the local optical potential method predicts cross sections that are less than the upper bound itemized in column 4. Secondly, the differences between the two theories' predictions are small. The optical potential cross section differs by less than 4% from the upper limit values over the entire temperature range, including the ultra-cold region.

One can also note that the optical potential method predicts cross sections

that are larger than the radiative association cross sections. At the same time the optical potential method provides a reliable upper bound for the total (RR) cross section.

In figure 2.4 we plot the total radiative relaxation cross section, obtained using the optical potential method, for the gas temperature range $1mK < T < 1K$.

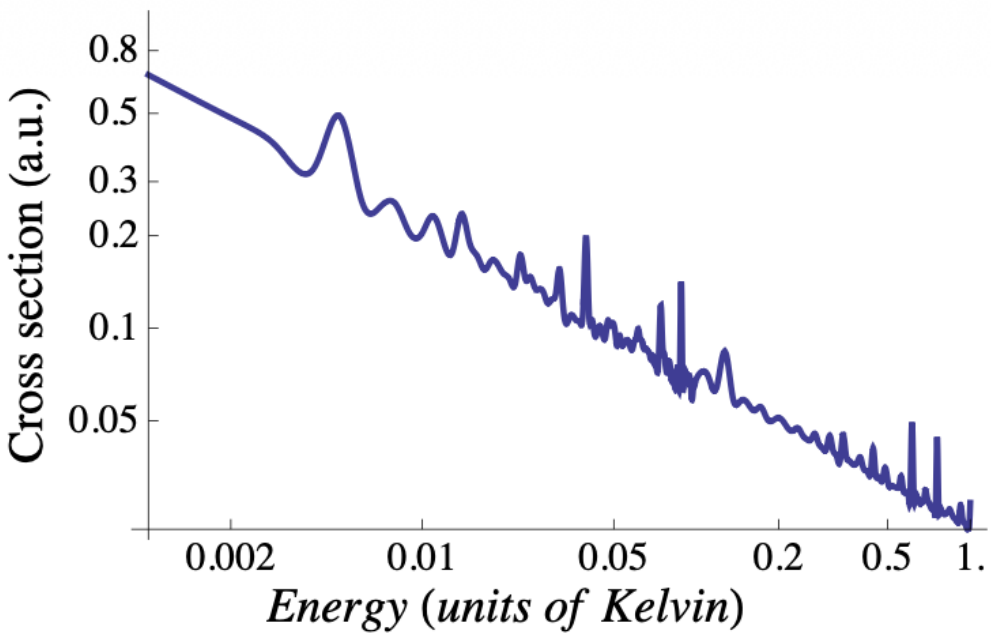


Figure 2.4: Total radiative relaxation cross section as a function of collision energy expressed as $E = 3/2\, k_B T$, where T is the temperature in Kelvin

Although the optical potential method provides a good approximation for the total radiative relaxation rate, calculation of the photon emission spectrum requires the use of the FGR method. use of the FGR method. In 2.5 we illus-

trate the association cross sections $\sigma_{n\,J}$ at $T = 1mK$ for the individual rho-vibrational levels as a function of the frequency of the emitted photon. The structure of the emission pattern is the result of the oscillation in the incoming wave as shown in 2.1

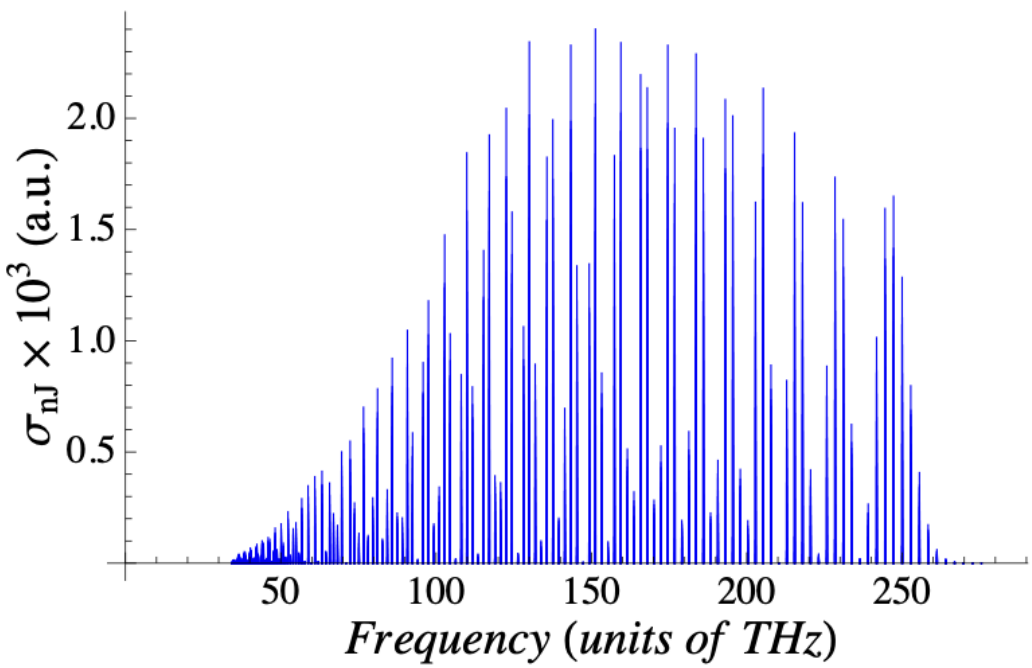


Figure 2.5: mission spectrum for the radiative association process at a gas temperature of 1mK.

In the limit $T \to 0$, we define a complex scattering length for the s–wave solution to (2.21)

$$a \equiv -\frac{1}{k}\tan\delta(k) = 10785 - i0.008842; \quad \text{as} \;\; k \to 0 \tag{2.28}$$

Therefore, the total RR cross section, according to the optical potential method,

has the limiting value

$$\sigma = \frac{4\pi}{k} 8.842 \times 10^{-3} \tag{2.29}$$

Defining the rate coefficient

$$k_{RR} \equiv \langle \nu\sigma \rangle \tag{2.30}$$

we use the results from[61] to show the values of $k_{RR} \approx 1.15 \times 10^{-14}$.
By looking at 2.4 we observe that k_{RR} varies slow, from $1.6 \times 10^{-15} cm^3 s^{-1}$ at 1mK to $2 \times 10^{-15} cm^3 s^{-1}$ at 1K.

We can summarize the current chapter as follows. Using both quantum and semi-classical methodologies, we calculated the rates for the radiative association and charge transfer in cold collisions of Yb^+ with Ca. We have shown that the optical potential method provides very good results in predictions for the total radiative relaxation rates, in the ultra-cold s-wave scattering regime In addition, we have computed and shown the cross section σ as a function of temperature K and frequency of the emitted photons.

2.5.1 Isotope Dependency

Different Yb isotopes $(^{170}Yb, ^{174}Yb, ^{176}Yb)$ differ only in nuclear mass, not in electronic structure. Thus, isotope dependence arises from nuclear motion. Following[65] we observe that radiative processes $Yb^+ + Ca \rightarrow Yb + Ca^+ + \hbar\omega$ and $Yb^+ + Ca \rightarrow YbCa^+ + \hbar\omega$ are both controlled by the overlap integral

$$\int_0^{R_0} dR\, \phi(R) D(R) \phi^n_{J=1}(R) \tag{2.31}$$

Because isotope mass shifts the bound-state energies, it leads to order-of-magnitude differences in rate constants between isotopes.

At ultra-low temperatures (μK and below), only the s-wave is present, and the rates obey the Wigner threshold law. In that case, if a Yb isotope supports a bound state just below threshold, the scattering length becomes very large, and the radiative association rate is strongly enhanced. The enhancement becomes isotope-specific. This explains the pronounced isotope separation seen in calculated rate curves as shown in[65]. So, radiative association shows strong isotope dependence, while the radiative charge transfer isotope dependence is smaller. From[66], we copy, with permission, the diagram showing isotopic dependency of the radiative association.

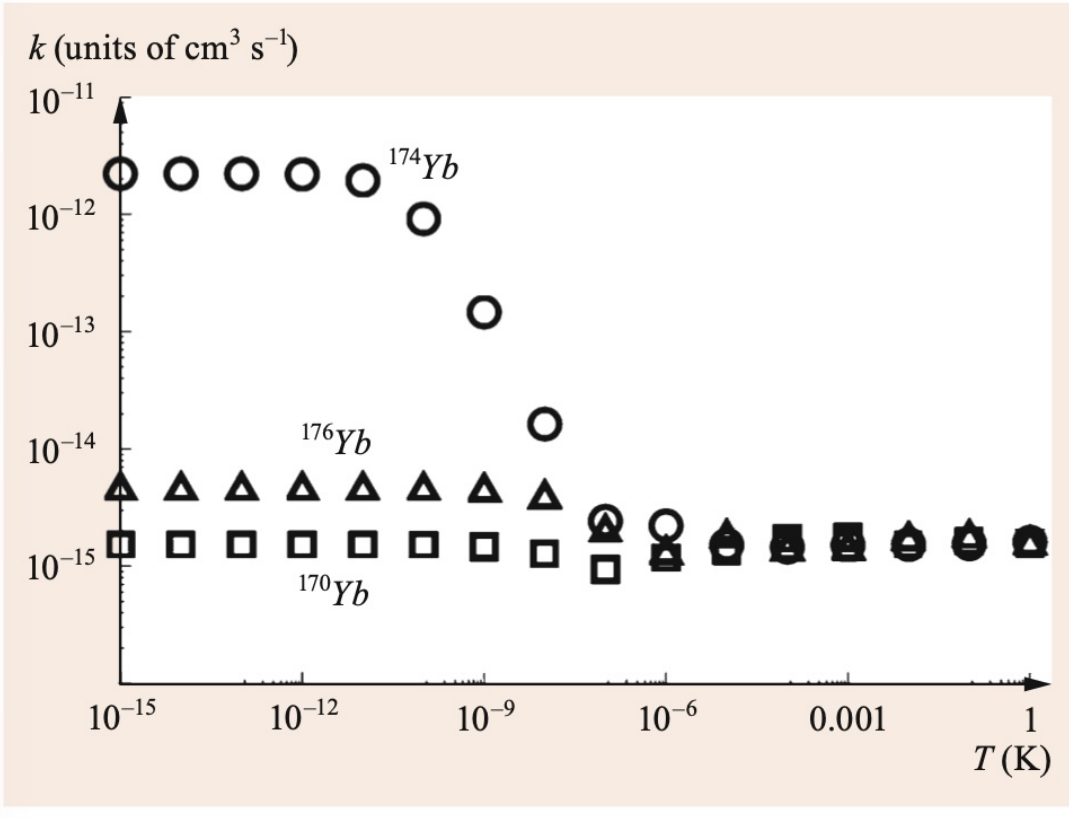


Figure 2.6: Isotopic dependence of radiative association into the $YbCa^+$ molecular ion

In the following sections, we focus on 2D systems and describe Resonant Charge Transfer.

CHAPTER 3

CT PROCESSES IN TWO DIMENSIONS

In this part of the thesis, we will consider the following resonant charge transfer process

$$H^+ + H \rightarrow H + H^+$$

i.e., the exchange of an electron in the collision of a proton with a hydrogen atom. This fundamental CT process has been studied extensively in three dimensions, but in this dissertation, we, for the first time, calculate the rate for (xx) in a two-dimensional substrate, i.e., both the electron and nuclei are constrained to move in two dimensions. The motivation for this study stems from advances in laboratory studies of 2D systems and the fabrication of quantum materials in 2D. Thus, fundamental collision processes must be investigated theoretically in 2 dimensions. 2D materials hold great promise for innovative applications. To advance this emerging technology, it is essential to gain a fundamental understanding of the processes that govern the behavior of matter at the atomic and molecular levels. In this thesis, we will examine the in-

teractions of ions with neutral matter in a 2D environment. Before we focus on the calculation of Eq. (xx), we review, below, some advances in the physics of 2D systems.

3.0.1 Overview of 2D material science

2D materials can exhibit unique topological properties that do not exist in the 3D case. An exciting and important feature in this context is the emergence of anyons[8]. Anyons are neither Fermions nor Bosons. The exchange of two anyons induces an arbitrary relative phase between the partners; an effect called fractional statistics. In 3D space, the relative phase can only take the integer values of 2π (Bosons) or π (Fermions).

3.0.2 On 2D Materials No-Go Theorems and Graphene

Landau and Pierls[9][10] initially argued that 2D materials could not exist. That argument posited that the displacement of atoms at finite temperatures induces thermal fluctuations on the order of the atomic spacing, thereby melting the 2D material above 0 K.

There is an even stronger statement against 2D materials, in the form of the Mermin-Wagner-Hohenberg (MWH) theorem[11][12]. In these papers, the authors show that 2D systems with short-range interactions are unstable and cannot exist. Using thermodynamic arguments, one can show that a 3D system is stable if its free energy is bounded from below and convex (upwards). In this way, the system can minimize the ground-state energy. Because higher-order

fluctuations are finite, the system is stable. The MWH argument posits that, in 2D systems, fluctuations around the ordered state (lattice in the 2D case) decorrelate over large distances, thus destroying the large-scale order. Therefore, it seemed that the possible existence and applications of the 2D materials were quite limited. The only 2D materials considered stable were thin molecular films, which could form on a solid's surface. Dash[13] predicted that 2D materials exist as a thin film either on the surface of the boundaries of 3D materials.

Despite those arguments, it has been speculated that stable 2D materials are possible[13]. The situation changed dramatically in 2004 when 2D carbon (graphene) was discovered and created in the laboratory[14;15] at the University of Manchester, Great Britain.

2D materials have qualitatively different topological properties[16]. Moreover, those properties do not depend on the type of microscopic interactions between particles. While this subject is beyond the scope of this thesis, it does show that the field of 2D materials is a fertile ground for both fundamental and applied research. Interestingly, it has been shown[17] that it takes 10 graphene layers for it to begin behaving like a 'regular' 3D material. The existence of graphene does not contradict the above-mentioned no-go theorem[12], but it does circumvent it[18]. In graphene, thermal oscillations produce ripples or bending of the graphene sheet. That way, graphene behaves like an elastic membrane that allows phonons to propagate in 2 dimensions and couple in a plane stretching along the transverse fluctuations out of the plane. These

phonons mediate a long-range interaction, thereby circumventing the Mermin-Wagner no-go theorem.

Additionally[14 19 20],2D materials exhibit several other interesting properties. Its quantum excitations behave in accordance with relativistic dynamics. In condensed matter systems, the solutions of the Schrödinger equation describe the material's electronic properties. In graphene, the electrons obey the non-relativistic Schrodinger equation, but interactions with the graphene lattice's periodic potential give rise to solutions referred to as quasiparticle excitations, and they behave in the same manner as massless Dirac fermions. Even at low energies, these quasiparticles are described by a Dirac equation in 2+1 dimensions.

3.0.3 EXCITONS

These excitations represent a wide class of intrinsic electronic excitations (Excitons)[21;22] in crystals of semiconductors and dielectrics. In one model, excitons are bound states of an electron and a hole, typically formed when an incident photon is absorbed, thereby exciting an electron from the valence to the conduction band. The attractive Coulomb interaction between the excited electron and the newly created hole binds them together to form a bound neutral compound system of the two charge carriers, similar to a hydrogen atom. The character of exciton motion depends on the strength of the exciton interaction with phonons. The electron-hole model is common in insulating and semiconducting crystals. Ya. I. Frenkel (1931) introduced excitons to explain

the light absorption in crystals, which does not lead to photoconductivity. In the Frenkel model, the exciton is considered as an electronic excitation of one crystal site with the energy close to, but a bit smaller than, that necessary for the excitation of a free electron. Due to the crystal's translational symmetry, the exciton can move along the lattice sites, transferring energy to electrically active or luminescent centers.

Similar excitations occur in 2D systems.[23;24]. 2D excitons promise to serve as building blocks for 2D electronics.[25]. Although exciton-based transistor action has been demonstrated in bulk semiconductor coupled quantum wells, the low operating temperature required limits their practical application. The emergence of 2D semiconductors with large exciton binding energies may lead to excitonic devices and circuits that operate at room temperature.[25].

3.0.4 GRAPHENE

Graphene exhibits similar properties[14], such as its Quantum Electrodynamics (QED) like electronic spectrum, electron tunneling. Another interesting consequence of the QED-like electron spectrum is the possibility of experimentally studying QED in curved space by controllable bending of a graphene sheet. Such a study may offer a possibility to address a certain class of cosmological problems.

At present, since the discovery of graphene, several new 2D materials have been identified[26]. As new 2D materials are discovered, researchers are investigating their potential applications. One of the most useful applications of

exotic materials is electronics. These new 2D materials promise that they can be used to build existing electronic components, but with potentially better performances[27][28].

Nanotubes are rolled-up graphene sheets, and depending on the application, graphene (as a sheet) is sometimes superior, sometimes inferior, and sometimes completely different. Another notable application is the emergence of high-temperature superconductivity in 2D materials[29].

2D Electronics

Driven by a demand for higher performance and lower consumption in electronic devices, 2D materials have found application in that arena[27],[28]. There is the hope that graphene can be used to build active electronic components, such as diodes, transistors (FET), and other electronic components. Alternatively, graphene can be used as a conductor in electronic components and batteries owing to its high electrical conductivity. Graphene itself is a zero-gap semiconductor. There is another unique opportunity in 2D semiconductors. In a 3D semiconductor, electronic states are buried deep inside the material. Unlike the surface state, electron states within the material are localized, i.e., their wave functions have exponentially decaying tails.

In 2D semiconductors, the electronic states exist on the surface. The surface states admit both localized and non-localized block-bulk states[30]. Another interesting application of 2D electronics is the LED elements in 2D[31]. Electronics research is closely related to research on new 2D materials. For example,

graphene itself is not compatible with silicon. However, there is a new material, black phosphorus[32], which is both compatible with silicon and it holds promise for future electronic[33] devices. This fact relies on two key properties: the first is that black phosphorus has higher mobility, the speed at which it can carry an electrical charge, than silicon. The other is that it has a finite bandgap, whereas graphene does not. So, in essence, in the presence of an electric field, black phosphorus can act as a semiconductor[34].

Quantum Computing

The concept of topological quantum computing[1] depends on the unique properties of 2D materials. It[1;2;3;4;5], relies on the unusual quantum properties of 2-dimensional (2D) matter. In principle, there are no theoretical objections to building a quantum computer; there are serious technical issues, such as noise and scaling limitations, for current qubit technologies that need to be overcome[43]. The topological approach promises a physically realizable quantum computer[2;3].

The current computer architecture is based on the so-called 'von Neumann' architecture. The algorithms based on that architecture have inherent limitations, and, as it currently stands, the architecture cannot overcome them.

There are two main classes of computer algorithms, P and NP. There are subclasses within those classes, but those two classes capture the essential behavior of classical algorithms. The algorithms belonging to class P are solvable in polynomial time, with regard to the size of the input. Those algo-

rithms belonging to class NP are those that require exponential time to complete[35].

Currently, the major unsolved problem in Computer Science is the P vs. NP problem[35]. While the hypothesis that $P \neq NP$ is generally accepted, no proof has been found. Applying quantum-mechanical principles in engineering disciplines, such as computer science[36], promises to yield a qualitatively more powerful quantum computer.

Now, all problems in the NP class (more precisely, NP-complete subclass) can be transformed into each other in polynomial time[37], therefore finding a polynomial-time solution for one of them would amount to having a polynomial-time solution to all problems in the NP class. So the goal is rather worthy.

It is currently known that quantum computers cannot solve NP problems. They could solve the class of the bounded-error, quantum, polynomial time (BQP) problems[38]. Even if the NP problems remain intractable, just the possibility that, according to Feynman[36], a quantum computer can be used to simulate the quantum processes, which will in itself be a step forward.

For example, Schor's algorithm[39] can provably solve integer factorization in polynomial time, as opposed to the best classical algorithm (quadratic sieve), which works in sub-exponential time[40]. Unfortunately, the integer factorization problem, as well as other BQP problems, do not belong to the class of NP-complete problems.

Interestingly, according to Abrams and Lloyd[41],[42], if the small nonlinear term

is added to Quantum Mechanics, quantum computers would be able to solve NP-complete problems in polynomial time.

ANYONS

Anyons are quasi-particles that can arise in 2D systems and whose quantum statistics in neither Fermionic nor Bosonic[44]. This fact has been proven[45], but arbitrary statistics are valid only in 2 dimensions[46]. After exchanging two identical particles, the wave function gains an arbitrary phase factor $\Psi = e^{i\theta}\Psi$, where $\theta = 2\pi\nu^s$. In contrast, for a 3D system, the exchange of two particles leads to either phase factor π or 0, for fermions and bosons, respectively.

In general, there are two kinds of Anyons: Abelian and non-Abelian. Abelian Anyons have been reliably detected in systems exhibiting the Fractional Quantum Hall Effect[47]. Whereas the computation with Abelian Anyons is theoretically possible[48], considerably more attention has been paid to non-Abelian Anyons. There are several possibilities for realizing non-Abelian statistics experimentally:

- A two-dimensional electron gas in a large magnetic field (the fractional quantum Hall effect)
- Rapidly rotating Bose-Einstein condensates[49]
- Frustrated magnets[50].
- One of the anyons candidates the Majorana fermion[51].

Majorana fermion (particle) is a particle that is its own antiparticle. A pair of localized Majorana states has been predicted to reside at the ends of a specially designed superconducting wire. Very recently (2020)[52] there is a possibility that Majorana particles has been observed for the first time. The other property of the Majorana fermion is that it is a non-Abelian anyon. This presents the possibility for the realization of fault-tolerant topological quantum computing[53].

Excitons and Trions

Other interesting quasiparticle phenomena are called excitons[23]. It is a bound state of a hole and an electron in a solid-state material. Excitons enable energy transport without charge transport. There are two types of excitons:

1. Frenkel excitons are found in organic molecular crystals[54] and have binding energies on order of $1eV$ and radii 10Å.

2. Wannier excitons are found in semiconductors[22] and have binding energies on the order of $1meV$ with radii 100Å.

Excitons have been observed in 2D materials[25] as well as 2D excitons in 3D materials. The promise is that these quasi-particles can enable the development of new electronic components, particularly optoelectronic ones.

Since the exciton system resembles either the hydrogen atom or the hydrogen molecule, it seems conceivable that the dynamics of the H_2^+ molecule is qualitatively similar and could be applied to the exciton systems as well.

A similar system is the trion, a bound state of three charged particles. It consists of either 2 electrons and a hole, or 2 holes and an electron, therefore resembling the hydrogen molecular ion.

CHAPTER 4

OVERVIEW OF CT THEORY FOR LOW ENERGY COLLISIONS IN 3D

4.1 MOLECULAR ORBITALS IN 2 D

In order to calculate the resonant charge transfer rate (RCT), we require the potential surfaces for the two lowest ground state Born-Oppenheimer (BO) that correlate to the degenerate asymptotic limits of the proton-hydrogen system. Many open-source and proprietary tools are available to calculate the latter in three dimensions; to the best of our knowledge, none of such tools are available for the 2D case. In this thesis, we use a first-principles approach to calculate accurate BO potential surfaces for application in our collision studies.

Prior to that discussion, we present a qualitative description and derive approximate solutions using the LCAO method to examine the gerade and ungerade potentials for the ground state of the H_2^+ ion.

4.1.1 Gerade and Ungerade Molecular Potentials

In diatomic molecules, gerade (German for even) and ungerade (German for odd) describe how the electronic wavefunction behaves under inversion symmetry, i.e., symmetry of the molecular orbital with respect to the inversion center. If the internuclear axis is placed through the center of the molecule, the inversion manifests with every coordinate point r with $-r$. These terms are meaningful only for molecules or systems that possess a center of inversion, points such that $r \to -r$ maps the molecule onto itself. The definition:

- Gerade(g): an orbital wavefunction $\Psi(r)$ is gerade if it is unchanged by inversion: $\Psi(-r) = \Psi(r)$. It has even parity. Example: $1s\sigma_g$
- Ungerade(u): an orbital wavefunction $\Psi(r)$ is ungerade if it changes sign under inversion: $\Psi(-r) = -\Psi(r)$. It has odd parity. Example $2p\sigma_u$.

4.2 Selection rules and consequences

Electric dipole transitions: transitions between electronic states follow the parity selection rule. The electric dipole operator is odd (Ungerade). Therefore, allowed transitions must change parity: $g \to u$. Transitions $g \to g$ or $u \to u$ are electric dipole forbidden (though allowed by higher multipoles or vibronic coupling).

Orbital interactions: orbitals can only mix (form nonzero overlap/hybridize) if they belong to the same symmetry species, including parity. Thus, g orbitals do not mix with u orbitals.

4.3 LCAO

The ground state expression gives amplitude for the 2-dimensional hydrogen atom[67]

$$\psi_{1s}(\mathbf{r}) = \frac{4}{\sqrt{2\pi}} e^{-2\,r} \tag{4.1}$$

where $r = |\mathbf{r}|$. In the LCAO method, an approximate description of the 2D gerade molecule H_2, for internuclear separation $\mathbf{R}$ is given by

$$\psi_g(\mathbf{r}, \mathbf{R}) = C(R)\Big(\psi_{1s}(\mathbf{r} - \mathbf{R}/2) + \psi_{1s}(\mathbf{r} + \mathbf{R}/2)\Big) \tag{4.2}$$

where $C(R)$ is a normalization factor so that

$$\int d^2\mathbf{r}\, |\psi_g(\mathbf{r}, \mathbf{R})|^2 = C^2(R)\Big(2 + 2\int d^2\mathbf{r}\, \psi_{1s}(\mathbf{r})\psi_{1s}(\mathbf{r} + \mathbf{R})\Big) = 1 \tag{4.3}$$

or,

$$C(R) = \frac{1}{\sqrt{2(1 + S(R))}} \text{ and } \quad S(R) \equiv \int d^2\mathbf{r}\, \psi_{1s}(\mathbf{r})\psi_{1s}(\mathbf{r} + \mathbf{R}) \tag{4.4}$$

We now evaluate the expectation values $\langle V \rangle$ of the potential

$$-\frac{1}{|\mathbf{r} - \mathbf{R}/2|} - \frac{1}{|\mathbf{r} + \mathbf{R}/2|}. \tag{4.5}$$

Inserting the above wave amplitude to evaluate the expectation value, we get

$$\langle V \rangle_g = -2\int \frac{d^2\mathbf{r}}{r}\, |\psi_g(\mathbf{r} + \mathbf{R}/2)|^2. \tag{4.6}$$

We use the virial theorem to estimate the gerade ground state expectation value for the kinetic energy operator

$$\langle KE \rangle_g = \frac{1}{2}\langle \mathbf{r} \cdot \nabla V \rangle \tag{4.7}$$

and since

$$\mathbf{r} \cdot \nabla V = \mathbf{r} \cdot \frac{(\mathbf{r} - \mathbf{R}/2)}{|\mathbf{r} - \mathbf{R}/2|^3} + \mathbf{r} \cdot \frac{(\mathbf{r} + \mathbf{R}/2)}{|\mathbf{r} + \mathbf{R}/2|^3} \tag{4.8}$$

we find

$$\langle KE \rangle_g = \int \frac{d^2\mathbf{r}}{r} \, |\psi_g(\mathbf{r} + \mathbf{R}/2)|^2 + \int \frac{d^2\mathbf{r}}{r^3} \, (\mathbf{r} \cdot \mathbf{R}/2) \, |\psi_g(\mathbf{r} + \mathbf{R}/2)|^2. \tag{4.9}$$

In the same manner, we obtain the BO energy for the ungerade BO state,

$$\psi_u(\mathbf{r}, \mathbf{R}) = \frac{1}{\sqrt{2(1 - S(R))}} \int d^2\mathbf{r} \left(\psi_{1s}(\mathbf{r} - \mathbf{R}/2) - \psi_{1s}(\mathbf{r} + \mathbf{R}/2) \right). \tag{4.10}$$

In Table 8.1, we present the accurate and LCAO values for the gerade state.

Graphical comparison of the accurate vs LCAO potential for the gerade state. The exact potential is shown in Red, and the approximation is shown in blue.

This is more readily observed in the enhanced plot.

The above graphs show the accurate gerade potentials (red dots) versus those obtained with the simple LCAO method described above. We observe that for $R > 10$, there is fair agreement; the agreement is poorer for $R < 10$. This is a reasonable result, as the LCAO, which includes only the ground-state orbital, is a poor description in the molecular region.

We now show the same comparison for the ungerade state in Table 8.2 we show the accurate vs LCAO values for the ungerade state.

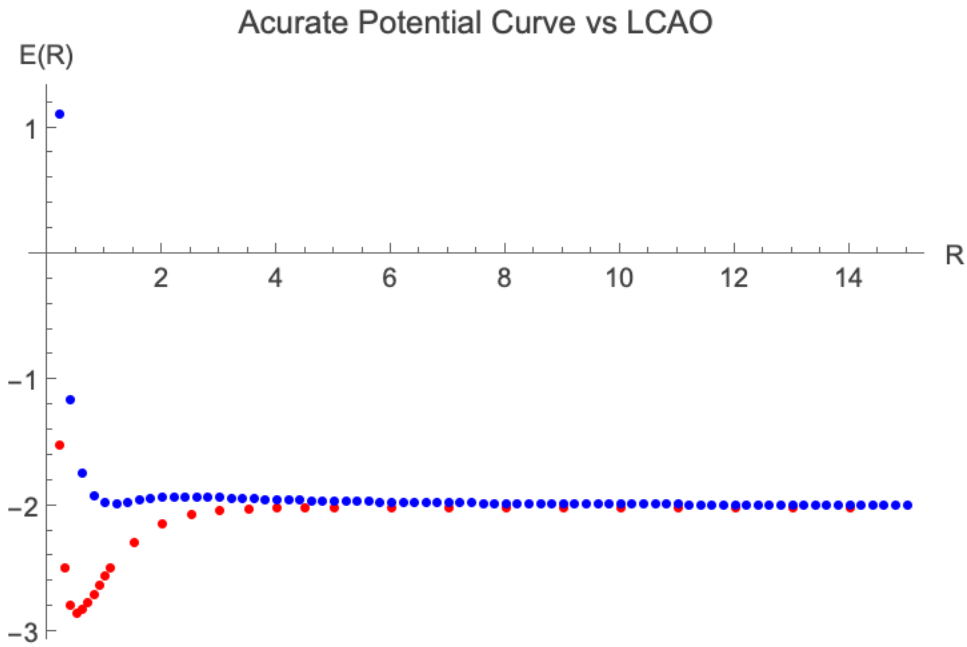


Figure 4.1: Accurate potential curve (red) vs LCAO approximation (blue) for the H_2^+ ground gerade state

Graphical comparison of the accurate and LCAO potentials for the ungerade state.

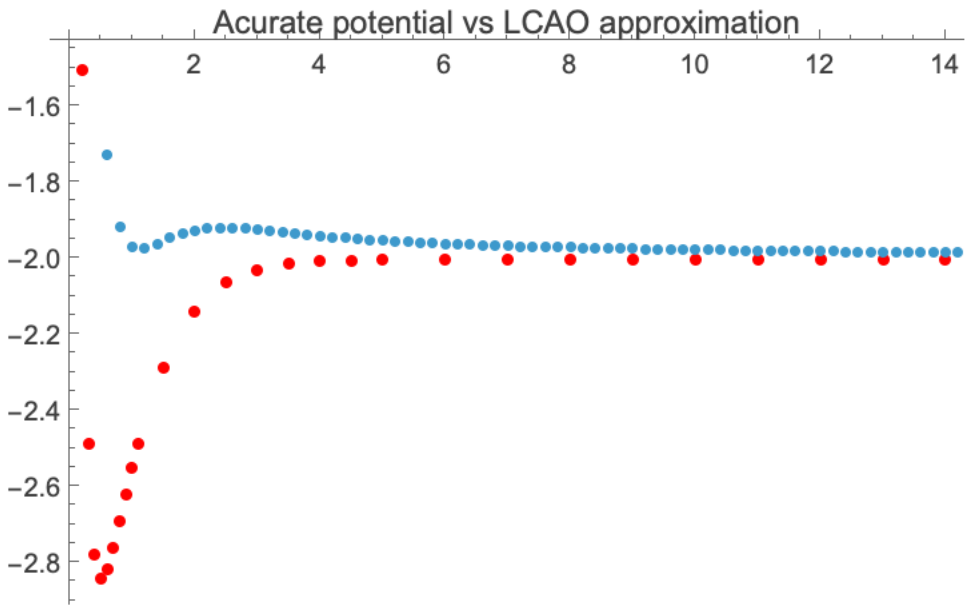


Figure 4.2: Accurate potential curve (red) vs LCAO approximation (blue) for the H_2^+ ground gerade state, enhanced plot

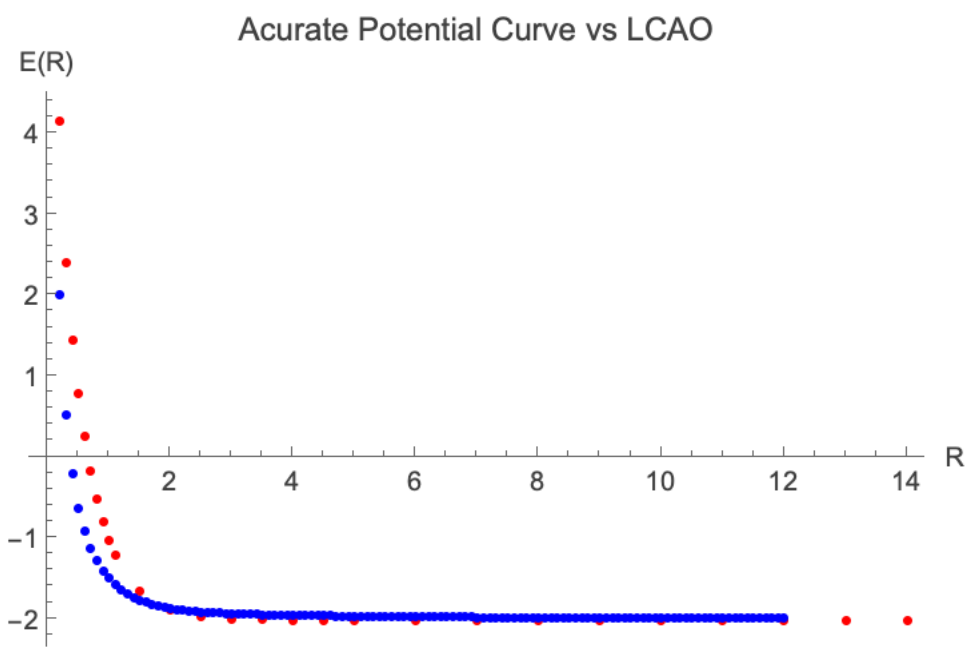


Figure 4.3: Accurate potential curve (red) vs LCAO approximation (blue) for the H_2^+ 1st ungerade state

Unlike for the gerade state, in the ungerade case the LCAO is quite a good approximation to the accurate potential.

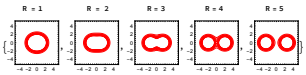
The reason is the symmetries of the gerade and ungerade states and the overlap integral (4.4). In the ungerade case we have:

$\psi_u(\mathbf{r}, \mathbf{R}) \approx \psi_{1s}(\mathbf{r} - \mathbf{R}/2) - \psi_{1s}(\mathbf{r} + \mathbf{R}/2)$ which makes $1 - S \approx 1$.

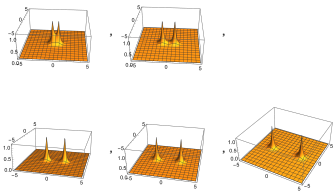
It follows that the orbital are close to being orthogonal, which makes LCAO approximation more accurate.

In the gerade case, we have $\psi_g(\mathbf{r}, \mathbf{R}) \approx \psi_{1s}(\mathbf{r} + \mathbf{R}/2) - \psi_{1s}(\mathbf{r} + \mathbf{R}/2)$. In this case, we have $1 + S > S$. In this case, the LCAO orbits are much less orthogonal, so the overlap error becomes much greater.

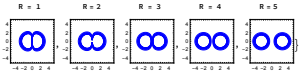
In the following figures, we show the atomic orbitals for the gerade and ungerade cases as a function of the internuclear distance R.



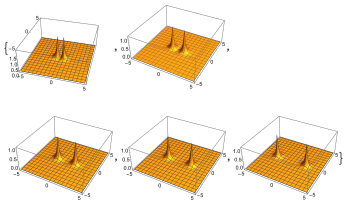
(a) Gerade Orbitals



(b) Gerade Orbitals in 3D



(c) Ungerade Orbitals



(d) Ungerade Orbitals 3D

Plot of LCAO potentials

Following our previous discussion, we have determined that our LCAO example provides only a rough qualitative description of the two-dimensional ground state potential surfaces for the H_2^+ molecule. To obtain more accurate data, we will introduce a semi-analytic procedure initially developed by Bates and his colleagues for the treatment of the three-dimensional H_2^+ molecular ion in the upcoming chapters.

In the next chapter, we will review this method as it applies to the three-dimensional system and provide a more precise description of the data presented in Ref.[68]. Subsequently, we will use a similar approach to generate the data for the two-dimensional case of H_2^+.

We will compare our results with those from a previous study (cite) and apply our findings in the calculation of the RCT process given by Eq. (xx) at ultracold temperatures.

CHAPTER 5

HYDROGEN MOLECULAR ION IN 3 DIMENSIONS (3D)

In this chapter, we look at the hydrogen molecular ion in 3D, which has been solved. We review analytic methods[68 69 70] used to obtain solutions of the time-independent Schrodinger equation for the hydrogen molecular ion.

As we will apply a similar method to treat the hydrogen molecular ions in 2D, we first review the methodology of Bates[68 69] used to calculate the lowest electronic eigenstates and eigenvalues for the H_2^+ system in 3D. *

The hydrogen molecular ion, and similar systems (such as HeH^+), are one of the few that allow a full analytic quantum mechanical description. Although its solutions do not exist in the closed form, they can be expressed as a convergent infinite series. Wasserman in his thesis[71] in 1963 reports that while the existence of H_2^+ has been verified, no discrete spectrum of the H_2^+ has been observed In the same thesis, Wasserman reports that the ionization potential of H_2^+ has been measured and shows good agreement with the theoreti-

*In this section we use atomic (Hartree) units, unless otherwise stated.

cal calculations. However, in 1989 Carrington, McNab and Montgomerie report[72] that the radio frequency hyperfine transitions of the H_2^+ have been measured. They note that the H_2^+ is highly reactive and therefore hard to keep in the lab.

As a three-body problem, containing two protons and an electron, the Schrodinger PDE can be separated by expressing it in an elliptical coordinate system. Unfortunately, there does not exist a closed-form representation for the eigenvalues, and one must resort to numerical methods in their calculation. Bates et al.[68][69] did not provide numerical codes used in the evaluation of eigenvalues. Instead, they used an expanded fraction method for calculating the latter with the desired level of precision. The paper[69] does not report on the details of those calculations, and so in this thesis we provide an alternative description. We will compare the results obtained here with those reported in[69]. Harmony with those reported in[69] will provide confidence in the fidelity of the analogous calculations in 2D.

We use Hartree Atomic Units (5.1), in all equations. In this system, the numerical values of the following fundamental constants are 1:

$$\begin{aligned}
&\text{Electron mass:} && m_e = 1 \\
&\text{Electron charge:} && e = 1 \\
&\text{Reduced Planck constant:} && \hbar = 1 \\
&\text{Coulomb's constant:} && \frac{1}{4\pi\epsilon_0} = 1
\end{aligned} \tag{5.1}$$

In addition, in equations, variables that represent vectors are typeset either in

bold or with an arrow above the symbol, $\vec{r}$. A caret $(\hat{})$ above a symbol indicates that the symbol is an operator.

The only instance in which we will avoid using atomic units is during the derivation that involves transforming the molecular Hamiltonian into a Jacobian coordinate system. Since certain terms will include the reduced mass, we aim to maintain the integrity of the units throughout this process.

5.1 Description

The starting point is the Schrodinger equation for the hydrogen molecular ion in 3D. We seek stationary states for non-relativistic motion, and neglect the spin degrees of freedom. With these assumptions the Schrodinger equation for such a system is given by:

$$\hat{H}\psi\left(\mathbf{R},\mathbf{x}\right) = E\,\psi\left(\mathbf{R},\mathbf{x}\right). \tag{5.2}$$

$\hat{H}$ represents total Hamiltonian and $\psi\left(\mathbf{R},\mathbf{x}\right)$ is an eigenstate wavefunction. It is a function of both the collective nuclear coordinates $\mathbf{R}$, and electronic coordinates $\mathbf{x}$. Of course, since we are considering the hydrogen molecular ion, $\mathbf{x}$ is the coordinate of a single electron.

The hydrogen molecular ion H_2^+ is composed of 2 protons and a single electron. Thus, as we are neglecting relativistic effects and spin, its Hamiltonian is given as:

$$\hat{H} = \hat{T}_n + \hat{T}_e + \hat{V}_{en} + \hat{V}_{nn}. \tag{5.3}$$

The terms in the Hamiltonian are as follows:

The kinetic energy of the protons:

$$‘\hat{T}_n = -\frac{\hbar^2}{2M}\sum_{i=1}^{2}\nabla^2_{R_i} = \frac{1}{2M}\sum_{i=1}^{2}\hat{P}_i^2 \tag{5.4}$$

The kinetic energy of the electron:

$$\hat{T}_e = -\frac{\hbar^2}{2m}\nabla^2_x = \frac{1}{2m}\hat{p}^2 \tag{5.5}$$

The potential energy between the electron and the nuclei, i.e. the total Coulomb electron - nuclei attraction:

$$\hat{V}_{en} = -\sum_{i=1}^{2}|\mathbf{R}_i - \mathbf{x}| \tag{5.6}$$

Potential energy (operators) between the nuclei - the total Coulomb proton - proton repulsion.

$$\hat{V}_{nn} = \frac{e^2}{|\mathbf{R}_1 - \mathbf{R}_2|} \tag{5.7}$$

The second term in the equation above represents the operator expression.

In the equations above, $\mathbf{R}_i,\ i = 1, 2$ and M represents the coordinates and mass of the nuclei (protons) $1, 2$ respectively, and $\mathbf{x}$ and m are the coordinates and mass of the electron, respectively. Coordinates are defined with respect to a common origin in a space fixed frame.

It is convenient to separate the motion of the center of mass from the motion of the point particles. A similar approach can be applied for the quantum case. We show in Appendix A that the motion of the center of mass will be a plane

wave, which possess a well defined momentum and which is not square integrable. That makes the position of the center of mass equally probable in the all space, and confirms with the Heisenberg uncertainty principle.

In the Appendix A we define $\chi(\mathbf{R})$ as the wavefunction representing the motion of the nuclei. We then obtain:

$$\chi(\mathbf{R}) = \frac{1}{(2\pi\hbar)^{3/2}} e^{\frac{i}{\hbar}\sqrt{2ME_{CM}}\mathbf{R}} \tag{5.8}$$

where E_{CM} is the energy of the center of the mass.

Therefore, we proceed to transform the Hamiltonian into a Jacobian coordinate system, following reference[73]. The outcome of this procedure will be the transformation of the wave function as $\psi(\mathbf{R}_i, \mathbf{r}) \rightarrow \psi(\mathbf{R}_{CM}, \mathbf{R}, \mathbf{r})$ where the $\mathbf{R}_{CM}$ is the coordinate of the center of the mass, R is the distance between nuclei and $\mathbf{r}$ is an electron coordinate. We choose the coordinates as:

$$\begin{aligned}
\mathbf{R}_{CM} &= \frac{M\,\mathbf{R}_1 + M\,\mathbf{R}_2 + m\,\mathbf{x}}{2M + m} \\
\mathbf{R} &= \mathbf{R}_1 - \mathbf{R}_2 \\
\mathbf{r} &= \mathbf{x} - \frac{\mathbf{R}_1 + \mathbf{R}_2}{2}
\end{aligned} \tag{5.9}$$

Now following[73],[74], we obtain the expression of the molecular Hamiltonian in

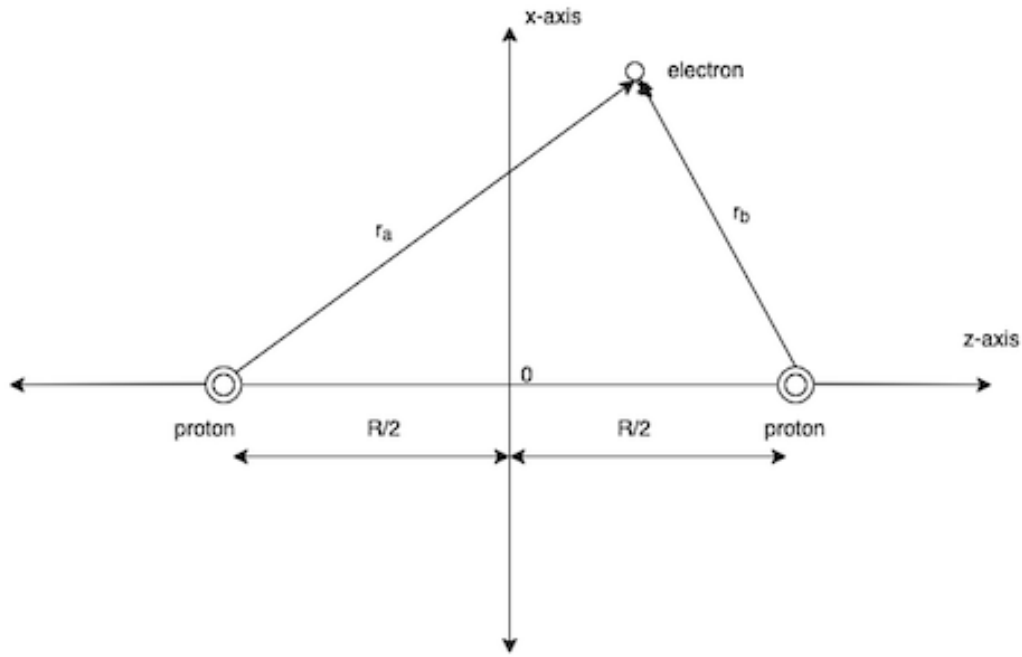


Figure 5.1: H_2^+ Ion, showing the position of the nuclei at distance R, and a coordinates of an electron

the new coordinates,

$H^{'} = T_{CM} + T_{KE} + H_{AD}$ where

Motion of the center of the mass which can be separated from the total energy of the nuclei:

$$T_{CM} = -\frac{1}{2(2M+m)}\nabla^2_{CM}$$

The relative motion of the nuclei

$$T_{KE} = -\frac{1}{2\mu}\nabla^2_R$$

where

$\mu = \dfrac{M}{2}$ is a reduced mass of the 'bare' nuclei

(5.10)

and

The adiabatic Hamiltonian:

$$H_{AD} = -\frac{1}{2m}\nabla^2_r - \frac{1}{4M}\nabla^2_r + \frac{1}{|R|} - \frac{1}{|r+R/2|} - \frac{1}{|r-R/2|} \tag{5.11}$$

The second term $\frac{1}{4M}\nabla^2_r$, in the expression for the H_{AD} represents the mass polarization term, and because $M >> m$, it can be neglected. This term represents the movement of the electron as the center of the mass is moving. In this case the mass polarization is the coupling of the two nuclei, since they are charged and interact with each other. The physical basis for the approximation is the fact that the $m_p \approx 1836\, m_e$, where m_p and m_e are the proton's and elec-

tron's mass respectively. Because of the mass difference, the center of the mass (whose mass is mostly mass of the nuclei) is moving much slower than the electron, and thus justifies neglecting this term.

Now we get for the Schrodinger equation:

$$T_{KE}\psi(\mathbf{R},\mathbf{r}) + H_{AD}\psi(\mathbf{R},\mathbf{r}) = E\,\psi(\mathbf{R},\mathbf{r}) \tag{5.12}$$

where E is the internal energy of the nuclei - electron system. To develop a working theory we use Born-Oppenheimer (BO) approximation. Using BO approximation we solve for the electronic Hamiltonian in the Coulomb field of the nuclei, where the nuclei are assumed to be stationary with respect to inertial frame. In this case, the electronic Hamiltonian is the H_{AD} from (5.11). We neglect the mass polarization term and express the electronic Hamiltonian as:

$$H_{AD} = T_e + V_{NN} + V_{Ne} \quad \text{or}$$
$$H_{AD} = \frac{1}{2m}\nabla_r^2 + \frac{1}{|R|} - \frac{1}{|r+R/2|} - \frac{1}{|r-R/2|} \tag{5.13}$$

The BO approximation neglects the motion of the nuclei when describing the electron motion in a molecule. The physical basis for the approximation is the same one as above, $m_p \approx 1836\,m_e$. In addition, due to their opposite charges, there is an attractive Coulomb force between proton and electron. But the magnitude of the acceleration is inversely proportional to the mass, so the acceleration of the electron is large and the acceleration of the proton is small, and can be neglected.

So at this step, we neglect the nuclear kinetic energy, effectively rendering the nuclei immobile. Following[70] we then solve for the motion of an electron in the Coulomb field of the nuclei, which leads to a following Schrodinger equation:

$$\left[-\frac{1}{2}\nabla_r^2 + V(\mathbf{R};\mathbf{r})\right] u(\mathbf{R};\mathbf{r}) = E(\mathbf{R})u(\mathbf{R};\mathbf{r}) \tag{5.14}$$

Now the equation (5.14) is the Schrodinger equation where $u(\mathbf{R};\mathbf{r})$ is the wavefunction for the electron with its coordinate **r** and parameter **R**. Electron is moving in the potential $V(\mathbf{R};\mathbf{r})$ with the coordinate $\mathbf{r}$. The electron's wavefunction $u(\mathbf{R},\mathbf{r})$ and the energy $E(\mathbf{R})$ both depend on the position of the nuclei $\mathbf{R}$ as a parameter. The last step in Born-Oppenheimer approximation is to solve the Schrodinger equation for the nuclei, using the above calculated $E(\mathbf{R})$ as a potential energy, which leads to a following equation.

$$\left[-\frac{1}{2}\nabla_R^2 + E(\mathbf{R})\right] v(\mathbf{R}) = \epsilon\, v(\mathbf{R}) \tag{5.15}$$

where R is the internuclear distance.

Once these operations are carried, the Born-Oppenheimer approximation states that the energy ϵ is a good approximation of the energy levels of the exact Schrodinger equation (5.2) As a result, the total molecular wave function is separated into the electronic and the nuclear wavefunction.

$$\psi(\mathbf{R}, \mathbf{r}) = u(\mathbf{R}; \mathbf{r}) v(\mathbf{R}) \tag{5.16}$$

In addition, each electronic eigenvalue $E_n(R)$ will give a rise to the electronic surface, known as Born-Oppenheimer surfaces.

One could proceed further for the complete treatment of the Born-Oppenheimer approximation, for example dealing with the Born-Oppenheimer off-diagonal correction[75], but it will go beyond the scope of this thesis .

At this moment, using BO approximation we have a Schrodinger equation for the electron, orbiting the two nuclei (each consisting of a single proton). Note that in the further text the ψ will denote the function $u(\mathbf{R}; \mathbf{r})$, since it is customary to denote the wave function as ψ.

5.2 Replicating the 3D Solution for the H_2^+ Molecular Ion

5.2.1 Solution Process

Following the seminal papers by Bates et. all.[68 69], we can re-express the equation as:

$$\left(-\frac{1}{2}\nabla^2 - \frac{1}{r_a} - \frac{1}{r_b}\right)\psi = E\,\psi \tag{5.17}$$

where $r_a = \left|\mathbf{r} - \frac{1}{2}\mathbf{R}\right|$ and $r_b = \left|\mathbf{r} + \frac{1}{2}\mathbf{R}\right|$ are distances of the electron from the nuclei and $\mathbf{R}$ is a distance between nuclei.

Due to the cylindrical symmetry of the problem, we choose the elliptical coor-

dinates. We follow[68], where authors chose the confocal elliptic co-ordinates $\lambda,\ \mu$, which transforms the coordinates r_a and r_b from (5.17) into:

$$r_a = \frac{R}{2}(\lambda + \mu)\,, \quad r_a = \frac{R}{2}(\lambda - \mu) \tag{5.18}$$

After applying the coordinate transformations (5.18) into the equation (5.17), we obtain another partial differential equations[68]:

$$\begin{aligned} &\frac{d}{d\lambda}\left\{(\lambda^2-1)\frac{d\psi}{d\lambda}\right\} + \frac{d}{d\mu}\left\{(1-\mu^2)\frac{d\psi}{d\mu}\right\} + \left\{\frac{1}{\lambda^2-1} + \frac{1}{1-\mu^2}\right\}\frac{d^2\psi}{d\phi^2} + \\ &+ \left\{\frac{1}{4}R^2E(\lambda^2-\mu^2) + 2R\lambda\right\}\psi = 0 \end{aligned} \tag{5.19}$$

We assume that the solution can be separated, in the product of the three functions of the single variable, λ, μ, ϕ. Under this assumption the solution can written as:

$$\psi(\lambda,\mu,\phi) = L(\lambda)M(\mu)e^{im\phi} \quad \text{where is} \quad p^2 = -\frac{1}{4}R^2E \tag{5.20}$$

In this equation the L, M are the functions of their respective coordinates, and m is an integer, either positive or negative. The next step is to insert the ansatz (5.20) into the Eq (5.19) . After some algebra, one observes that the equation (5.19) separates into the system of two ordinary, coupled differential equations. an

$$\frac{d}{d\mu}\left\{(1-\mu^2)\frac{dM}{d\mu}\right\}+\left\{-A+p^2\mu^2-\frac{m^2}{1-\mu^2}\right\}M=0 \tag{5.21}$$

$$\frac{d}{d\lambda}\left\{(\lambda^2-1)\frac{dL}{d\lambda}\right\}+\left\{A+2R\lambda-p^2\lambda^2-\frac{m^2}{\lambda^2-1}\right\}L=0 \tag{5.22}$$

In[68], Bates et al. state that they used the method of successive approximation to find the coefficients in the series. It was tacitly assumed that the successive approximation themselves converge. The approximation is terminated either when the desired precision is achieved of when the certain number of approximation is met. However the details of the numerical calculation are not provided in the paper.

5.2.2 Differential Operator Spectrum

We can view the equations for M and L as the eigenvalue operator equations $D^1\,M(\mu)\,=\,A\,M(\mu)$ and $D^2\,L(\lambda)\,=\,-A\,L(\lambda)$. Since these equations model a real, physically realizable system, we assume that operators $D^{1,2}$ are compact on l^2 space, therefore each $D^{1,2}$ operator satisfies:

$$\|D_n^{1,2}\rightarrow D^{1,2}\| \tag{5.23}$$

where $D_n^{1,2}\,=\,P_nD^{1,2}P_n$ So with these assumptions, operators $D^{1,2}$ can be approximated by the finite dimensional matrices.

5.2.3 TOWARDS THE SOLUTION

Here we replicate the solution of Bates et. all.[68 69], by using a more of a brute force approach. The idea is to numerically calculate the spectrum of a differential operator to a desired precision We expressed equations (5.21),(5.22) as a matrix eigenvalue problem. The system (5.21),(5.22) of the two ODEs then depend on the two parameters, total energy E, and the separation constant A. The problem is reduced to finding those values of E and A for which the solution to the differential equation satisfy the requisite boundary conditions. Since it is a Sturm-Liouville problem, or more precisely a system of two Sturm-Liouville problems, our approach is fundamentally the same approach as taken by Bates. but it is more practical, as it can better accommodate the existing numerical software. There exist excellent numerical packages (Lapack)[76] for computing the matrix eigenvalues, and they greatly simplify the actual computation. The Fortran code in D uses Lapack library to compute eigenvalues.

We express both function $M(\mu)$ and $L(\lambda)$ using the set of orthogonal functions which satisfy the boundary conditions. We then follow the method of Robyn's. Once we replace the assumed solution into the equation and after some algebra, we obtain effectively two uncoupled, infinite systems of equations, where the unknowns are the coefficients in the solution series. The solution vector depends on the two parameters, p, which is the function of energy and A which is the separation constant.

The next step is rather simple. For functions $M(\mu)$ and $L(\lambda)$, we need to find the functional dependency of the parameters p and A, so that solution exists. The intersections of the two curves $p_1 = f(A_1)$ and $p_2 = g(A_2)$ provides the value of p and A when the condition for the solutions exists. The desired precision is obtained by choosing the size number of terms in the series.

The remaining issue is that even with the modern fast computers and optimized numerical libraries[76], calculating eigenvalues is a slow process. The complexity of the QR decomposition algorithms is $O(n^3)$[77], which make the QR factorization feasible by today's computer systems, but fairly slow for large matrices.

5.3 Series Expansions and Eigenvalue Determinant

Here we describe in details the method describe in the previous section.

5.3.1 L Equation

Start with:

$$\begin{aligned}
&\frac{d}{d\lambda}\left\{\left(\lambda^2-1\right)\frac{d\Lambda}{d\lambda}\right\}+\left\{A-p^2\lambda^2+2R\lambda\right\}\Lambda=0\Rightarrow\\
&\left(\lambda^2-1\right)\frac{d^2\Lambda}{d\lambda^2}+2\lambda\frac{d\Lambda}{d\lambda}+\left\{A-p^2\lambda^2+2R\lambda\right\}\Lambda=0
\end{aligned}\tag{5.24}$$

For practical reasons we first shift the L equation b $x=\lambda-1$ to obtain:

$$\frac{d}{d\,x}\left\{x(x+2)\frac{d\,\Lambda}{d\,x}\right\}+\left\{-p^2\,x^2+(-2p^2+2R)x-p^2+2R+A\right\}\Lambda=0$$

$$x(x+2)\frac{d^2\,\Lambda}{d\,x^2}+(2x+2)\frac{d\,\Lambda}{d\,x}+\left\{-p^2\,x^2+(-2p^2+2R)x-p^2+2R+A\right\}\Lambda=0 \tag{5.25}$$

Now we assume that the solution exists and that it can be represented as the sum of Laguerre polynomials:

$$\Lambda(x)=e^{-px}\sum_{n=0}^{\infty}c_n\,L_n(x) \tag{5.26}$$

Inserting the ansatz (5.26) into (5.25), applying the properties of Laguerre's polynomial some algebra we obtain.

$$\begin{aligned}
&-2p\sum_{n=0}^{\infty}c_n\,n(2n+1)L_n+2p\sum_{n=0}^{\infty}c_{n+1}\,n(n+1)L_{n+1}+2p\sum_{n=0}^{\infty}c_{n-1}\,n^2L_{n-1}+\\
&+(2p-1)\sum_{n=0}^{\infty}c_{n-1}n(2n-1)L_{n-1}-(2p-1)\sum_{n=0}^{\infty}c_nn^2L_n-\\
&-(2p-1)\sum_{n=0}^{\infty}c_{n-2}n(n-1)L_{n-2}+(-2p+2R)\sum_{n=0}^{\infty}c_n(2n+1)L_n-\\
&-(-2p+2R)\sum_{n=0}^{\infty}c_{n+1}(n+1)L_{n+1}-(-2p+2R)\sum_{n=0}^{\infty}c_{n-1}nL_{n-1}+\\
&+(-4p+1)\sum_{n=0}^{\infty}c_n\,n\,L_n+(4p+1)\sum_{n=0}^{\infty}c_{n-1}\,n\,L_{n-1}+\\
&+(-2p-p^2+2R+A)\sum_{n=0}^{\infty}c_n\,L_n=0
\end{aligned} \tag{5.27}$$

The next step is to multiply the equation (5.27) with $e^{-x}L_m(x)$ and integrate

from 0 to ∞ using the orthogonality property of the Laguerre's polynomials: $\int_0^\infty dx\, e^{-x} L_n(x) L_m(x) \;=\; \delta_{mn}$ to obtain the recurrence equations for the unknown coefficients c_n

$$
\begin{aligned}
&-2p\,n(2n+1)c_n + 2p\,n(n+1)c_{n+1} + 2p\,n^2 c_{n-1} + (2p-1)n(2n-1)c_{n-1} - \\
&-(2p-1)n^2 c_n - (2p-1)n(n-1)c_{n-2} + (-2p+2R)(2n+1)c_n - \\
&-(-2p+2R)(n+1)c_{n+1} - (-2p+2R)nc_{n-1} + (-4p+1)\,n\,c_n + \\
&+(4p+1)\,n\,c_{n-1} + (-2p-p^2+2R+A), c_n = 0
\end{aligned}
\tag{5.28}
$$

This expression (5.28) is a system of linear equations, where the unknowns are the coefficients c_n. The coefficients multiplying the unknowns c_n in (5.28) form the band matrix.

5.3.2 M EQUATION

We use the same approach for the M equation.

$$
\frac{d}{d\,\mu}\left\{\left(1-\mu^2\right)\frac{d\,M}{d\,\mu}\right\} + \left(-A + p^2\mu^2\right) M(\mu) = 0 \tag{5.29}
$$

We express the $M(\mu)$ as the series of Legendre polynomials.

$$
M(\mu) = \sum_{n=0}^{\infty} f_n\, P_n(\mu) \tag{5.30}
$$

where

$$\int_{-1}^{1} P_m(x)P_n(x)dx = \frac{2}{2m+1}\delta_{m,n} \tag{5.31}$$

Substituting (5.30) into (5.29), multiplying by $P_m(\mu)$, and integrating from -1 to 1 we obtain

$$\sum_{n=0}^{\infty}\Big\{-\int_{1}^{1} P_m(\mu)n(n+1)P_n(\mu)d\mu - A\int_{1}^{1} P_m(\mu)P_n(\mu)d\mu +$$

$$+\, p^2\int_{1}^{1} P_m(\mu)\mu^2 P_n(\mu)d\mu\Big\} = 0 \quad \text{or} \tag{5.32}$$

$$\sum_{n=0}^{\infty}\Big\{\frac{-2}{2n+1}n(n+1)\delta_{m,n} - A\frac{2}{2n+1}\delta_{m,n} + p^2\int_{1}^{1} P_m(\mu)\mu^2 P_n(\mu)d\mu\Big\} = 0$$

Taking advantage of the orthogonality of Legendre Polynomials (5.31), we obtain a system of linear equations, where the coefficients form a band matrix.

$$\sum_{n=0}^{\infty}\Big\{\frac{-2n(n+1)}{2n+1}\delta_{m,n} - A\frac{2}{2n+1}\delta_{m,n} +$$

$$+p^2\frac{2}{2n+1}\left[\frac{(n+1)^2}{(2n+1)(2n+3)} + \frac{n^2}{(2n+1)(2n-1)}\right]\delta_{m,n} +$$

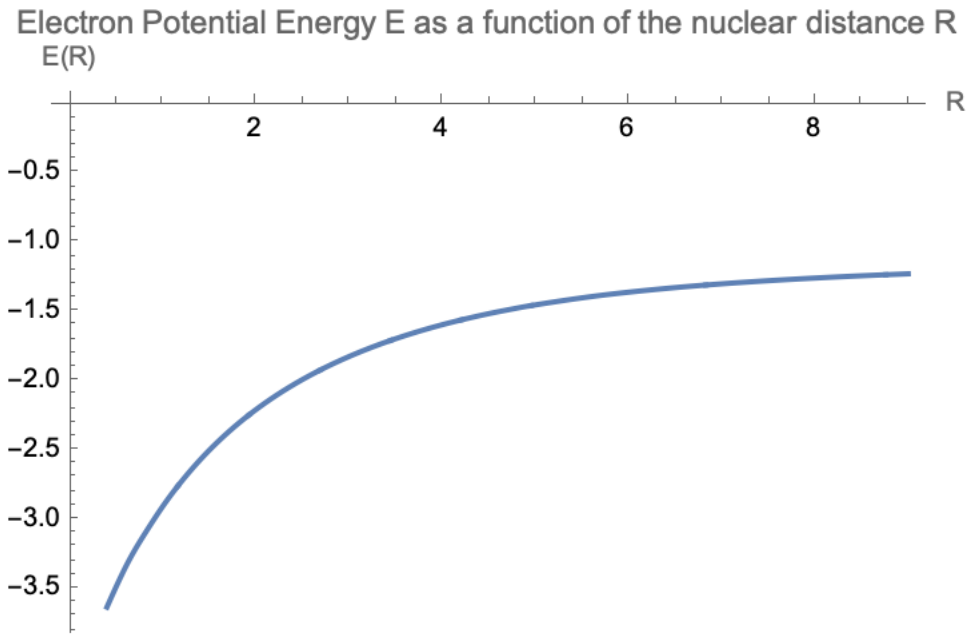
$$p^2\frac{2(n+1)(n+2)}{(2n+1)(2n+3)(2n+5)}\delta_{m,n+2} + p^2\frac{2\,n(n-1)}{(2n+1)(2n-1)(2n-3)}\delta_{m,n-2}\Big\} = 0 \tag{5.33}$$

In B we present the Mathematica code for calculating electron energies of the molecular H_2^+ ion in 3 dimensions using Mathematica eigenvalue solver.

5.3.3 NUMERICAL CALCULATION

Equations (5.28) and (5.33) constitute the couple system of linear equation. Each series is parameterized by 2 parameters, p and A , as well as the parameter R, the distance between nuclei. We select the values of R as shown in table 5.1. We tacitly assume that the series (5.26) (5.29) converge, moreover, we assume that they converge uniformly. Therefore, we truncate the series at some index $i = K$, chosen to achieve the desired accuracy. Now the coefficient matrices are dimensions $K \times K$ and each matrix has K eigenvalues, and the lowest eigenvalue corresponds to the lowest value of p, thus to the lowest energy.

In next step, for each value of R, we calculate the eigenvalues, sort them, and pick the lowest (or the i-th one from the bottom), and view them as functions $A = A(p)$. So for each of the equations (5.28) and (5.33), we choose the interval $[0, 2 * R]$ and divide it into N points $p_i, i = 1..N$. For each point, we calculate, sort and pick desired the eigenvalue of the matrix M (L). That way we obtain two functions $A_M = A_M(p)$ and $A_L = A_L(p)$, evaluated at points p_i. We interpolate these two functions, using Wolfram's Mathematica 3-rd order Spline interpolation. As a side note, for a small subset of values R we used the higher order Spline interpolation but we were unable to observe the difference in the results obtained. The intersection of these two curves in the value of p for which both $M(\mu)$ and $L(\lambda)$ equations have results satisfying the boundary conditions. Now we repeat the calculation for the next value of



R.

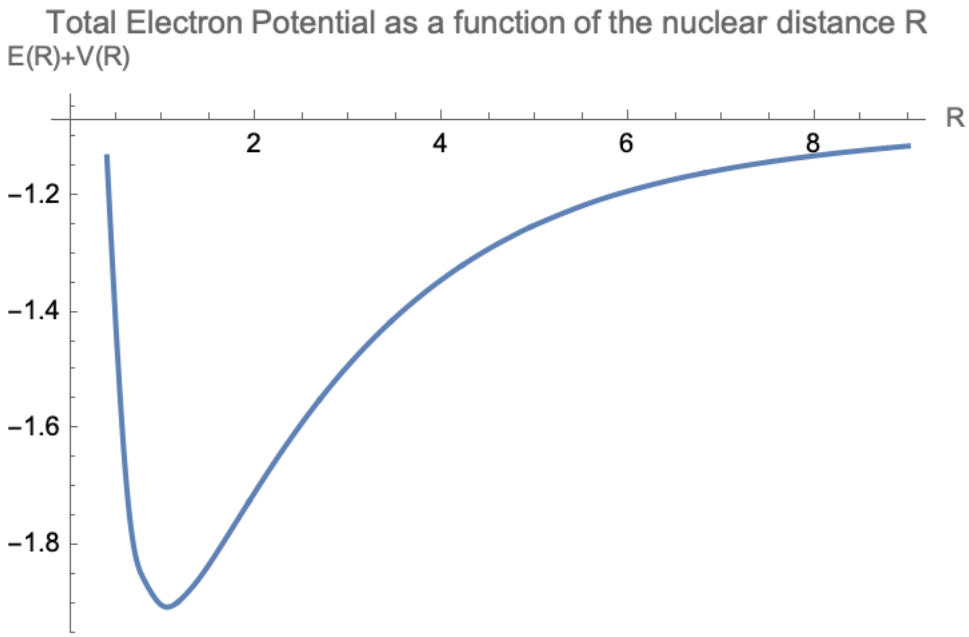
After all this exercise, we get the function $E = E(R)$, from (5.14) which is the electron energy in the BO approximation. Adding the potential energy between nuclei is $V(R) = 1/R$, we obtain the potential energy curve. The paper by Bates does not explicitly use this curve to calculate the vibrational energies, but we will, in our 2D case.

For illustration purposes we present some of the results below.

Plot of the ground state energies $E(R)$ and $E(R) + V(R)$ where $V(R) = \frac{1}{R}$ represents the potential energy of the nuclei.

Electron energy as an eigenvalue of the equations (5.28) and (5.33).

Total energy, calculated by adding the potential energy of the nuclei $V(R) =$



$1/R$.

Electron energy as an eigenvalue of the equations (5.28) and (5.33).

Table comparing the electron energies obtain using our method vs method applied by Bates et al. We chose a subset of values to indicate the overall trend. The $1s\sigma_g$ and $2p\sigma_u$ represent the ground Gerade and Ungerade state respectively. In the table below we used atomic units for both R and $E(R)$.

For reference we use the values obtained by Bates[68 69]. Valued obtained by our method are is good alignment with the referent values.

Table 5.1: Electron Energy, $1s\sigma_g$ state,

R	-E(R) (Our Method)	-E(R) (Bates)
0.4	3.63220	3.601570
0.6	3.34296	3.34301
0.8	3.10895	3.10895
1.0	2.90357	2.90356
1.2	2.7246154	2.72461
1.4	2.5685373	2.56853
1.6	2.4318740	2.43186
1.8	2.3116176	2.31162
2.0	2.2052681	2.20525
2.2	2.1107695	2.11076
2.4	2.0264405	2.02642
2.6	1.9508966	1.95090
2.8	1.88299	1.88299
3.0	1.82179	1.82178
3.2	1.76648	1.76647
3.4	1.71640	1.71639
3.6	1.67097	1.67097
3.8	1.62970	1.62971
4.0	1.59216	1.59216
4.2	1.55799	1.557998
4.4	1.52685	1.52685
4.6	1.49844	1.49844
4.8	1.47252	1.47252
5.0	1.44884	1.44884

Table 5.2: Electron Energy, $2p\sigma_u$ state

R	-E(R) (Our Method)	-E(R) (Bates)
0.4	-1.3456	1.02158
0.6	-1.0487	1.04861
0.8	-1.08552	1.08548
1.0	-1.12961	1.12962
1.2	-1.17719	1.1772
1.4	-1.22415	1.22415
1.6	-1.26723	1.26721
1.8	-1.30446	1.30446
2.0	-1.33507	1.13507
2.2	-1.35912	1.35912
2.4	-1.37715	1.37715
2.6	-1.38992	1.38991
2.8	-1.39823	1.39822
3.0	-1.40284	1.40285
3.2	-1.40443	1.40443
3.4	-1.40359	1.40358
3.6	-1.40082	1.40082
3.8	-1.39655	1.39654
4.0	-1.3911	1.39110
4.2	-1.38477	1.38478
4.4	-1.37778	1.37779
4.6	-1.37033	1.36255
4.8	-1.36255	1.36255
5.0	-1.35458	1.35458

CHAPTER 6

NON-RELATIVISTIC SOLUTION OF THE HYDROGEN MOLECULAR ION IN 2 SPATIAL DIMENSIONS

6.1 EXISTING SOLUTIONS

We apply similar method used in the 3D case of the H_2^+ ion to find the solution in 2 dimensions. The problem itself is analogous to the 3D problem, but leads to the Mathieu's function as the solution for the radial problem.

There are not too many papers dealing with solving the H_2^+ but here is some relevant, and recent work in this problem. In addition to the Bates paper[68], there are other, more recent papers[78 79],[80] which deal with the solution of the Schrodinger equation and spectrum of the H_2^+ molecule in 3 dimensions. As with other work in this problem, the paper relies on the work done by Bates et all.[68]. The solution in[78] agrees well with our solution. One can also apply approximate methods, but they should be considered inferior to the analytical solution, we provide here.

Analogous to the 3D case, we rely on Born-Oppenheimer (BO) approximation, in order to provide an analytical solution. The same justification for using the BO approximation in 3D molecular systems applies to the 2D problems as well, since the masses of the nuclei and electron(s) remain unchanged. Following the same BO approximation as in Chapter 1, and using atomic units, the Schrodinger equation for the H_2^+ molecules is given by equation (6.1) below.

$$\left(-\frac{1}{2}\nabla^2 - \frac{1}{r_a} - \frac{1}{r_b}\right)\psi = E\,\psi \tag{6.1}$$

The equation (6.1) is superficially similar to the equation (5.17) and the diagram 5.1, but in this case, the r_a and r_b represent the nuclei coordinates in 2 dimensions.

Following the[81], The electron wavefunction depends on only 3 quantum numbers, principal quantum number n, angular quantum number l and spin s. Again we ignore the spin degrees of freedom, considering the electron to be the regular 3 dimensional particles, where only its orbit is restricted to 2 dimensions. Therefore the spin magnetic moment of the electron is not affected by the 2D restriction and the spin vector can point in any direction in 3D space. Also the orbit can have different shapes, thus there remains the need for an orbital quantum number. However, there can be one one orientation of the orbit, thus we do not consider the magnetic quantum number. This reasoning agrees with the solution of the electron wavefunction to the hydrogen

atom in 2 dimensions[82] where electron wavefunction solution only depends on the principal and orbital quantum numbers.

The geometry of the 2D problem leads to choosing the elliptical coordinates (as in 3D problem), with two coordinates, μ and λ denoting the position of the electron in 2D plane.[83]

6.2 Exact solution of the electron energies of the H_2^+ molecule

The goal is to provide the exact solution to the wavefunction of the H_2^+ electron, for a given definition of exact. However even for this, relatively simple problem, it is impossible to find a closed form solution. So the solution is exact, in the sense that it can be done to an arbitrary precision.

As illustrated by the figure 6.1, we express equation (6.1) in the elliptical coordinates, and by setting the x axis to be perpendicular to the inter-nuclear axis, we have the nuclei at: $z = \pm\frac{R}{2}$, R being the distance between nuclei. So in 2D elliptic coordinates, λ, μ we have:

$$r_a = \frac{R}{2}(\lambda + \mu) \qquad r_b = \frac{R}{2}(\lambda - \mu) \tag{6.2}$$

so

$$\lambda = (r_a + r_b)/R; \qquad \mu = (r_a - r_b)/R \tag{6.3}$$

where $\lambda \in [1, \infty]$, $\quad \mu \in [-1, 1]$ (6.4)

Now if we plug in the (6.2) into (6.1) we get by using ∇ in 2D elliptic cylindrical coordinates[84] .

$$\nabla^2 = \frac{4}{R^2(\lambda^2-\mu^2)}\left[\sqrt{\lambda^2-1}\frac{\partial}{\partial\lambda}\left(\sqrt{\lambda^2-1}\frac{\partial\psi}{\partial\lambda}\right)+\sqrt{1-\mu^2}\frac{\partial}{\partial\mu}\left(\sqrt{1-\mu^2}\frac{\partial\psi}{\partial\mu}\right)\right] = \frac{4}{R^2(\lambda^2-\mu^2)}\left[(\lambda^2-1)\frac{\partial^2\psi}{\partial\lambda^2}+\lambda\frac{\partial\psi}{\partial\lambda}+(1-\mu^2)\frac{\partial^2\psi}{\partial\mu^2}-\mu\frac{\partial\psi}{\partial\mu}\right] \tag{6.5}$$

So the equation transforms to (with $\psi = \psi(\lambda,\mu)$:

$$-\frac{1}{2}\frac{4}{R^2(\lambda^2-\mu^2)}\left[(\lambda^2-1)\frac{\partial^2\psi}{\partial\lambda^2}+\lambda\frac{\partial\psi}{\partial\lambda}+(1-\mu^2)\frac{\partial^2\psi}{\partial\mu^2}-\mu\frac{\partial\psi}{\partial\mu}\right]-\frac{4}{R}\frac{\lambda}{\lambda^2-\mu^2}\psi = E\psi \tag{6.6}$$

We assume that the total electronic wavefunction can be written as the product of two functions:

$$\psi(\lambda,\mu) = F(\mu)L(\lambda) \tag{6.7}$$

it follows:

$$\frac{2}{R^2(\lambda^2-\mu^2)}\left[(\lambda^2-1)F\frac{\partial^2 L}{\partial\lambda^2}+\lambda F\frac{\partial L}{\partial\lambda}+(1-\mu^2)L\frac{\partial^2 F}{\partial\mu^2}-\mu L\frac{\partial F}{\partial\mu}\right]+\frac{4}{R}\frac{\lambda}{\lambda^2-\mu^2}FL+EFL=0 \Rightarrow$$

$$(\lambda^2-1)\frac{1}{L}\frac{\partial^2 L}{\partial\lambda^2}+\frac{\lambda}{L}\frac{\partial L}{\partial\lambda}+(1-\mu^2)\frac{1}{F}\frac{\partial^2 F}{\partial\mu^2}-\frac{\mu}{F}\frac{\partial M}{\partial\mu}+2R\lambda+\frac{R^2}{2}E(\lambda^2-\mu^2)=0 \tag{6.8}$$

We can conclude that both equations for λ and μ must be equal to the separa-

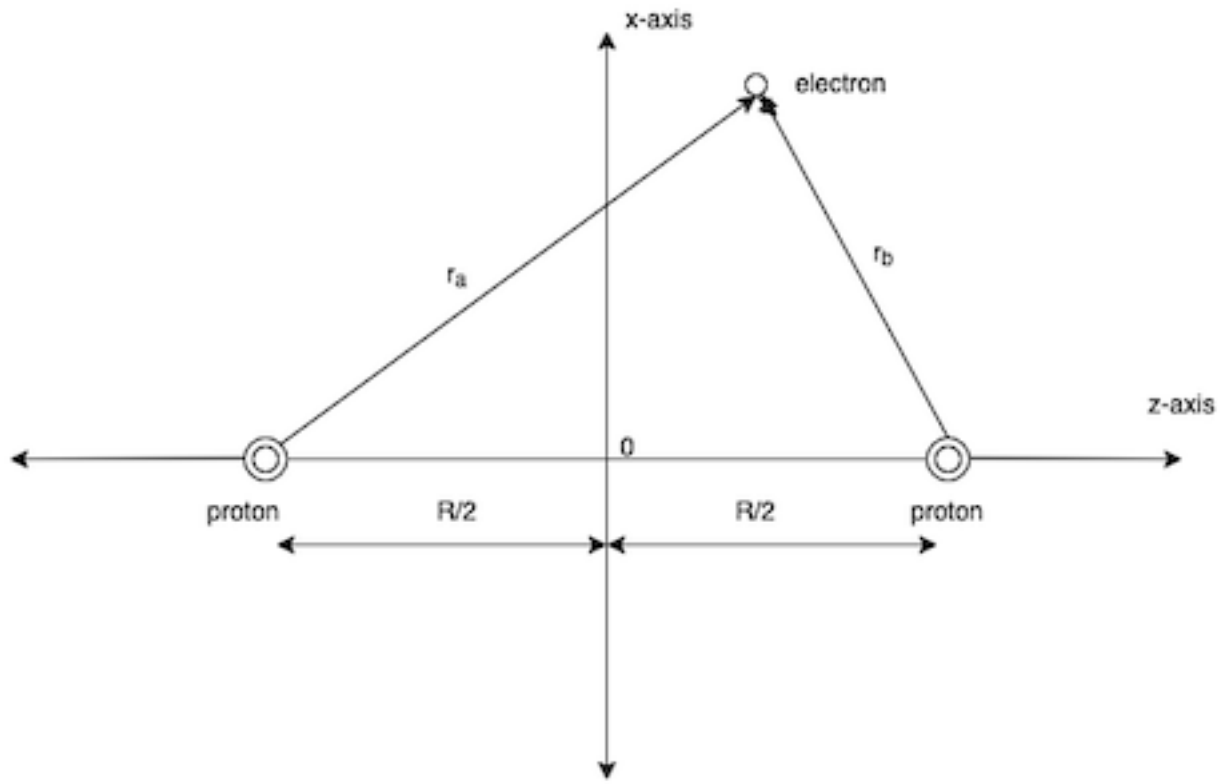


Figure 6.1: Hydrogen Ion in 2 dimension

tion constant A.

$$
\begin{aligned}
&(\lambda^2-1)\frac{\partial^2 L}{\partial\lambda^2}+\lambda\frac{\partial L}{\partial\lambda}+\left(2R\lambda-\frac{R^2}{2}E\lambda^2\right)L=AL\\
&(1-\mu^2)\frac{\partial^2 F}{\partial\mu^2}-\mu\frac{\partial F}{\partial\mu}-\frac{R^2}{2}E\mu^2F=-AF
\end{aligned}
\tag{6.9}
$$

where A is the separation constant.

6.2.1 Algebraic Solution

M Equation

Using the substitution $\mu = \cos x$ and $F(\mu) = M(x)$ we get a new equation:

$$
\frac{d^2M}{dx^2}+\left[-A+\frac{p^2}{2}+\frac{p^2}{2}\cos(2x)\right]M=0 \tag{6.10}
$$

The equation (6.10) can be written as:

$$
\frac{d^2M}{dx^2}+\left[w-2q\cos(2x)\right]M=0 \tag{6.11}
$$

where

$$
\begin{aligned}
&w=-A+\frac{p^2}{2}\\
&q=-\frac{p^2}{2}
\end{aligned}
\tag{6.12}
$$

The equation (6.11) is a form of a Mathieu's equation[85].

From the geometry of the problem we conclude that $M(\mu)$ must be an even

function. Following[86] we look for the solution as class I and class II, which has even and odd eigenvalue functions as:

$$V_0 = \cfrac{2}{V_2 - \cfrac{1}{V_4 - \cfrac{1}{V_6 - ...}}} \tag{6.13}$$

$$V_1 - 1 = \cfrac{1}{V_3 - \cfrac{1}{V_5 - \cfrac{1}{V_7 - ...}}} \tag{6.14}$$

where

$$V_m = \frac{w - m^2}{q} \tag{6.15}$$

The equations 6.13 and 6.14 provide the first gen equation for the even and odd solution respectively.

F EQUATION

$$(\lambda^2 - 1)\frac{\partial^2 F}{\partial \lambda^2} + \lambda \frac{\partial F}{\partial \lambda} + \left(2R\lambda - \frac{R^2}{2}E\lambda^2\right) F = -AF \tag{6.16}$$

We observe that both equations for the functions $M(\mu)$ and $F(\lambda)$ are either exact match or related to the Mathieu's equations. In general the Mathieu's equation represents the standing wave on an elliptical drum, (2D space), and the solution for the time independent Schrodinger equation is in general a

standing wave, in 2D in this case. So it is plausible that these types of solutions are similar.

There are actually two ways to try to solve this equation. One would be a brute force, by using some orthogonal polynomial (Laguerre) and looking for the eigenvalues.

The Laguerre approach seems to be numerically unstable. The algorithm is fast but the results are only good to the order of magnitude.

The second is a bit more algebraic, and likely more accurate.

Following[68] and[87] we could look for the solution in the form of:

$$L(\lambda) = (\lambda + 1)^{\sigma} e^{-p\lambda} \sum_{n=0}^{\infty} a_n x^n \tag{6.17}$$

where we use the substitution

$$\sigma = \frac{R}{p} - \frac{1}{2} \quad \text{and} \quad x = \frac{\lambda - 1}{\lambda + 1} \tag{6.18}$$

Substituting (6.18) into (6.17) and after some formidable algebra, we get a recurrence relation:

$$\alpha_n a_{n+1} - \beta_n a_n + \gamma_n a_{n-1} = 0 \quad n \geq 0 \tag{6.19}$$

with

$$\alpha_n = (n+1)\left(n + \frac{1}{2}\right)$$

$$\beta_n = \left[2n^2 + (4p - 2\sigma)n - A + p^2 - 2p\sigma - \frac{\sigma}{2}\right] \tag{6.20}$$

$$\gamma_n = (n-1)\left(n - 2\sigma - \frac{1}{2}\right) + \sigma\left(\sigma - \frac{1}{2}\right)$$

and if follows that

$$\frac{a_n}{a_{n-1}} = F_n \quad \text{where} \quad F_n = \cfrac{\gamma_n}{\beta_n - \cfrac{\alpha_n \gamma_{n+1}}{\beta_{n+1} - \cfrac{\alpha_{n+1}\gamma_{n+1}}{\beta_{n+1} - \ldots}}} \tag{6.21}$$

Since $a_{-1} = 0$ we have

$$\frac{\beta_0}{\alpha_0} = F_1 \tag{6.22}$$

The Mathematica code is in Appendix C. The energies calculated and the plot of the potential for the states $1s_g^+$ and $2p_u^+$ are in the tables below.

The states $1s_g$ and $2p_u^+$ represent the Gerade ground state and the Ungerade state respectively.

For comparison we used paper by Patil[78]. Our result are very close to the results obtained by Patil. Also one can observe that in 2 dimensions the max depth of the potential well of the H_2^+ ground state is approx $V(R = 0.5) = -2.83822$ a.u. compared to the $V(R = 0.4) = -3.63220$ in the 3D case, which we showed in table 5.1 Also the shape of the potential well is more compact in the 2D case, for the ground $1s_g^+$ gerade state. For the ungerade $2p_u^+$ state the situation is even more different. The potential curve of the $2p_u^+$ state does not really have a minimum for and does not allow a bound state.

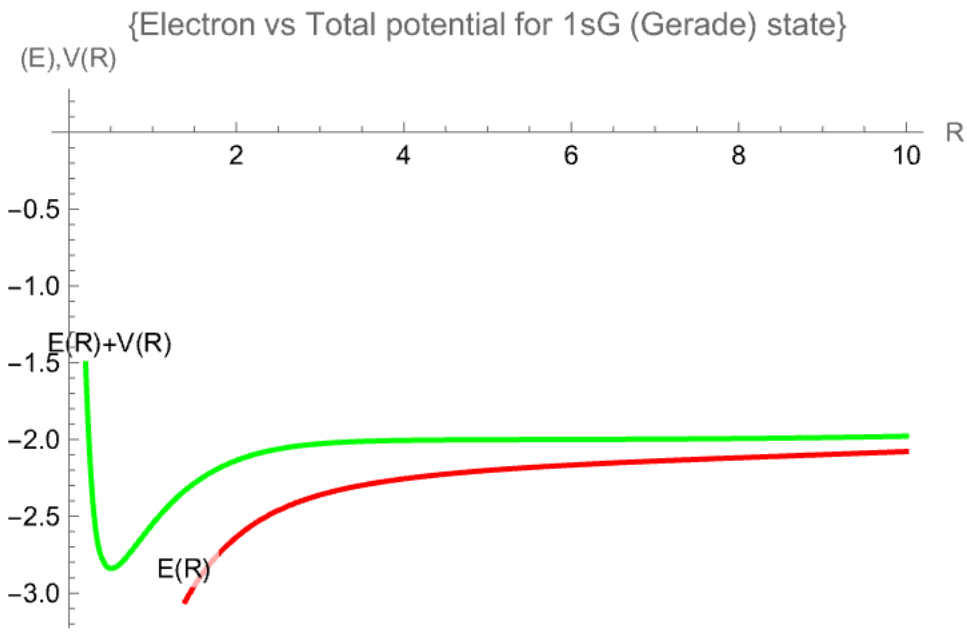


Figure 6.2: H_2^+ Energy and Potential curves for the $1s_g^+$ state

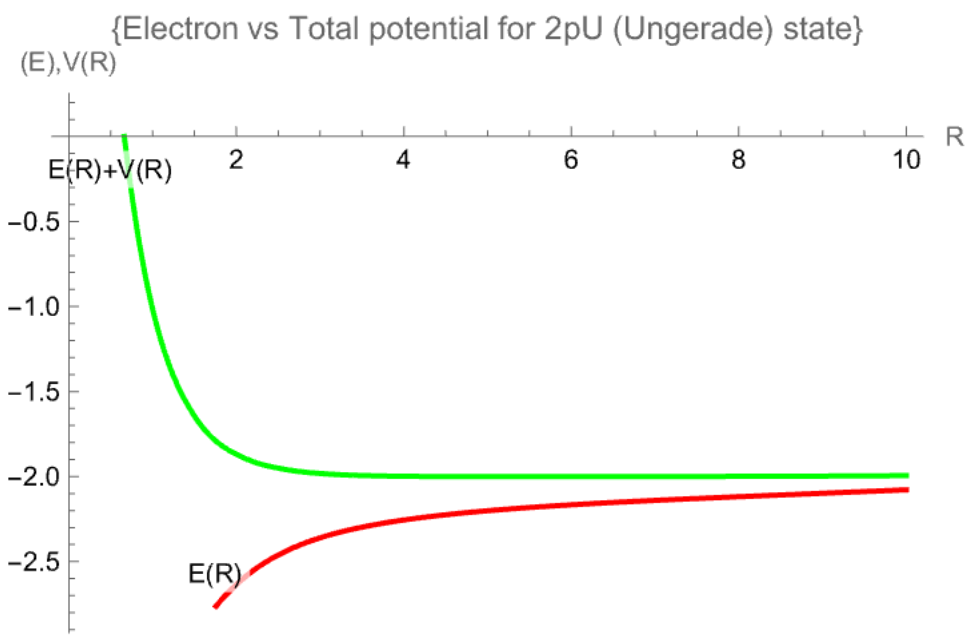


Figure 6.3: H_2^+ Energy and Potential curves for the $2p_u^+$ state

Table 6.1: The separation constant A, electronic energy E

total energy $E + \frac{1}{R}$, for the state $1s_g$ some values of the inter-nuclear separation R

R	E	$E + \frac{1}{R}$	A	p
0.2	-6.50361	-1.50361	19.2757	0.360655
0.3	-5.81983	-2.48649	175.774	0.511754
0.4	-5.27552	-2.77552	198.466	0.649647
0.5	-4.83822	-2.83822	218.834	0.777674
0.6	-4.48148	-2.81482	30.6031	0.898146
0.7	-4.18616	-2.75759	255.124	1.01272
0.8	-3.93852	-2.68852	271.721	1.12264
0.9	-3.72861	-2.6175	287.554	1.22886
1.	-3.54905	-2.54905	302.754	1.33211
1.1	-3.39429	-2.4852	317.417	1.43302
1.5	-2.95083	-2.28416	371.99	1.822
2.	-2.63638	-2.13638	433.864	2.29625
2.5	-2.4621	-2.0621	491.12	2.77382
3.	-2.36086	-2.02752	545.139	3.25943
3.5	-2.29797	-2.01225	596.728	3.75167
4.	-2.25565	-2.00565	646.407	4.24796
4.5	-2.22496	-2.00274	694.536	4.74634
5.	-2.20134	-2.00134	741.373	5.24564
6	-2.16637	-1.9997	16608.	6.24457
7	-2.14043	-1.99757	919.107	7.24158
8	-2.11874	-1.99374	1003.68	8.23405
9	-2.09856	-1.98745	1086.12	9.21909
10	-2.07822	-1.97822	1166.79	10.1937

Table 6.2: The separation constant A, electronic energy E

total energy $E + \frac{1}{R}$, for the state $2p_u^+$ for some values of the inter-nuclear separation R

R	E	$E + \frac{1}{R}$	A	p
0.2	-0.841481	4.15852	33.4451	0.129729
0.3	-0.911818	2.42152	39.6245	0.202563
0.4	-1.03317	1.46683	44.6816	0.287495
0.5	-1.20111	0.798885	49.0907	0.387478
0.6	-1.39266	0.274004	53.0632	0.500679
0.7	-1.58212	-0.153551	56.7159	0.622591
0.8	-1.75267	-0.502667	395.73	0.748901
0.9	-1.89725	-0.786142	417.783	0.876577
1.	-2.01521	-1.01521	66.3746	1.0038
1.1	-2.10899	-1.1999	459.254	1.12957
1.5	-2.31161	-1.64494	534.677	1.61262
2.	-2.36744	-1.86744	619.719	2.17598
2.5	-2.35029	-1.95029	698.046	2.7101
3.	-2.31526	-1.98193	111.724	3.2278
3.5	-2.2797	-1.99399	841.77	3.73673
4.	-2.24845	-1.99845	909.103	4.24118
4.5	-2.22222	-2.	974.188	4.74341
5.	-2.20044	-2.00044	1037.4	5.24457
6	-2.16704	-2.00038	1159.29	6.24554
7	-2.14278	-1.99992	1276.36	7.24556
8	-2.12404	-1.99904	1389.63	8.24435
9	-2.10851	-1.9974	1499.83	9.24093
10	-2.09461	-1.99461	1607.45	10.2338

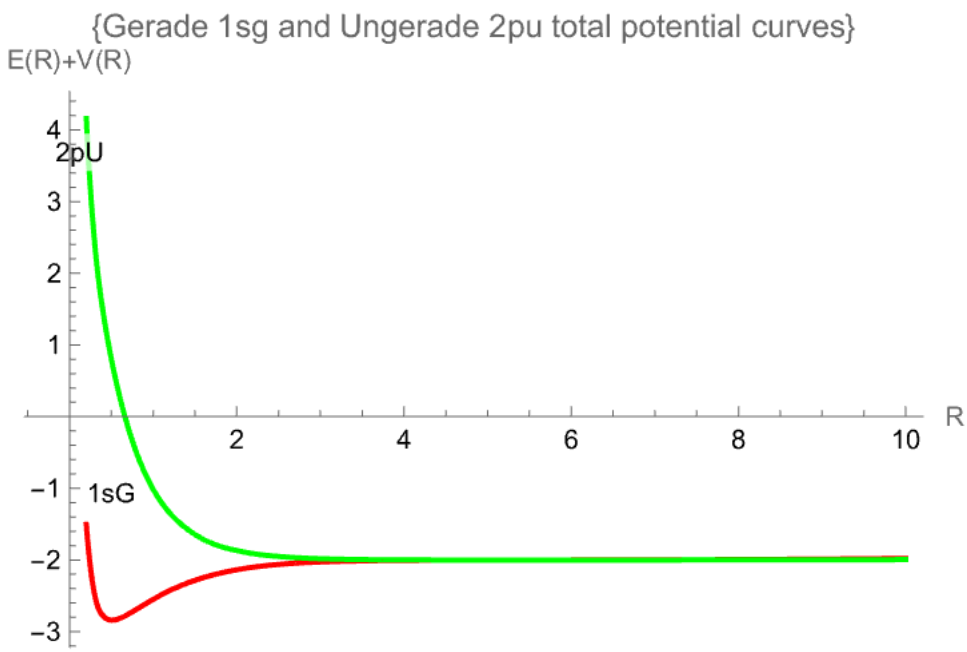


Figure 6.4: H_2^+ Potential curves for the $1s_g^+$ and $2p_u^+$ states

CHAPTER 7

ELASTIC SCATTERING IN TWO DIMENSIONS

Here, we address scattering in two dimensions and derive an expression for the elastic scattering cross section.

The quantum scattering in 3 dimensions is described in numerous textbooks and papers, so we omit explicitly referencing them here.

Scattering in 2 dimensions follows the same approach as in 3D, with some subtle differences. We start with the Schrodinger equation in 2 dimensions, using atomic units:

$$\left(-\frac{1}{2}\nabla^2 + V(\mathbf{r})\right)\psi(\mathbf{r}) = E\psi(\mathbf{r}) \tag{7.1}$$

where ∇ is the 2D Laplacian: $\nabla^2 = \frac{\partial^2 x}{\partial x^2} + \frac{\partial^2 y}{\partial y^2}$

Switching to polar coordinates and using polar coordinates 2D Laplacian:

$$\left[-\frac{\partial^2}{\partial r^2} - \frac{1}{r}\frac{\partial}{\partial r} - \frac{1}{r^2}\frac{\partial^2}{\partial \phi^2} + V(r)\right]\psi(r,\phi) = 2E\ \psi(r,\phi) \tag{7.2}$$

Using separation of variables: $\psi(r,\phi) = F(r)T(\phi)$, we end up with 2 ordinary differential equations. We set $k^2 = 2E$

$$\begin{aligned} &\frac{d^2F(r)}{dr^2} + \frac{1}{r}\frac{dF(r)}{dr} + (k^2 - 2V(r) - \frac{m^2}{r^2})F(r) = 0 \\ &\frac{d^2T(\phi)}{d\phi^2} + m^2T(\phi) = 0 \end{aligned} \tag{7.3}$$

Note that the m is a quantum number, not the mass.

The function $T(\phi)$ must be single-valued and symmetric about the incident beam, taken to be centered at x. In that case, m must be an integer, and the solution must be symmetric along the x-axis. Ergo, the normalized angular function is:

$$T(\phi) = \frac{1}{\sqrt{\pi}}e^{im\phi} \tag{7.4}$$

where m is an integer, and we are using atomic units.

The radial equation $F(r)$ is:

$$\frac{d^2F}{dr^2} + \frac{1}{r}\frac{dF}{dr} + \left(k^2 - 2V(r) - \frac{m^2}{r^2}\right)F = 0 \tag{7.5}$$

Since scattering is a localized process, from the physics of the process and by examining equation (7.5), we can identify 2 regions: the far-away region and the scattering region.

FAR AWAY REGION, INCOMING PARTICLE

For the regions far away from the scattering area where $V(r) \to 0$, we have a free space Schrodinger equation with $k^2 = E$;

$$\frac{d^2F(r)}{dr^2} + \frac{1}{r}\frac{dFr(r)}{dr} + \left(k^2 - \frac{m^2}{r^2}\right)F(r) = 0 \tag{7.6}$$

Here we have an incoming particle (or particle beam) with the uniform momentum $\vec{p}$. So the general solution of the equation (7.5) in the incoming plane wave: $F(x) = e^{i\ kx}$, where $k = \hbar\, p$.

The way to solve the equation (7.6) is to expand the incident wave in terms of the Bessel function. This is the Partial-Wave Expansion method. We get:

$$e^{i\ kx} = e^{i\ kr\cos\theta} = \sum_{m=0}^{\infty} \epsilon_m ß^m \cos(m\theta) R_m(kr) \tag{7.7}$$

where the most general form of $R_m(kr)$ is the combination of cylindrical Bessel functions:

$$R_m(kr) = A_m J_m(kr) + B_m N_m(kr) \tag{7.8}$$

We shall subsequently show the asymptotic form of the Bessel functions.

FAR AWAY REGION, SCATTERED PARTICLE

Here we consider the region where $V(r) \to 0$

In this region, the solution to the wave equation (7.5) looks like the combina-

tion of the incoming plus the scattered outgoing wave.

$$R(r,\theta) = e^{ikx} + f(\theta)\frac{e^{ikr}}{\sqrt{r}} \tag{7.9}$$

Now we can recognize the equation (7.5) as a 2D Helmholtz equation whose solution can be expressed as:

$$\psi(\mathbf{r}) = \sum_m C_m H_m^+(kr) e^{im\phi} \tag{7.10}$$

where $H_m^+(kr)$ is the Hankel Function . And for the $r \to \infty$ we get approximate behaviour:

$$H_m(kr) \xrightarrow[r\to\infty]{} \sqrt{\frac{2}{\pi kr}} exp\left[i\left(kr - \frac{(m+1/2)\pi}{2}\right)\right] \sim e^{ikx} + f_m(\theta)\frac{e^{ikr}}{\sqrt{r}} \tag{7.11}$$

where the term $f(\theta)$ is the amplitude of the scattered wave relative to the incident wave. In the remote region, we have an incident plane wave propagating along the axis, plus the scattered outgoing wave.

The scattering amplitude f_m is, in the most general case, related to the phase shift δ_m as $f(\theta) \sim e^{i\delta(\theta)} - 1$.

In the often-used partial wave expansion, the scattering amplitude and the phase shift are related as:

$$f_m = \frac{e^{2i\delta_m}}{2ik} = \frac{e^{i\delta_m}\sin(\delta_m)}{k} \tag{7.12}$$

In general, the shape of the $f(\theta)$ will depend on the symmetry of the system.

The actual $f(\theta)$ is not observable; however, it is used to compute differential

cross section which is observable.

$$\frac{d\sigma}{d\theta} = \frac{1}{k}\left|f(\theta)\right|^2 \tag{7.13}$$

SCATTERING REGION

In this case, we consider the scattering area, where $V(r) \neq 0$. In this case, we use the potential function $V(R)$ obtained in Chapter 2 and numerically solve the 2D Schrodinger equation. To obtain the phase shift, we match the wave function in the scattering region to the wavefunction in the far-away region and use the optical theorem.

In these regions, the radial Schrodinger equation $F(r)$ has the form:

$$\frac{d^2F}{dr^2} + \frac{1}{r}\frac{dF}{dr} + \left(k^2 - 2V(r) - \frac{m^2}{r^2}\right)F = 0 \tag{7.14}$$

SCATTERING PROCESS

Now we expand:

$$e^{ikx} = e^{ikr\cos\theta} = \sum_m \epsilon_m i^m \cos(m\theta) J_m(kr) \tag{7.15}$$

where $\epsilon_m = 2$ for $m \neq 0$ and $\epsilon_0 = 1$.

Note: This looks very similar to the 3D case. The main difference is that in the 2D case, the solution is a sum of cylindrical Bessel functions $J_m(kr)$ and $N_m(kr)$. In the 3D case, the solution is the sum of the spherical Bessel functions.

Using asymptotic expressions for $r \to \infty 0$:

$$\begin{aligned} J_m(kr) &\xrightarrow{r\to\infty}= \sqrt{\frac{2}{\pi kr}} \cos\left(kr - \frac{m\pi}{2} - \frac{\pi}{4}\right) \\ N_m(kr) &\xrightarrow{r\to\infty}= \sqrt{\frac{2}{\pi kr}} \sin\left(kr - \frac{m\pi}{2} - \frac{\pi}{4}\right) \end{aligned} \tag{7.16}$$

The most general solution of the radial part in the absence of potential is:

$$R(r) = A_m J_m(kr) + B_m N_m(kr) \tag{7.17}$$

For $r \to \infty$ the (7.17) has asymptotic form

$$R(r) = A_m \frac{1}{\sqrt{kr}} \cos\left[kr - \pi\frac{m+1/2}{2} + \delta_m\right] \tag{7.18}$$

Now equating equations (7.18) and and using (7.15),(7.16) we obtain:

$$\begin{aligned} &\sum_m \epsilon_m i^m \sqrt{\frac{2}{\pi kr}} \cos(m\theta) \cos\left[kr - \pi\frac{m+1/2}{2}\right] + f(\theta)\frac{e^{ikr}}{\sqrt{r}} = \\ &\sum_m A_m \frac{1}{\sqrt{kr}} \cos\left[kr - \pi\frac{m+1/2}{2} + \delta_m\right] \cos(m\theta) \end{aligned} \tag{7.19}$$

Now this can be rewritten as:

$$\begin{aligned} &\sum_m \epsilon_m i^m \sqrt{\frac{2\cos(m\theta)}{\pi kr}} \left[e^{i\left(kr - \pi\frac{m+1/2}{2}\right)} + e^{-i\left(kr - \pi\frac{m+1/2}{2}\right)}\right] + f(\theta)\frac{e^{ikr}}{\sqrt{r}} = \\ &\sum_m A_m \frac{\cos(m\theta)}{\sqrt{kr}} \left[e^{i\left(kr - \pi\frac{m+1/2}{2} + \delta_m\right)} + e^{-i\left(kr - \pi\frac{m+1/2}{2} + \delta_m\right)}\right] \end{aligned} \tag{7.20}$$

Now coefficients with e^{ikr} and e^{-ikr} must be equal. After some algebra and following[88] we get:

$$A_m = 2\epsilon_m \frac{i^m}{\sqrt{2\pi k}} e^{i\delta_m} \tag{7.21}$$

So the scattering amplitude is:

$$f(\theta) = \sqrt{\frac{1}{2\pi i k}} \sum_{m=0}^{\infty} \epsilon_m \cos(m\theta) \left[e^{2i\delta_m} - 1 \right] \tag{7.22}$$

7.1 Elastic Scattering

In the PSS (perturbed stationary state) approximation, and after factoring out the center of mass coordinates, the system wave amplitude is given by the expression

$$\psi_g(\mathbf{R})\phi_g(\mathbf{R},\mathbf{r}) + \psi_u(\mathbf{R})\phi_u(\mathbf{R},\mathbf{r}) \tag{7.23}$$

where $\phi_{u,g}(\mathbf{R},\mathbf{r})$ are, respectively, the gerade and ungerade electronic ground BO eigenstates and $\psi_g(\mathbf{R}), \psi_u(\mathbf{R})$ are the gerade and un-gerade amplitudes for the relative motion of the atom and projectile. The amplitudes satisfy the following

$$-\frac{1}{2\mu}\nabla^{\mathbf{2}}\psi_{u,g}(\mathbf{R}) + V_{u,g}(R)\psi_{u,g}(\mathbf{R}) = E\,\psi_{u,g}(\mathbf{R}) \tag{7.24}$$

where $E = k^2/2\mu$ is the collision energy, and $V_{u,g}$ are the Born-Oppenheimer potentials for the gerade and ungerade electronic states respectively. Here ∇^2 is the two-dimensional Laplacian, and because $V_{g,u}(R)$ are rotationally invariant, we can express the amplitudes as a partial wave expansion

$$\psi_{g,u}(\mathbf{R}) = \sum_{m=-\infty}^{m=\infty} e^{im\phi} F_m^{g,u}(R) \tag{7.25}$$

where the radial functions $F_m(R)$ obey

$$F_m''(R) + \frac{F_m'}{R} - \frac{m^2}{R^2}F_m(R) + (k^2 - 2\mu V_{g,u}(R))F_m(R) = 0 \tag{7.26}$$

where we suppressed the g, u superscripts in the radial functions, and the prime notation denotes derivatives with respect to R.

In the asymptotic region as $R \to \infty$ and $V_{g,u} \to 0$, Eq. (7.26) tends to the Bessel equation for integer order, and the radial solutions tend to

$$F_m(R) \to a_m \, J_{|m|}(kR) + b_m H^{(1)}_{|m|}(kR) \tag{7.27}$$

where $J_m(kR)$ is the regular Bessel function and $H^1_m(kR)$ is a Hankel function of the first kind, and a_m, b_m are constants. We impose the boundary condition

$$\psi(\mathbf{R}) \to e^{i\,kR\,\cos\phi} + f(\phi)\,\frac{e^{i\,kR}}{\sqrt{R}} \tag{7.28}$$

Using the identity

$$e^{ikR\cos\phi} = \sum_{m=-\infty}^{m=\infty} -i^{-|m|} J_{|m|}(k\,R) e^{im\phi} \tag{7.29}$$

we set $a_m = -i^{-|m|}$ and inserting into (7.28) we get another equation

$$\psi(\mathbf{R} \to e^{i\,kR\,\cos\phi} + \frac{e^{ikR}}{\sqrt{R}} \sqrt{\frac{2}{\pi\,k}} \sum_m (-1)^{m+1} e^{-i\frac{\pi}{4}} e^{im\phi} b_m \tag{7.30}$$

where we used the asymptotic form

$$H^{(1)}_m \to \sqrt{\frac{2}{\pi k R}}\, e^{\left[i\,(k\,R - \frac{m\pi}{2} - \frac{\pi}{4})\right]}. \tag{7.31}$$

We can solve for the coefficients $b_{|m|}$ by matching the radial logarithmic derivative

$$y_m \equiv \frac{F'_m}{F_m} \tag{7.32}$$

evaluated at a cutoff distance R_c so that the ratio

$$i^{-|m|}\,\frac{J'_{|m|} - y_m\,J_{|m|}}{H'^{(1)}_{|m|} - y_m\,H^{(1)}_{|m|}} \tag{7.33}$$

tends to a constant value $b_{|m|}$ at $R > R_c$.

Using the expression

$$\mathbf{j} = Re\left[\frac{-i}{\mu}\,\psi^*(\mathbf{R})\nabla\psi^*(\mathbf{R})\right] \tag{7.34}$$

where ∇ is the two-dimensional gradient operator for the probability current, we obtain for the incident current along the $z-$ direction

$$j_{inc} = \frac{k}{\mu} \tag{7.35}$$

and for the leading order term of outgoing radial current

$$j_r = \frac{k}{R\mu} f^*(\phi) f(\phi) \tag{7.36}$$

Defining the cross section as the ratio of particle flux scattered into the region $R\,d\phi$ centered at angle ϕ with the incident current j_{inc}, we get

$$d\sigma \equiv |f(\phi)|^2\,d\phi \tag{7.37}$$

Comparing equation (7.28) with the equation (7.30) we find that

$$f(\phi) = \sqrt{\frac{2}{\pi k}}\sum_m (-1)^{(m+1)} e^{-i\frac{\pi}{4}}\,e^{im\phi}\,b_{|m|} \tag{7.38}$$

Thus

$$\sigma = \int_0^{2\pi} d\phi\,\frac{d\sigma}{d\phi} = \int_0^{2\pi} d\phi |f(\phi)|^2 = \frac{4}{k}\sum_m |b_{|m|}|^2 \tag{7.39}$$

Alternatively, the constant b_m is sometimes expressed as a phase shift.

CHAPTER 8

RESONANT CHARGE TRANSFER TWO DIMENSIONS

8.1 THEORY

8.1.1 LONG RANGE BEHAVIOR OF $E(R)$

In the calculation for the phase shifts, or b_m, we use the accurate g, u potential curves calculated using the method outlined in Chap. 6.

However, for large internuclear separations $R \approx> 13a_0$, that numerical method becomes unstable. Instead, we match those calculated curves with the analytic asymptotic form described below. Although the LCAO approximation given above correctly predicts the asymptotic energy $E(R) \to -2$ as $R \to \infty$, it does not accurately predict the long range behavior of $E(R)$. To that end we note that as $R \to \infty$,

$$V(\mathbf{r}, \mathbf{R}) + 1/R \to -\frac{1}{r} - \frac{z}{R^2} + \frac{x^2 - 2z^2}{2R^3} + \dots \tag{8.1}$$

where z is the electronic coordinate along the direction $\mathbf{R}$, and we ignored

terms that contain higher inverse powers of R. Thus the Hamiltonian

$$H \approx H_0 - \frac{z}{R^2} + \frac{x^2 - 2z^2}{2R^3} + \dots \tag{8.2}$$

where H_0 is the Hamiltonian for the 2D hydrogen atom. Making use of perturbation theory,

$$E(R) \approx E_0 + E^{(1)}(R) + E^{(2)}(R) + \cdots \tag{8.3}$$

and

$$E^{(1)} = -\frac{\langle \psi_{1s}|z|\psi_{1s}\rangle}{R^2} + \frac{\langle \psi_{1s}|(x^2 - 2z^2)|\psi_{1s}\rangle}{2R^3} = 0 \tag{8.4}$$

The lowest non-vanishing contribution arises from second-order perturbation theory

$$E^{(2)} = -\frac{C_4}{R^4}$$

$$C_4 = \sum_n \frac{|\langle \psi_{1s}|z|\psi_n\rangle|^2}{E_{1s} - E_n}$$

The polarization constant $C_4 = 0.082$ was given in[67].

8.2 Resonant Charge Transfer Cross Sections

Consider the amplitude given by expression (7.27)

$$F(\mathbf{R}, \mathbf{r}) = \psi_g(\mathbf{R})\phi_g(\mathbf{R}, \mathbf{r}) + \psi_u(\mathbf{R})\phi_u(\mathbf{R}, \mathbf{r}) \tag{8.5}$$

In the limit as $R \to \infty$,

$$F \to \psi_g(\mathbf{R})\frac{1}{\sqrt{2}}(\phi_a + \phi_b) + \psi_u(\mathbf{R})\frac{1}{\sqrt{2}}(\phi_a - \phi_b) \tag{8.6}$$

where

$$\phi_a \equiv \psi_{1s}(\mathbf{r} - \mathbf{R}/2) \quad \phi_b \equiv \psi_{1s}(\mathbf{r} + \mathbf{R}/2) \tag{8.7}$$

Now imposing the boundary conditions

$$\psi_{g,u}(\mathbf{R}) \to e^{ikR\cos\phi} + f_{g,u}(\phi)\frac{e^{ikR}}{\sqrt{R}} \tag{8.8}$$

we get

$$F \to \sqrt{2}e^{ikR\cos\phi}\phi_a + \frac{1}{\sqrt{2}}(f_g(\phi) + f_u(\phi))\phi_a + \frac{1}{\sqrt{2}}(f_g(\phi) - f_u(\phi))\phi_b \tag{8.9}$$

Due to the H nuclei being equal, the charge-transfer probability is governed by the phase difference. So the ratio of flux coming into the ϕ_a channel to that going into the ϕ_b channel (i.e. resonant charge transfer) is given by

$$\sigma_{RCT} = \frac{1}{4}\int_0^{2\pi} |f_g(\phi) - f_u(\phi)|^2 = \frac{1}{k}\sum_m \left| b_{|m|}^{(g)} - b_{|m|}^{(u)} \right|^2 \tag{8.10}$$

where $b_{|m|}^{(g,u)}$ are the elastic scattering coefficients for the gerade and ungerade potentials, respectively.

charge transfer cross section. We compute it in the Wolfram Mathematica code listed in Appendix E.

The following is the plot of the charge transfer cross section in a.u. vs energy. We did a calculation for the range of $10\mu eV$ to $1meV$ which corresponds to temperatures from $0.11K$ to $11K$.

We also calculated the resonant charge transfer cross for each of the first 5 partial waves. The following graphs show the cross section for the same energy (temperature) range.

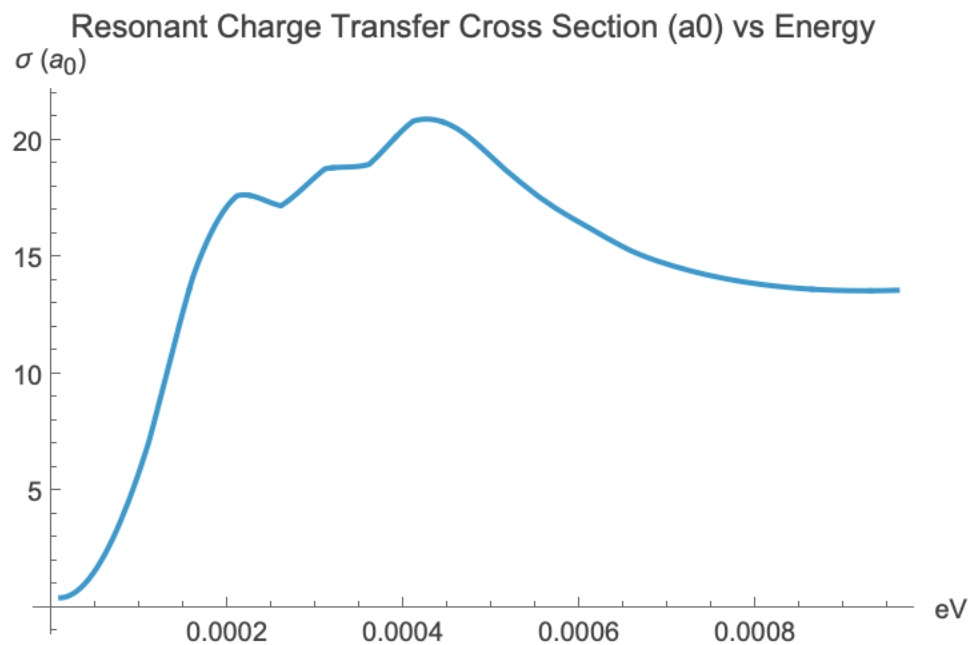


Figure 8.1: Charge Transfer Cross Section corresponding to the first 5 partial waves

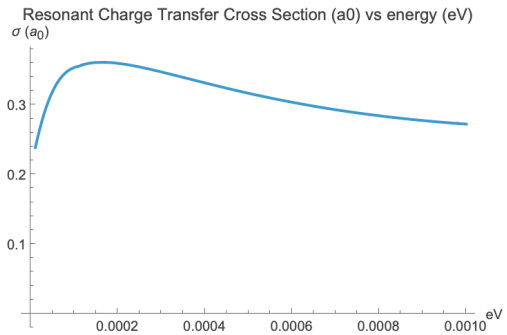
8.3 Anyons

Here we briefly discuss the anyon case, i.e since the two H_2^+ nuclei are confined in 2D, they obey neither Fermi-Dirac nor Bose-Einstein statistics.

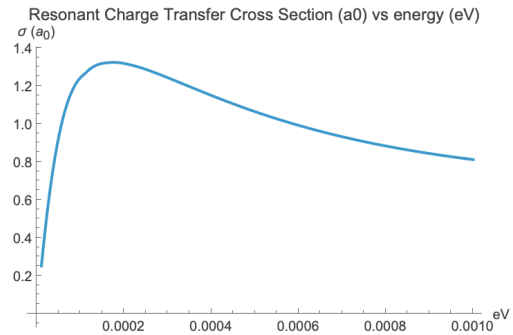
The preceding form of the 2D Schrodinger equation and its solution in 2D remain the same. For two identical particles, in polar coordinates we impose:

$$\Psi(r, \theta + 2\pi) = e^{i2\pi\alpha}\Psi(r, \theta) \tag{8.11}$$

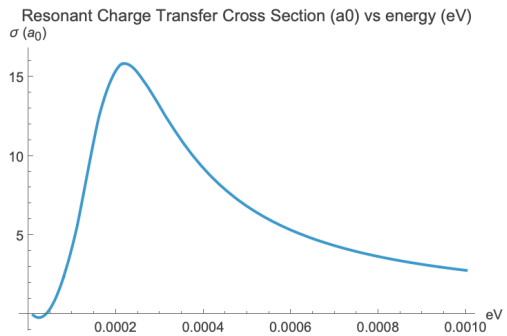
where we have for $\alpha = 0$ - bosons, $\alpha = 1/2$ - fermions, $\alpha =$ any other value - anyons.



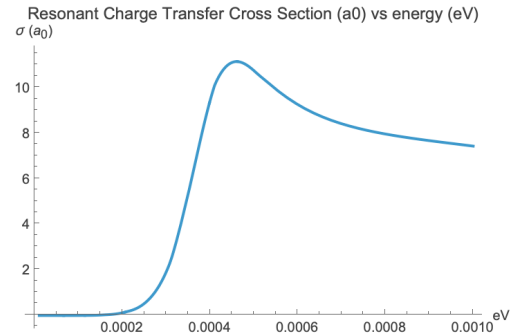
(a) Resonant Charge Transfer Cross Section, 1st partial wave



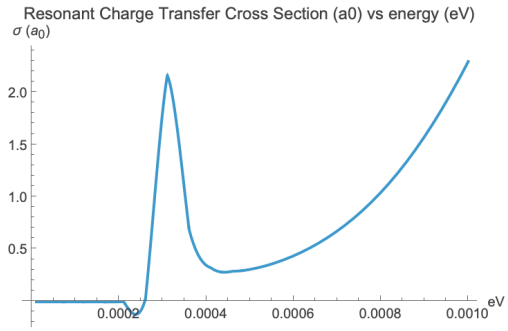
(b) Resonant Charge Transfer Cross Section, 2nd partial wave



(c) Resonant Charge Transfer Cross Section, 3rd partial wave



(d) Resonant Charge Transfer Cross Section, 4th partial wave



(e) Resonant Charge Transfer Cross Section, 5th partial wave

Resonant Charge Transfer Cross Section for each of the first 5 partial waves

In the anyon case, the partial waves becomes fractional, i.e. the phase shift and scattering matrix elements become:

$$\delta_m(k), S_m(k) \to \delta_{m+\alpha}(k), S_{m+\alpha}(k) \tag{8.12}$$

and the scattering amplitude becomes fractional

$$f_\alpha(\theta) = \frac{1}{\sqrt{2\pi k}} \sum_{m=-\infty}^{\infty} (S_{m+\alpha} - 1)\, e^{im\theta} \tag{8.13}$$

Finally the scattering cross section changes to:

$$\sigma(k;\alpha) = \frac{4}{k} \sum_{m=-\infty}^{+\infty} \sin^2(\delta_{m+\beta}) \tag{8.14}$$

The charge transfer cross section becomes:

$$\sigma_{CT}(k;\alpha) = \frac{4}{k} \sum_{m=-\infty}^{+\infty} \left|\delta^+_{m+\beta} - \delta^-_{m+\beta}\right| \tag{8.15}$$

Note that protons are not anyons in 3D. To make them behave as anyons we need 2D topological medium or quantum Hall effect.

The rest of the computation would be same, taking into account

Table 8.1: Accurate LCAO Energy as Function or R (internuclear distance) for the gerade state

R	E(R) + V(R) accurate	E(R) + V(R) LCAO
0.2	-1.50328	1.11882
0.3	-2.48639	-0.436367
0.4	-2.77548	-1.14571
0.5	-2.8382	-1.51849
0.6	-2.81481	-1.72704
0.7	-2.75759	-1.84613
0.8	-2.68852	-1.91333
0.9	-2.61749 -1.94936	
1.	-2.54904	-1.96642
1.1	-2.4852	-1.97198
1.5	-2.28416	-1.95159
2.	-2.13638	-1.9237
2.5	-2.0621	-1.91727
3.	-2.02753	-1.9224
3.5	-2.01226	-1.93066
4.	-2.00568	-1.93856
4.5	-2.00284	-1.94523
5.	-2.00156	-1.95066
6	-2.00065	-1.95882
7	-2.00036	-1.96462
8	-2.00023	-1.96898
9	-2.00016	-1.97238
10	-2.00012	-1.97512
11	-2.00009	-1.97736
12	-2.00006	-1.97924

Table 8.2: Accurate and LCAO Energy as Function or R (internuclear distance) for the ungerade state

R	E(R) + V(R) accurate	E(R) + V(R) LCAO
0.2	4.15868	2.01244
0.3	2.42148	0.534181
0.4	1.46676	-0.186921
0.5	0.798837	-0.617186
0.6	0.273983	-0.905274
0.7	-0.153558	-1.11241
0.8	-0.502669	-1.26834
0.9	-0.786143	-1.38934
1.	-1.01521	-1.48521
1.1	-1.1999	-1.56225
1.5	-1.64494	-1.75286
2.	-1.86744	-1.85654
2.5	-1.95029	-1.90027
3.	-1.98193	-1.92094
3.5	-1.99399	-1.93255
4.	-1.99845	-1.94038
4.5	-2.	-1.94638
5.	-2.00046	-1.95129
6	-2.00049	-1.95897
7	-2.00034	-1.96465
8	-2.00023	-1.96899
9	-2.00016	-1.97239
10	-2.00012	-1.97512
11	-2.00009	-1.97736
12	-2.00007	-1.97923

CHAPTER 9

CONCLUSION

In this thesis, we considered the 2 dimensional system embedded in the 3 dimensional world. We investigated the scattering processes for particles in 2 dimensions. In Introduction we researched and discussed the application of the 2 dimensional materials, current and some possible future ones. We also discussed the applications of the 2D processes for Quantum Computing.

9.1 COMPARISON WITH 3D CASE

The paper by Dalgarno and M R C McDowell[89] reports that the charge transfer cross sections decrease from $2 \times 10^{-14} cm^2$ at an impact energy of 1 eV to $0.3 \times 10^{-14} cm^2$ at 1000 eV and the diffusion cross sections lead to values of the mobility of H- ions in atomic hydrogen decreasing from $3.5\, cm^2\, volt^{-1}\, sec^{-1}$ at 100°K to $1.8\, cm^2\, volt^{-1}\, sec^{-1}$ at 600°K.

Also the paper by B. Zygelman et all[61] in the ultra-cold limit shows values of $A(R)$ in the order of $10^6\, s^{-1}$.

In our case, we did calculation for the ultracold limit, from $10meV$ to $1eV$. The values for the cross section computed are show figure 8.1

In this thesis, we considered the 2 dimensional system embedded in the 3 dimensional world. We investigated the scattering processes for particles in 2 dimensions. In Introduction we researched and discussed the application of the 2 dimensional materials, current and some possible future ones. We also discussed the applications of the 2D processes for Quantum Computing.

In 2D, the difference in potential between the gerade and ungerade states $X^2\Sigma^+$, $A^2\Sigma^+$ respectively is smaller than in the 3D case, so the probability of a transition is greater.

Following the Introduction we focused on the hydrogen molecular ion H_2^+. We recalculated the existing result in 3 dimension. Then we applied the same technique to calculate the energy levels of the hydrogen molecular ion,H_2^+ in 2 dimensions. Our approach produced results quickly but the accuracy was lacking. We used an existing approach to increase the accuracy of the calculation. In chapters 3, 4 we used the results from the chapter 2 to calculate the scattering length, then compute dipole moment and Einstein's A coefficients as function of the inter-nuclear distance for the Radiative Quenching.

The results obtain are different from the 3D case. The potential curves for the $A^2\Sigma^+$ and $X^2\Sigma^+$ are similar but the potential difference between states is smaller.

The computed value of the Einstein's coefficients is different than in the 3D

case, indicating that excited state of the H_2^+ is quite unstable. In addition the potential well of the lowest $X^2\Sigma^+$ state is quite shallow, so in 2D case even the H_2^+ is less stable than its 3D counterpart.

The references contain books and articles used to support the above investigation and calculation.

P.S.

Copying from one place is plagiarism, copying from multiple places is research.

APPENDIX A

CENTER OF MASS AS A PLANE WAVE

Take the isolated system of 2 interacting spinless particles. The particles are of masses m_1 and m_2 and are located at positions $\mathbf{r}_1$ and $\mathbf{r}_2$. The potential energy of the particles depends only on their relative position $\mathbf{r}_1 - \mathbf{r}_2$. The Langrangian of this system can be written as:

$$L = T - V = \frac{1}{2}m_1 \cdot r_1^2 + \frac{1}{2}m_2 \cdot r_2^2 + V(\mathbf{r}_1 - \mathbf{r_2}) = \frac{1}{2}M \cdot R^2 + \frac{1}{2}\mu \cdot r^2 + V(\mathbf{r}) \quad \text{(A.1)}$$

where

$M = m_1 + m_2$ is a total mass of the system

$\mu = \dfrac{m_1 m_2}{m_1 + m_2}$ is a reduced mass of the system

$\mathbf{R} = \dfrac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}$ are the coordinates of the center of the mass of the system, and

$\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$ are the relative coordinates.

(A.2)

with:

$p_i = \dfrac{\partial L}{\partial q_i}$ we define

$$\mathbf{P} = M \cdot \mathbf{R} \quad \text{and } \mathbf{p} = \mu \cdot \mathbf{r} \text{ and we obtain} \tag{A.3}$$

$\cdot\mathbf{P} = 0 \quad \cdot\mathbf{p} = -\nabla V \mathbf{r}$

Now using the variables $\mathbf{P}, \mathbf{R}, \mathbf{p}, \mathbf{r}$ we get for the Hamiltonian:

$$H = \frac{P^2}{2M} + \frac{p^2}{2\mu} + V(\mathbf{R}) = H_{CM} + H_r \tag{A.4}$$

The first term represents the motion of the center of the mass, which is a uniform inertial motion, since $\cdot\mathbf{P} = 0$. The other terms represent the energy associated with the motion relative to the center of the mass. Now if we choose the inertial frame in which the center of the mass is at rest, we get the Hamiltonian to reduce to H_r, representing the motion of a single, fictitious particle in an external potential.

Now transitioning to a quantum case. The variables $H, \mathbf{P}, \mathbf{R}, \mathbf{p}, \mathbf{r}$ become operators, with the usual commutation relations:

$$\begin{gathered} [R_i, P_j] = [r_i, p_j] = i\delta_{ij} \text{ and } [R_i, p_j] = [r_i, P_j] = 0 \implies \\ [H_{cm}, H_r] = 0,\ [H_r, H] = 0\, [H_{CM}, H] = 0 \end{gathered} \tag{A.5}$$

Since all the Hamiltonians commute, there exists a common eigenbasis in some state space E of H, H_{CM}, H_r. We can express the space E as a tensor product of spaces E_R and E_{cm} with $E = E_{CM} \otimes E_r$. The operators $\mathbf{P}, \mathbf{R}$ operate in space E_{CM} and operators $\mathbf{p}, \mathbf{r}$ operate in E_r space. From the La-

grangian and the Hamiltonian above we observe that the motion of the center of mass and the relative motion are completely independent of each other.

$$H_{CM}\left|\chi\right\rangle = E_{CM}Ket\chi \quad H_r\left|\omega\right\rangle = E_rKet\omega$$

$$H\left|\phi\right\rangle = (H_{CM}+H_r)(\left|\chi\right\rangle \otimes Ket\omega) = H_{CM}(\left|\chi\right\rangle \otimes Ket\omega) + H_r(\left|\chi\right\rangle \otimes Ket\omega) = (E_{CM}+E_r)(\left|\chi\right\rangle \otimes Ket\omega \tag{A.6}$$

Expressing the expressions above in coordinate representation, we have:

$$\phi(\mathbf{R},\mathbf{r}) = \chi(\mathbf{R})\omega(\mathbf{r})$$

$$H_{CM}\chi(\mathbf{R}) = E_{CM}\chi(\mathbf{R}) \quad H_r\omega(\mathbf{r}) = E_r\omega(\mathbf{r}) \tag{A.7}$$

$$-\frac{\hbar_R^2}{2M}\nabla^2\chi(\mathbf{R}) = E_{CM}\chi(\mathbf{R}) \quad -\frac{\hbar^2}{2M}\nabla_r^2\omega(\mathbf{r}) = E_r\omega(\mathbf{r})$$

So we can solve for $\chi(\mathbf{R})$ to obtain:

$$\chi(\mathbf{R}) = \frac{1}{(2\pi\hbar)^{3/2}}e^{\frac{i}{\hbar}\sqrt{2ME_{CM}}\mathbf{R}} \tag{A.8}$$

APPENDIX B

MATHEMATICA CODE FOR CALCULATING THE EIGENVALUES FOR THE H2 ION IN 3 DIMENSIONS

This is a fairly simple program which illustrates the power of Mathematica (and similar) mathematical applications. Following the derivation in chapter 1, we are left with two differential equations $M(\mu)$ and $L(\lambda)$. Each equation depends on two parameters $p^2 = -\frac{1}{4}R^2E$ where E is the electron energy and separation constant A.

We then create a array of values of R (nuclear distance), and for each value or F calculate eigenvalues of the equations (5.28) and (5.33), as a functions of p and A. Interpolate and the intersection of the functions of p and A provides an eigenvalue.

The last step is to add nuclear potential energy $1/R$, tabulate and plot the function.

```
(* Matrix dimensions *)
numCoefficents := 10;

numInterPoints := 20;

d[m_, n_] := KroneckerDelta[m, n];

lP[n_, x_] := LegendreP[n, x];

mSum[n_, x_] := (-n (n + 1) lP[n, x] + p^2 x^2 lP[n, x]);
```

$$\text{mSumExpanded}[m_, n_] := \left(\frac{-2\,n\,(n+1)}{2\,n+1} + p^2\,\frac{2\,(2\,n^2+2\,n-1)}{(2\,n+3)\,(2\,n+1)\,(2\,n-1)}\right) d[m, n] + \left(p^2\,\frac{2\,(n+1)\,(n+2)}{(2\,n+1)\,(2\,n+3)\,(2\,n+5)}\right) d[m, n+2] + \left(p^2\,\frac{2\,n\,(n-1)}{(2\,n+1)\,(2\,n-1)\,(2\,n-3)}\right) d[m, n-2];$$

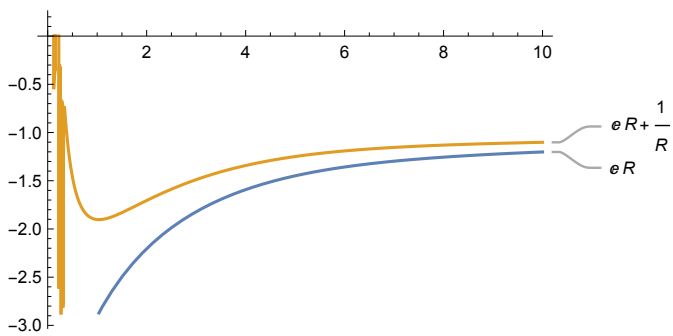
$$\begin{aligned}\text{lSumExpanded}[m_, n_] :=\ & \left(-2\,p\,n\,(2\,n+1) - (2\,p-1)\,n^2 + (-2\,p+2\,R)\,(2\,n+1) + (-4\,p+1)\,n + \left(-2\,p-p^2+2\,R\right)\right) \\ & d[m, n] + \left(2\,p\,n\,(n+1) - (-2\,p+2\,R)\,(n+1)\right) d[m, n+1] + \\ & \left(2\,p\,n^2 + (2\,p-1)\,n\,(2\,n-1) - (-2\,p+2\,R)\,n + (4\,p-3)\,n\right) d[m, n-1] + \\ & \left(-(2\,p-1)\,n\,(n-1)\right) d[m, n-2];\end{aligned}$$

```
tableEofR = {};


allRs = Table[i, {i, 0.2, 5, 0.2}];
allRs  = Join[allRs, Table[i, {i, 5.5, 9, 0.5}]];

CalcE[radius_, max_, np_] := Module[{matrixM, matrixL, ee, q,
   pe, rm, pMax, pMin, pStep, eigenM, eigenL, pgrid, mFunc, lFunc},

    (* m = rows, n = columns.
     Sums go from 0 (zero), matrix indices go from 1
   *)


  (* M matrix *)
  matrixM = Table[0, {i, 1, max}];
  Do[
   matrixM[[m + 1]] = Table[mSumExpanded[m, n] (2 m + 1)/2, {n, 0, max - 1}],
   {m, 0, max - 1}
  ];
```

```
(* L Matrix  For Power series solution *)

matrixL = Table[0, {i, 1, max}];

Do[
 matrixL[[m + 1]] = Table[lSumExpanded[m, n], {n, 0, max - 1}],
 {m, 0, max - 1}
] ;

q = 2.;
pMax = (2. q radius)/2. + 1.;
pMin = radius/2.;
pStep = (pMax - pMin)/np;

(*Output *)

(* {p, eigenvalue @ p } *)
eigenM = Table[0, {i, 1, np}, {j, 1, 2}];
eigenL = Table[0, {i, 1, np}, {j, 1, 2}];
pgrid = Table[pMin + (i - 1) * pStep, {i, 1, np}];

rm[x_] := If[Im[x] ≠ 0, 10^-99, x];

Do[
 eigenM[[i]][[2]] =
  Sort[Map[rm, Eigenvalues[N[matrixM /. p → pgrid[[i]], 10.]]], Greater][[1]];
 eigenM[[i]][[1]] = pgrid[[i]];
 eigenL[[i]][[2]] = Sort[Map[rm,
     (Eigenvalues[N[matrixL /. {R → radius, p → pgrid[[i]]}, 10.]] (-1))]][[1]];
 eigenL[[i]][[1]] = pgrid[[i]],
 {i, 1, np}
] ;

(* Now Interpolation *)

mFunc = (eigenM // Interpolation) ;
lFunc = (eigenL // Interpolation) ;

pe = x /. FindRoot[mFunc[x] == lFunc[x], {x, radius}];
```

```
  ee = -4 (pe/radius)^2;
  ee
 ]


(* Syntax:
    [E(R), E(R)+1/R] = CalcE[1,numCoefficents, numInterPoints]
*)

Energies := CalcE[#, numCoefficents, numInterPoints] & ;
Plot[{Energies[r], Energies[r] + 1/r}, {r, 0.1, 10},
 AxesOrigin → {0, 0}, PlotLabels → {E (R), E (R) + 1/R}]
```

Out[117]=



APPENDIX C

HYDROGEN ION IN 2 DIMENSION EQUATION, MATHEMATICA CODE

```
(*UsingZhu, Xiongpaper*)
(* Distance between nuclei*)

(**)

(* L equation and coefficients *)
σ[r_]:=r/p - 1/2;
α[n_]:=(n + 1) (n + 1/2);
β[n_, r_]:=2n^2 + (4p - 2σ[r])n - A + p^2 - 2pσ[r] - σ[r]/2;
γ[n_, r_]:=(n - 1) (n - 2σ[r] - 1/2) + σ[r] (σ[r] - 1/2);

(* Create a continued fraction equation for the coefficients of L equation *)

lCoeff2[n_, r_]:=β[0,r]/α[0] + Simplify[ContinuedFractionK[-α[i - 1]γ[i, r], β[i, r], {i, 1, n}]];
```

```
(* M equation and coefficients *)

w = -A + p^2/2; q = -p^2/4;
v[n_]:=(w-n^2)/q;
(* Evaluate as a continued fraction *)
mCoeffV0[n_]:=v[0] + 2Simplify[ContinuedFractionK[-1, v[2i], {i, 1, n}]];

mCoeffV1[n_]:=v[1] - 1 + Simplify[ContinuedFractionK[-1, v[2i + 1], {i, 1, n}]];

(*Parameters :
n : Depthofthecontinuedfraction,
R : Internucleardistance,
gerade : flagindicatingwhetherwearecalculatinggeradeorungeradestates
output :
R, electronenergy, totalenergy, separationconstant, andpvalue
*)
energySingleR[n_, R_, gerade_]:=
Module[{asAndps, pe, energy, aa},
(*Now find p and A *)
If[gerade==1,
asAndps = NSolve[{lCoeff2[n, R] == 0,
mCoeffV0[n] == 0}, {A, p}, Reals],
```

```
asAndps = NSolve[{lCoeff2[n, R] == 0,
mCoeffV1[n] == 0}, {A, p}, Reals]];
aa = Sort[asAndps[[All, 1, 2]], Greater][[1]];
pe = Sort[asAndps[[All, 2, 2]], Greater][[1]];
energy = (-2)/R^2 pe^2;
{R, energy, energy + 1/R, aa, pe}
]

allRs = Table[0.2 + i * 0.1, {i, 0, 9}];
allRs = Join[allRs, Table[1 + .5 * i, {i, 1, 8}]];
allRs = Join[allRs, Table[5 + i, {i, 1, 5}]];

energiesGerade = Map[energySingleR[3, #1, 1]&, allRs];
energiesUngerade = Map[energySingleR[3, #1, 2]&, allRs];
Prepend [energiesGerade, {R, E, "E+1/r", A, p}] //MatrixForm
Prepend [energiesUngerade, {R, E, "E+1/r", A, p}] //MatrixForm

eG = Interpolation[Transpose[{energiesGerade[[All, 1]], energiesGerade[[All, 2]]}]];
eGTotal = Interpolation[Transpose[{energiesGerade[[All, 1]], energiesGerade[[All, 3]]}]];
Plot[{eG[x], eGTotal[x]}, {x, 0.2, 10}, PlotStyle->{Red, Green}, AxesLabel → {R, "(E),V(R)"},
AxesOrigin → {0, 0}, PlotLabel->{"Electron vs Total potential for 1sG (Gerade) state"},
PlotLabels->Placed[{"E(R)", "E(R)+V(R)"}, {Left}], PlotRange->Automatic]
```

```
eU = Interpolation[Transpose[{energiesUngerade[[All, 1]], energiesGerade[[All, 2]]}]];
eUTotal = Interpolation[Transpose[{energiesUngerade[[All, 1]], energiesUngerade[[All, 3]]}]];
Plot[{eU[x], eUTotal[x]}, {x, 0.2, 10}, PlotStyle->{Red, Green}, AxesLabel → {R, "(E),V(R)"},
AxesOrigin → {0, 0}, PlotLabel->{"Electron vs Total potential for 2pU (Ungerade) state"},
PlotLabels->Placed[{"E(R)", "E(R)+V(R)"}, {Left}], PlotRange->Automatic]

Plot[{eGTotal[x], eUTotal[x]}, {x, 0.2, 10}, PlotStyle->{Red, Green}, AxesLabel → {R, "E(R)+V(R)"
AxesOrigin → {0, 0}, PlotLabel->{"Gerade 1sg and Ungerade 2pu total potential curves"},
PlotLabels->Placed[{"1sG", "2pU"}, {Above, Left}], PlotRange->Full]
```

APPENDIX D

FORTRAN CODE FOR CALCULATING ENERGIES FOR THE 2D PROBLEM

To run the numerical calculations I used gfortan 5.2.0, running on 2.2 GHz, Intel Code 7, Macbook Pro 15" Laptop (mid 2014). The numerial library used was Lapack, version 3.6.0, compiled on the same machine using the gfortran above.

This is Fortran 90 code for calculating even and odd eigenvalues and consequently the energies.

```
! To Compile
! gfortran -O4 -o h2Plus h2plus.f90 hpsort.f90 -L ./ -l lapacke -l lapack -l blas -l
program main

implicit none

interface
```

```
subroutine M_Matrix(mMatrix,max,p)
integer :: max
double precision :: p
double precision,dimension(:,:) :: mMatrix
end subroutine

subroutine L_Matrix(lMatrix,max,R,p)
integer :: max
double precision :: R,p
double precision,dimension(:,:) :: lMatrix
end subroutine

subroutine CalculateEigenvalues(c,matrix,max,wr,wi)
character*1 :: c
double precision,dimension(:,:) :: matrix
integer :: max,np,count
double precision,dimension(:) :: wr,wi
end subroutine

subroutine hpsort(n,rra)
integer :: n
double precision, dimension(:) :: rra
```

```
end subroutine

end interface

! Declare local variables
integer,parameter :: max = 500   ! matrix dimesions
integer,parameter :: np = 1000  ! number of p points
integer,parameter :: state = 0   ! 0=ground state, 1=1st excited, etc..
integer,parameter :: numRadii = 1
double precision,dimension(numRadii) :: Radii
double precision,dimension(max,max) :: mMatrix,lMatrix
double precision,dimension(np) :: pGrid
double precision :: R  ! Nuclei distance
double precision,dimension(max,np) :: mEig, lEig
double precision,dimension(max) :: wr,wi
double precision :: pStep, pMin, pMax
integer :: count,i,j, index
double precision :: p,rr
character(len=6) :: fmt
character(len=16) :: rOut

i = 1;
rr = 20.0d0
```

```
do while( (rr <= 5.0d0) .and. (i <= numRadii))
Radii(i) = rr
  rr = rr + 0.2d0
i = i + 1
end do

do while(i <= numRadii)
  Radii(i) = rr
rr = rr + 0.5d0
  i = i + 1
end do

do j = 1, numRadii
R = Radii(j)

print *,"R = ", R

pMin = 0
pMax = 2.d0*R + 1.d0
pStep = (pMax - pMin) / dble(np)

count = 1;
p = pMin
```

```
do while( count <= np )
! Initialize M  matrix
call M_Matrix(mMatrix,max,p)

!Calculate eigenvalues for M matrirx
call CalculateEigenvalues('m',mMatrix,max,wr,wi)
   !  sort
do i = 1,max
if( wi(i) /= 0) then
wr(i) = -1.0d99
end if
end do

call hpsort(max,wr)
do i = 1,max
mEig(i,count) = wr(max-i+1)
end do

! And L Matrix
call L_Matrix(lMatrix,max,R,p)
call CalculateEigenvalues('l',lMatrix,max,wr,wi)
do i = 1,max
if( wi(i) /= 0) then
```

```
wr(i) = 1.0d99
end if
end do
call hpsort(max,wr)
   ! wr contains eigenvalues from largest to smallest
do i = 1,max
lEig(i,count) = wr(i)
end do

pGrid(count) = p
p = p + pStep
count = count + 1
end do

! Now write the eigenvalues to file
if( R < 10.0d0 ) then
fmt = "(F3.1)"
else
fmt = "(F4.1)"
end if
write(rOut, fmt) R
open(unit=11,file="Eigenvalues-Even-M-R="//trim(rOut)//".dat")
open(unit=12,file="Eigenvalues-Even-L-R="//trim(rOut)//".dat")
```

```
!  Todo: Excited states
index = 1
do i=1,np
write(11,*) pGrid(i), mEig(1,i),mEig(2,i), mEig(3,i), mEig(4,i), mEig(5,i)
write(12,*) pGrid(i), lEig(1,i),lEig(2,i), lEig(3,i), lEig(4,i), lEig(5,i)
end do
close(11)
close(12)
end do

end  ! main

subroutine CalculateEigenvalues(id,matrix,max,wr,wi)
! Calculate and return the eigenvalues
implicit none

character*1 ::id
double precision,dimension(:,:) :: matrix
integer :: max
double precision,dimension(:) :: wr,wi
!local
double precision,dimension(max,max) :: z
double precision,dimension(:), allocatable :: work
```

```
integer :: info,i
double precision,dimension(1,1) :: dummy
integer :: lda,lwork
double precision :: temp

lda = max
lwork = max * 11

allocate( work(lwork) )

call DGEEV('N','N',max,matrix,lda,wr,wi,dummy,1,dummy,1,work,lwork,info)

deallocate(work)

if( info /= 0 ) then
print *, "Error in info , info = ", info
call exit(1)
end if

end subroutine

subroutine M_Matrix(mMatrix,max,p)
! create the M matrix
```

```
implicit none
interface
   function Delta(m,k)
     double precision :: Delta
     integer, intent(in) :: m,k
   end function Delta
end interface

integer :: max
double precision :: p, val
  double precision,dimension(:,:) :: mMatrix
integer :: m,k

do m = 0,max-1
  do k = 0,max-1
!     Even state
val = ( (-1)*((2.0d0 * k)**2) + ( (p**2.0)/2.0d0) ) * Delta(m,k) + &
!     For the Odd state use
!     val = ( (-1)*((2.0d0 * k + 1)**2) + ( (p**2.0)/4.0d0) ) * Delta(m,k) + &
      ( (p**2)/4.0d0 ) * ( Delta(m,k+1) + Delta(m,k-1) )
mMatrix(m+1,k+1) = val
end do
end do
```

```
!     Even state
mMatrix(1,2) = 2 * mMatrix(1,2)
!     For the Odd state use
! mMatrix(1,1) = -1 + (p**2.)/4.0d0

end subroutine

subroutine L_Matrix(lMatrix,max,R,p)
! create the L matrix
implicit none
interface
   function Delta(m,k)
     double precision :: Delta
     integer, intent(in) :: m,k
   end function Delta

  function SubSum(m,k)
      double precision :: SubSum
      integer, intent(in) :: m,k
   end function SubSum
end interface
```

```
integer :: max
double precision :: p,R,val
double precision, parameter :: PI = 3.14159265358979323846264 3d0
  double precision,dimension(:,:) :: lMatrix
integer :: m,k

do m = 0,max-1
do k = 0,max-1
      val = ((-2.d0)*p*k*(2*k+1) - (2*p-1)*(k**2) - 4*p*k + &
(2*R-p)*(2*k+1)-p -(p**2)+2*R) * Delta(m,k) + &
      (2*p*k*(k+1) - (2*R-p)*(k+1))*Delta(m,k+1) + &
      (2.0d0*p*(k**2) + (2.0d0*p - 1)*k*(2.0d0*k - 1) +(4.0d0*p - 2)*k-&
(2*R - p)*k)*Delta(m,k-1) - &
      (2.0d0*p - 1)*k*(k - 1)*Delta(m,k-2) + SubSum(m,k)
lMatrix(m+1,k+1) = (-1)*val
end do
end do

end subroutine

function SubSum(m,k)
implicit none
interface
```

```
   function Delta(m,k)
     double precision :: Delta
     integer, intent(in) :: m,k
   end function Delta
end interface

double precision :: SubSum
integer, intent(in) :: m,k
integer :: i

SubSum = 0.0d0

do i = 0,k-1
   SubSum = SubSum + Delta(m,i)
end do

end function SubSum

function Delta(m,n)
implicit none
double precision :: Delta
integer, intent(in) :: m,n
```

```
if (m == n) then
    Delta = 1.d0
else
    Delta = 0.d0
end if
end function Delta
```

Appendix E

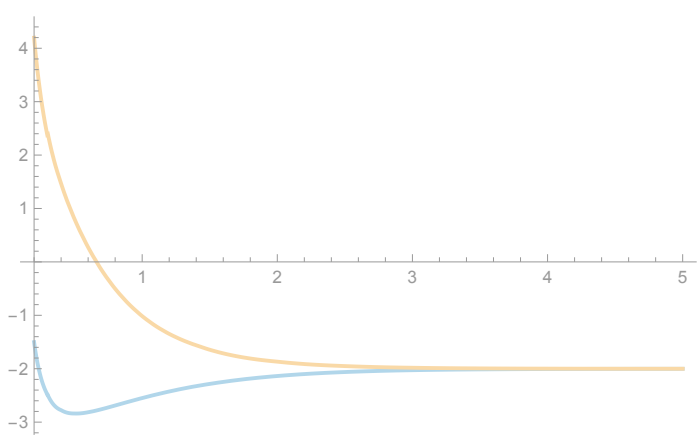
Scattering Length Code

In[1]:=

```
(* this routine needs to run Vpots.nb *)

geradeData =
  Import["/Users/zelimir/Work/Physics-Thesis/thesis-2/ZelimirOverleaf/
     MathematicaCodeFinal/gerade1sV3.mx"];
geradePotential = Table[{geradeData[[i, 1]], geradeData[[i, 3]]}, {i, 1, 27}];
ungeradeData =
  Import["/Users/zelimir/Work/Physics-Thesis/thesis-2/ZelimirOverleaf/
     MathematicaCodeFinal/ungerade1sV3.mx"];
ungeradePotential = Table[{ungeradeData[[i, 1]], ungeradeData[[i, 3]]}, {i, 1, 27}];
long[R_] := -2.0 - 0.082 / R^4;
geradefun = Interpolation[geradePotential];
ungeradefun = Interpolation[ungeradePotential];
(* Fit to a model *)
gdata = Drop[geradePotential, {5, 27}];
udata = Drop[ungeradePotential, {5, 27}];
model[R_, C1_, C2_, C3_] := C1 Exp[-C2 R] / R + C3;
gnlm = NonlinearModelFit[gdata, model[R, C1, C2, C3], {C1, C2, C3}, R];
unlm = NonlinearModelFit[udata, model[R, C1, C2, C3], {C1, C2, C3}, R];
gshort[R_] := model[R, C1, C2, C3] /. gnlm["BestFitParameters"];
ushort[R_] := model[R, C1, C2, C3] /. unlm["BestFitParameters"];
gpot[R_] = Piecewise[
   {{gshort[R], R ≤ 0.3}, {geradefun[R], 0.3 < R < 13.0}, {long[R], R ≥ 13.0}}];
upot[R_] = Piecewise[
   {{ushort[R], R ≤ 0.3}, {ungeradefun[R], 0.3 < R < 13.0}, {long[R], R ≥ 13.0}}];
Plot[{gpot[R], upot[R]}, {R, 0.2, 5}, PlotRange → All]

(*e=0.1/27.21;*)
μ = 1866.0 / 2.0;
(*k=√(2 μ e);*)
kB = 0.695; (* Boltzman in AU, cm^-1/K *)
(*λ=2π/k*)
a0 = 5.29188^-11;

r0 = 0.01;
rend = 100;

(*For a single partial wave m, energy e, and range r0 → rend*)
computePhaseShifts[m_, e_, r0_, rend_] :=
  Module[{k, eqG, eqU, solsG, solsU, fg, fu},
   k = √(2 μ e);
   equG := yg''[r] + yg'[r] / r - 2 μ (gpot[r] + 2) yg[r] - m^2 / r^2 yg[r] + 2 μ e yg[r];
   equU := yu''[r] + yu'[r] / r - 2 μ (upot[r] + 2) yu[r] - m^2 / r^2 yu[r] + 2 μ e yu[r];
```

```
    solsG = Flatten[NDSolve[
       {equG == 0, yg[r0] == 0, yg'[r0] == 1}, yg, {r, r0, rend}, MaxSteps → 200 000]];
    solsU = Flatten[NDSolve[
       {equU == 0, yu[r0] == 0, yu'[r0] == 1}, yu, {r, r0, rend}, MaxSteps → 200 000]];
    fg[r_] = yg[r] /. solsG;
    fu[r_] = yu[r] /. solsU;
    ylogG[r_] := fg'[r] / fg[r];
    ylogU[r_] := fu'[r] / fu[r];
    sl[n_, r_] := BesselJ[n, k r];
    cl[n_, r_] := HankelH1[n, k r];
    slp[n_, r_] := Derivative[0, 1][sl][n, r];
    clp[n_, r_] := Derivative[0, 1][cl][n, r];
    bG[n_, r_] :=
     (-I) ^ (Abs[n]) ( slp[n, r] - ylogG[r] × sl[n, r]) / ( clp[n, r] - ylogG[r] × cl[n, r]);
    bU[n_, r_] :=
     (-I) ^ (Abs[n]) ( slp[n, r] - ylogU[r] × sl[n, r]) / ( clp[n, r] - ylogU[r] × cl[n, r]);
    If[m == 0,
     Return[{Abs[N[bG[m, rend]]], Abs[N[bG[m, rend] - bU[m, rend]]]^2}],
     Return[{2 Abs[N[bG[m, rend]]], 2 Abs[N[bG[m, rend] - bU[m, rend]]]^2}]]
   ];
(*computePhaseShifts[10,0.003,0.1,100];*)
```

Out[18]=

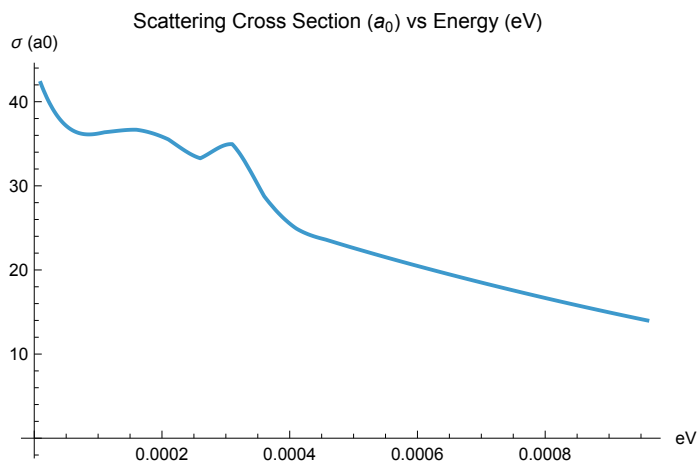


```
(* e = collision energy in eV, r0, rend - R range *)
computeCrossSect[eV_, r0_, rend_, mMin_, mMax_] :=
  Module[{allBs, eH, kk, crossSection},
   eH = eV / 27.21;  (* energy Hartree *)
   kk = Sqrt[2 μ eH];
   (* m - partial waves *)
   allBs = Table[{m, computePhaseShifts[m, eH, r0, rend]}, {m, mMin, mMax}];
   crossSection = 1/kk Total[allBs, {1}][[2]];
   {eV, crossSection}
  ];

(* Up to this partial wave *)
mMax = 5;

crossSection =
  Table[computeCrossSect[e, r0, rend, 0, mMax], {e, 0.00001, .001, 0.00005}];
```

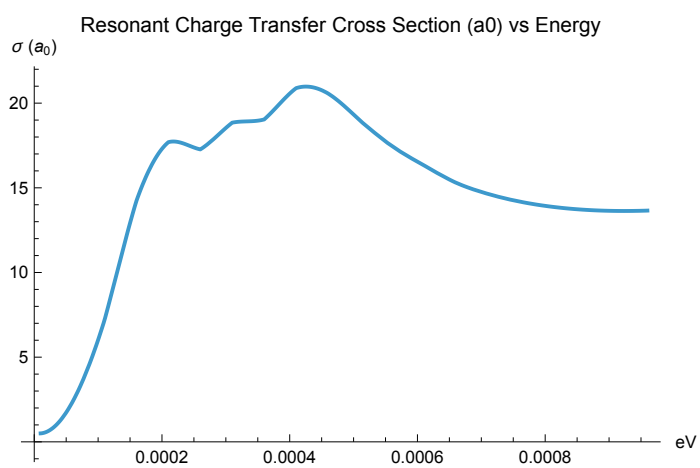
In[29]:=

```
crossSectionGraphSC =
  Interpolation[{crossSection⟦All, 1⟧, crossSection⟦All, 2⟧⟦All, 1⟧} // Transpose];
crossSectionGraphCT =
  Interpolation[{crossSection⟦All, 1⟧, crossSection⟦All, 2⟧⟦All, 2⟧} // Transpose];

start = crossSection⟦1⟧⟦1⟧
end = crossSection⟦Length[crossSection]⟧⟦1⟧
Plot[crossSectionGraphSC[e], {e, start, end},
 AxesOrigin → {0, 0}, AxesLabel → {"eV", "σ (a0)"},
 PlotLabel → "Scattering Cross Section (a0) vs Energy (eV) "]
Plot[crossSectionGraphCT[e], {e, start, end},
 AxesOrigin → {0, 0}, AxesLabel → {"eV", "σ (a0)"},
 PlotLabel → "Resonant Charge Transfer Cross Section (a0) vs Energy"]
```

Out[33]=



Out[34]=



Appendix F

Einstein A Coefficient

In[ ]:=

```
(* Radiative Quenching, Chapter 3 *)
(* This one uses my paper *)

(*electron mass *)
m_e = 1;
(* Proton Mass *)
m_p = 1836;
(* Reduced Mass *)
μ = m_p/2;
hbar = 1;
hbarSI = 1.054571817*^-34;
cLight = 137.036 ;(*SpeedOfLight, atomic units*)
cLightSI = 299 792 458 ;(*SpeedOfLight, m/s*)
oneSec = 2.418884326509*^-17; (* 1 sec SI ↔ 1 sec AU *)
aBorh = 5.29177*^-11;  (* Si ↔ AU *)

(* RawData format R,E, E + 1/R,A,p *)

vSg1RawData =
  Import["/Users/zelimir/Work/Physics-Thesis/thesis-2/ZelimirOverleaf/
     MathematicaCodeFinal/gerade1sV3.mx"];
vSu2RawData =
  Import["/Users/zelimir/Work/Physics-Thesis/thesis-2/ZelimirOverleaf/
     MathematicaCodeFinal/ungerade1sV3.mx"];


vSg1Data = vSg1RawData[[1 ;; 44]];
vSu2Data = vSu2RawData[[1 ;; 44]];

(* Calculate the Transition Dipole Moment *)
(* Upper Limit for integration *)
maxR = vSg1Data[[Length[vSg1Data]][[1]];
Print["maxR=", maxR]
maxR = 20;
maxIndex = 40;

Print["R=", vSg1Data[[maxIndex, 1]], ", A=",
  vSg1Data[[maxIndex, 4]], ", p=", vSg1Data[[maxIndex, 5]]];
Print["R=", vSu2Data[[maxIndex, 1]], ", A=",
  vSu2Data[[maxIndex, 4]], ", p=", vSu2Data[[maxIndex, 5]]];
```

In[ ]:=

```
(* Load Potential curves for 1sg, 2sg states
"R,E, E + 1/R,A,p *)
(* 1sg state, potential curve *)
vG = Interpolation[
   Transpose[{vSg1Data[[All, 1]], vSg1Data[[All, 3]]}], InterpolationOrder → 3];
(* 2sg state, potential curve *)
vU = Interpolation[
   Transpose[{vSu2Data[[All, 1]], vSu2Data[[All, 3]]}], InterpolationOrder → 3];

deltaV[r_] := Abs[vG[r] - vU[r]];
deltaV[20]
Plot[{vG[r], vU[r]}, {r, 0.2, maxR}, PlotRange → Full, PlotLegends → {"V_G", "V_U"},
 PlotLabel → "Potential curver for the Gerade and Ungerade state"]
```

In[ ]:=

```
(* Compute the Dipole momement D(r) for the single value of R *)
singleDR[R_, a1_, p1_, a2_, p2_, maxDistance_] :=
  Module[{norm1, norm2, dInt, l1, l2, m1, m2, lmNorm1, lmNorm2},
   (* 1s wavefunction *)
   l1 = NDSolveValue[{(ξ^2 - 1) L''[ξ] + ξ * L'[ξ] + (a1 + 2 R ξ - p1^2 ξ^2) L[ξ] == 0 ,
      L[1.001] == 0, L'[1.001] == 1 }, L, {ξ, 1.001, maxDistance}];
   m1 = NDSolveValue[{(1 - η^2) M''[η] - η * M'[η] - (-a1 + p1^2 η^2) M[η] == 0,
      M[0.999] == 0, M'[0.999] == 1}, M, {η, -.999, .999}];

   (* 2s wavefunction *)
   l2 = NDSolveValue[{(ξ^2 - 1) L''[ξ] + ξ * L'[ξ] + (a2 + 2 R ξ - p2^2 ξ^2) L[ξ] == 0 ,
      L[1.001] == 0, L'[1.001] == 1 }, L, {ξ, 1.001, maxDistance}];
   m2 = NDSolveValue[{(1 - η^2) M''[η] - η * M'[η] - (-a2 + p2^2 η^2) M[η] == 0,
      M[0.999] == 0, M'[0.999] == 1}, M, {η, -.999, .999}];

   Print["R=", R];
   (* Compute <ψ|ψ> = <ψ1s|ψ1s> + <ψ2s|ψ2s> *)
   (* <ψ1s|ψ1s> *)
   norm1 = Sqrt[NIntegrate[R^2/4 Abs[l1[ξ]]^2 Abs[ m1[η]]^2 (ξ^2 - η^2)/(Sqrt[ξ^2 - 1] Sqrt[1 - η^2]),
      {η, -.999, .999}, {ξ, 1.001, maxDistance}]];
   (* <ψ2s|ψ2s> *)
```

```
    norm2 = Sqrt[NIntegrate[R^2/4 Abs[l2[ξ]]^2 Abs[m2[η]]^2 (ξ^2 - η^2)/(Sqrt[ξ^2 - 1] Sqrt[1 - η^2]),
       {η, -.999, .999}, {ξ, 1.001, maxDistance}]];

    dInt = (R/2)^3 1/(norm1 norm2) NIntegrate[l2[ξ] × l1[ξ] × m1[η] × m2[η] (ξ η (ξ^2 - η^2))/(Sqrt[ξ^2 - 1] Sqrt[1 - η^2]),
       {η, -0.999, 0.999}, {ξ, 1.001, maxDistance}];
    {R, dInt }
   ];

singleDR[vSg1Data[[24]][[1]], vSg1Data[[19]][[4]],
 vSg1Data[[19]][[5]], vSu2Data[[19]][[4]], vSu2Data[[19]][[5]], 5]
(*singleDR[vSg1Data[[14]][[1]],vSg1Data[[19]][[4]],
  vSg1Data[[19]][[5]],vSu2Data[[19]][[4]],vSu2Data[[19]][[5]],5]
 singleDR[vSg1Data[[15]][[1]],vSg1Data[[20]][[4]],
  vSg1Data[[20]][[5]],vSu2Data[[20]][[4]],vSu2Data[[20]][[5]],5]*)
```

In[ ]:=

```
(* Get all Rs, and coefficients A and p and calculate D(R) for all Rs *)
(* RawData format R,E, E + 1/R,A,p *)
inputData = Transpose[
   {vSg1Data[[1 ;; maxIndex, 1]], vSg1Data[[1 ;; maxIndex, 4]], vSg1Data[[1 ;; maxIndex, 5]],
    vSu2Data[[1 ;; maxIndex, 4]], vSu2Data[[1 ;; maxIndex, 5]]}];

allDR = Map[singleDR[#[[1]], #[[2]], #[[3]], #[[4]], #[[5]], maxR] &, inputData];
```

In[ ]:=
```
(* Dipole moment in SI units *)
drAUtoSI = 8.478*^-30; (* Cm *)
allDRSI = allDR;
allDRSI[[All, 2]] = allDR[[All, 2]] * drAUtoSI;
allDRSI

dR =
  Interpolation[Transpose[{allDR[[All, 1]], allDR[[All, 2]]}], InterpolationOrder → 3];
Plot[dR[r], {r, 0.1, 7}, PlotLabel → "D(R)",
 AxesLabel → {"R(a0)", "D(R)(in a.u.)"}, PlotRange → Full]

dRSI = Interpolation[
   Transpose[{allDRSI[[All, 1]], allDRSI[[All, 2]]}], InterpolationOrder → 3];
Plot[dRSI[r], {r, 0.1, maxR}, PlotLabel → "D(R)",
 AxesLabel → {"R(a0)", "D(R) (units of C m)"}, PlotRange → Full, Axes → True]
```

In[ ]:=

$$\text{aR[R_]} := \frac{4}{3\,\text{hbar}\,\text{cLight}^3}\,(\text{dR[R]})^2\left(\frac{\text{vU[R]} - \text{vG[R]}}{\text{hbar}}\right)^3;$$

```
aR[1]

aRAUtoSI = 4.1341*^16;

(* Einstein A coefficient in SI  and a.u. units *)
```

$$\text{aRSI2[R_]} := \frac{4}{3\,\text{hbarSI}\,\text{cLightSI}^3}\,(\text{dRSI[R]})^2\left(\frac{(\text{vU[R]} - \text{vG[R]})\,4.35974{*}\text{^}{-18}}{\text{hbarSI}}\right)^3;$$

```
Plot[aRSI2[R], {R, 0.1, 1}, AxesLabel → {"R (a.u)", "A(R)s^-1"},
 PlotLabel → "Einstein's A coefficient for Σg -> Σu transition"]

Plot[aRSI2[R] * 10^6, {R, 0.1, 1}, AxesLabel → {"R (a.u.)", "A(R)10^6s^-1"},
 PlotLabel → {"Einstein's A coefficient for g -> u transition"}]
```

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CURRICULUM VITAE

PROFILE

Software engineer with over 18+ years experience in Software Engineering, Business to Business applications, Object Oriented design and methodologies. Highly Proficient in C/C++, UNIX and .NET platform. Up to date knowledge of distributed systems, Algorithms, Computational Mathematics, transaction processing, relational database and network engineering. Strong distributed system design, system programming background, excellent knowledge of mission critical applications, networks, protocols.

KEY SKILLS

• Operating Systems: UNIX (Linux, UNIX , Windows .NET, Win32, Embedded Systems

• Methodologies and concepts: Object Oriented analysis and design, Client/Server and N-tier programming, Design of Distributed Systems

• Real life experience of scaling and optimizing large systems, designing for reliability and scalability

• Throughout knowledge of C++, STL programming practices and paradigms. Early adopter of new C++ standards such as C++03, C++11.

• Excellent knowledge of full range of Microsoft technologies, and recent expertise in C# ,.NET, ASP.NET platform and SQL server (complete .NET stack), Excel RTD Server and Excel Extensions

• Deep knowledge and experience with: distributed systems, Web services, concurrent programming, threads, real time programming, embedded systems, TCP/IP and network protocols

• Experience with database theory, database programming, ORACLE 9i and 10g administration, mySQL

• Implemented Classification, Data analytics, Machine Learning solutions, using NumPY, Tensor Flow and Keras

• Theoretical background: Applied Mathematics, software engineering, digital signal processing applications, statistical process analysis, network analysis, Working experience in writing technical design specification, overseeing the development and testing process, and providing a mentorship role to junior developers and interns.

• Knowledge of practical software engineering techniques, such as basic waterfall and incremental, extreme programming and fast prototyping.

• Programming and Development tools: Gnu, SVN, MS Visual Studio, .NET, ATL, PERL, Mason, Various Unix shells, Purify, Quantify, Matlab

PROFESSIONAL EXPERIENCE, Full Time and Consulting

Tableau Software, Seattle, WA 2017-present

Senior Software Engineer

• Implemented Performance improvement of the existing query pipeline

• Participated in the Implemented of classification network for error tracking and prediction • Migrated several server components from Windows to Linux

• General feature development

• Mentored and supported junior team member.

Microsoft Corp., Redmond, WA 2016-2017

Senior Software Engineer

• Refactoring of the current code base

• Design and implementation of the local Cache subsystem

• Improved the current build and deployment process

Rocket Gaming System, Las Vegas, NV 2013-2014

Senior Math Software Engineer - Consulting

• Design and Implementation of the Game algorithms in C++

• Analyze and define the game characteristics, define and implement the mathematical model

• Refactoring, Performance and Usability improvements of the existing software and software modules.

• Programming the Game simulations of the game math models in C++ and C#.

• Improved the current methodology and development process

Bloomberg Inc, New York, NY 2012-2013

Senior Software Engineer

• Lead the design and implementation of custom Excel function handling high frequency sell side trading and data flow. The application consists of C++ TD server, and C#/.NET Excel UI extension.

• Designed and implemented the data flow, algorithms and Excel UI for the Bloomberg specific financial functions.

• Worked with other teams to implement end to end, high frequency trading and data flow applications. The applications are the real time Excel exten-

sions, passing large data sets to customer trading models, and executing customers orders in real time.

• Performance improvement, refactoring of existing applications,

Translattice, Santa Clara. CA 2010 – 2011

Senior Software Engineer

• Implementation of the Paxos algorithm for distributed DB operation on Postgres DB engine in C.

• Design and Implementation of distributed transactions, distributed table sharding for the Postgres DB

• Design and Implementation of the underlying multithreaded and multi process communication / replication functionality.

• Performance improvements, refactoring existing code, Using C for the database engine and C++ for the tools.

MySpace.com, Seattle, WA 2007 – 2010

Senior Software Engineer/Architect

• Lead the design and implementation of MySpace Messaging system, which included internal messaging and email server and client. The project integrated new messaging, email, comments, WebIm into the existing MySpace framework. Project was based on .NET platform, using C#, ASP.NET, SQL Server . • Designed and developed message passing system supporting transaction like operation.

• Developed Rest based services, using full .NET stack, from SQL Server as a back end DB to the C#/.NET as middle tier.

• Significantly refactored and the existing design, to provide the transition path to the new architecture

• Engineered and implemented the Web Service interoperability solutions between MySpace back end systems and external partners.

• Providing the design for the transition to multi level application, highly scalable system.

textbfAmazon.com. Seattle, WA 2004 – 2007

Senior Software Engineer (III)

• Leading the design of the next generation high performance, distributed software deployment system.

• Designed and developed (as a lead in the team) distributed and multithread set of C++ applications to effectively manage, update and monitor over 30,000+ hosts. The whole system consisted of several server side applications, each on separate server farm, set of client apps and daemons, persistence subsystem (Database) and a data streaming subsystem. The amount of data stored was second only to customer data within Amazon.

• Used several C++ libraries, multithread, multiprocessing methodologies and applications Justified and implemented specific solutions, related to scaling, performance improvement, operational support of the software configuration, distribution and release management

• Developed and programmed on the application server stack, from database (Oracle 9i and 10g) to application servers, Web servers, end host tools

• Improved, refactored, and significantly increased functionality of the existing application by improving performance, availability and usability

• Lead a team, provided designed review, support to junior team members

textbfPACKETEER Inc., Kelowna, BC 2002 – 2004

Software Engineer

• Implementation of compression algorithms for IP, TCP and HTTP based traffic in C.

• Member of the design and team for of the next generation TCP traffic-processing server. The application was developed in C for kernel drivers and C++ for the higher layers.

• Linux network stack programming and customization in C.

• Implementation of algorithm for traffic classification, supporting XML and SOAP protocols in C++.

• Troubleshooting of network and network based applications.

• Design of performance critical system and application

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EDUCATION

PhD Physics (Close to completion) University of Nevada Las Vegas, Las Vegas, NV

Master of Applied Mathematics University of Washington, Seattle, WA

Master of Engineering Simon Fraser University, Vancouver, BC, Canada

B.Sc. Hon in Electrical Engineering School of Electrical Engineering, University of Sarajevo